

Operations Research

From Real Decisions to Sensitivity Analysis

A textbook treatment of Lectures 1 to 5

Based on the lecture notes of El Mehdi Saad

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How to read this book

Each concept in this book is presented in the same rhythm.

1. **Motivation and intuition.** Why do we care? What problem does the idea solve, and what picture should you hold in your head?
2. **Formal statement.** The precise definition or theorem, set in a coloured box so it is easy to find when revising.
3. **Proof.** Every claim is proved. Longer proofs are split into labelled steps, and the one idea that makes the proof work is isolated in a *Key insight* box.
4. **Interpretation and discussion.** What the result really says, where it applies, and which misreadings to avoid.
5. **Examples.** From “trivially small” to “genuinely useful.”
6. **Techniques, classic problems and gotchas.** Computational shortcuts, proof patterns, exam favourites with one-line solutions, and the mistakes everyone makes once.

Margin notes carry the informal, “say it out loud” version of what is happening in the main text. Every major section closes with a summary box.

Prerequisites. Comfort with vectors, matrices, systems of linear equations, row reduction and the notions of rank and linear independence. Chapter 2 recalls exactly what is needed as it goes.

Chapter 1

From Decisions to Models

1.1 What Operations Research Is

Motivation

Every organisation makes decisions under limits: a train operator has a fixed fleet, a delivery company has a fixed number of vans, a hospital has a fixed number of nurses. The recurring question is

How should we use limited resources in the best possible way?

Operations Research (OR) answers this question with three ingredients: *mathematical models* that describe the situation, *data* that fill in the numbers, and *algorithms* that find good decisions. The word “research” in the name is historical: the field was born from research *on operations*, that is, from scientists studying how real systems actually behaved.

A little history: Blackett’s Circus

In 1940 German aircraft attacked Britain at night. Radar could locate an aircraft in the dark, but detection alone did not produce a hit. Between the radar and the gun sat a chain of steps: operators interpreted the echo, transmitted the information, a “predictor” estimated the future position, and only then was the gun aimed and fired. The aircraft moved, the information was imperfect, and every step took time. A small error or delay anywhere in the chain made the shot miss.

The birth of OR: do not study the radar alone, study the whole chain from detection to the fired shot.

In August 1940 the physicist Patrick Blackett joined Anti-Aircraft Command as scientific adviser and assembled a team of scientists from many disciplines, nicknamed *Blackett’s Circus*. Their method was the template for everything that followed:

observe the operation $\longrightarrow$ collect and analyse data $\longrightarrow$ improve how the system works.

Similar wartime teams studied convoy protection, aircraft patrols and military supply. After the war, these practical problems met new mathematics: John von Neumann’s game theory and, in 1947, George Dantzig’s *simplex method* for linear programming. From the 1950s onward OR spread to business, industry and public services: transport (which routes, which timetables?), healthcare (how many staff, how to cut waiting times?), energy (how much to produce and store?). In Morocco, for instance, ONCF must plan timetables and trains to serve passengers, and Amana must organise parcel routes to reduce travel while meeting deadlines.

The three ingredients of every model

Definition 1.1.1 (Ingredients of a decision model). A decision model consists of

- **decision variables:** the quantities we are free to choose;
- **constraints:** the conditions that limit our choices;
- **an objective:** a quantity we want to minimise or maximise.

Interpretation & discussion

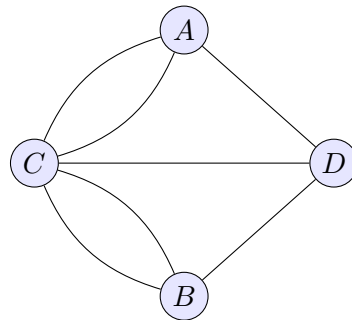
Every modelling exercise in this course, without exception, begins by answering three questions in plain language before writing a single formula: *What can we decide? What do we want to improve? What limits must we respect?* If you cannot answer these in words, you are not ready to write equations.

An earlier precursor: the Königsberg bridges

Long before wartime OR, in 18th-century Königsberg, seven bridges connected four land areas across the river Pregel. Citizens asked: *can you take a walk that crosses every bridge exactly once?* Start and finish anywhere you like.

Euler's idea (1736) was to keep only the connections: land areas become points, bridges become links. Four land areas become four points; seven bridges become seven links. The question becomes: *can we traverse every link exactly once?*

Euler's move: throw away everything except what is connected to what.



This abstraction is powerful enough that we can settle the question completely. The relevant notions (graph, degree, path) are introduced properly in Section 1.6; we state the theorem here because it is the historical origin of the whole “model it as a graph” philosophy, and its proof is short and beautiful. Feel free to read it now or after Section 1.6.

Theorem 1.1.2 (Euler, 1736). *Let G be a finite connected graph (multiple edges between two vertices are allowed). A walk that uses every edge exactly once exists if and only if the number of vertices with odd degree is 0 or 2. If it is 0, every such walk starts and ends at the same vertex; if it is 2, every such walk starts at one odd vertex and ends at the other.*

Proof. Step 1 (necessity). Suppose such a walk W exists. Each time W passes through a vertex v , it arrives on one edge and leaves on another, consuming two edges at v . If v is neither the first nor the last vertex of W , all edges at v are consumed in such pairs, so $\deg v$ is even. If v is the first vertex but not the last, one extra edge (the first) is consumed alone, so $\deg v$ is odd; similarly for the last vertex. If the first and last vertices coincide, the first and last edges pair up and

that vertex also has even degree. Hence the odd-degree vertices are exactly the endpoints when they differ, giving 0 or 2 odd vertices.

Step 2 (sufficiency, all degrees even). Assume every degree is even. Start at any vertex v_0 and walk along unused edges as long as possible, never reusing an edge. Whenever we *enter* a vertex $v \neq v_0$ we have used an odd number of its edges, so an unused one remains and we can leave. Therefore the walk can only get stuck at v_0 : it is a closed trail T_1 . If T_1 uses all edges we are done. Otherwise, because G is connected, some vertex u on T_1 has an unused edge. Removing the edges of T_1 leaves every degree even, so the same argument from u produces a closed trail T_2 of unused edges starting and ending at u . Splice T_2 into T_1 at u to obtain a longer closed trail. Repeat; since the edge set is finite the process ends with a closed trail using all edges.

Step 3 (sufficiency, exactly two odd vertices p, q). Add a new edge $\{p, q\}$. Now every degree is even, so by Step 2 there is a closed trail using all edges. Delete the added edge from that trail: what remains is a walk from p to q using every original edge exactly once. $\square$

Key insight

Count *pairs*. Passing through a vertex uses two of its edges, so parity of the degree is the only thing that matters. Euler's theorem is the first example of a recurring OR theme: a necessary condition (parity) that turns out to be sufficient as well, giving a complete characterisation.

Example 1.1.3 (Königsberg has no such walk). In the graph above, the degrees are $\deg A = 3$, $\deg B = 3$, $\deg C = 5$, $\deg D = 3$. Four vertices have odd degree, so by Theorem 1.1.2 no walk crosses every bridge exactly once. Two centuries of citizens failed not for lack of cleverness but because the task was impossible, and the proof took one line once the right model was chosen.

Modelling: translating a real decision

Definition 1.1.4 (Mathematical modelling). *Mathematical modelling* means translating a real-world problem into mathematical terms while keeping the features that matter for the decision. The translation table is

Real-world question	Mathematical ingredient
What can we decide?	Decision variables
What do we want to improve?	Objective function
What limits must we respect?	Constraints

Optimisation then finds the best decision allowed by the model. Afterwards we interpret the result and check whether it makes sense in reality. Two points deserve emphasis.

- **The model is a choice.** The same real problem can be modelled in several ways, and the choice affects both how hard the model is to solve and how useful the answer is.
- **Start simple.** Keep the essential decisions, constraints and objective. A simple model is easier to solve and to understand; extra detail (truck capacities, uncertain demand, ...) is worth adding *only if it changes the*

decision. Choose the criterion carefully too: the shortest route need not be the fastest. When several objectives compete, set priorities, use weights, or compare trade-offs.

Example 1.1.5 (Fertiliser delivery, OCP-inspired). Two plants (A and B) must supply two customers (1 and 2) with the same fertiliser this week. Either plant can serve either customer. Plant A has 700 t available, plant B has 500 t; customer 1 needs 600 t, customer 2 needs 400 t. Transport costs in MAD per tonne are

MAD/t	Customer 1	Customer 2	Available
Plant A	100	180	700 t
Plant B	140	80	500 t
Demand	600 t	400 t	

What can we decide? How many tonnes to send on each route: $x_{A1}, x_{A2}, x_{B1}, x_{B2}$. *What do we want to improve?* Total transport cost. *What limits?* Meet every order without exceeding any plant's stock. The model:

$$\begin{aligned}
 \min \quad & 100x_{A1} + 180x_{A2} + 140x_{B1} + 80x_{B2} \\
 \text{s.t.} \quad & x_{A1} + x_{A2} \leq 700, \quad x_{B1} + x_{B2} \leq 500 \quad (\text{available supply}) \\
 & x_{A1} + x_{B1} = 600, \quad x_{A2} + x_{B2} = 400 \quad (\text{meet demand}) \\
 & x_{A1}, x_{A2}, x_{B1}, x_{B2} \geq 0 \quad (\text{no negative shipments}).
 \end{aligned}$$

We return to this model in Section 1.5, where we will prove which plan is best.

From a model to a working solution

The full OR workflow has four steps: (1) **model** (decisions, constraints, objective, assumptions); (2) **choose an algorithm**, balancing solution quality, speed and implementation effort; (3) **validate**, through theoretical guarantees and tests on realistic data; (4) **deploy**, building reliable software that is monitored and maintained. If validation fails, revisit the model or the algorithm. This course concentrates on steps 1 and 2: modelling, then solution methods, starting with linear programming and continuing with graph algorithms.

Summary: What Operations Research is

- OR uses models, data and algorithms to make better decisions under limited resources.
- Every model has decision variables, constraints and an objective. Say them in words first.
- Modelling is a choice: start simple, add detail only when it changes the decision, choose the criterion with care.
- Euler's bridges problem is the prototype: strip the situation to its structure, and an impossible-looking question becomes a one-line parity argument.

1.2 Mathematical Programs

Motivation

Once decisions, constraints and objective are identified, every model in this course has the same abstract shape: *choose the best element of an allowed set*. Giving this shape a name lets us talk about feasibility, optimality and the different things that can go wrong before we specialise to any particular type of model.

Definition 1.2.1 (Mathematical program). A *mathematical program* is a problem of the form

$$\min_{x \in X} f(x),$$

where

- x is the *decision*, a vector of quantities we can choose;
- X is the *feasible set*, the set of all choices allowed by the constraints; an $x \in X$ is called *feasible*;
- $f(x)$ is the *objective value* of the choice x .

An *optimal solution* is a feasible $x^* \in X$ with $f(x^*) \leq f(x)$ for every $x \in X$; the number $f(x^*)$ is the *optimal value*. Maximisation problems $\max_{x \in X} p(x)$ are defined in the same way with the inequality reversed.

A maximisation problem is never a genuinely different object, as the next proposition shows.

Proposition 1.2.2 (Max is min of the negative). *For any set X and function $p : X \rightarrow \mathbb{R}$, the point x^* maximises p over X if and only if it minimises $-p$ over X , and $\max_X p = -\min_X (-p)$ whenever either side exists.*

Maximising profit is minimising loss. Flipping the sign lets every theorem be stated once.

Proof. For all $x \in X$: $p(x^*) \geq p(x) \iff -p(x^*) \leq -p(x)$. So x^* is a maximiser of p exactly when it is a minimiser of $-p$. Evaluating gives $\max_X p = p(x^*) = -(-p(x^*)) = -\min_X (-p)$. $\square$

The domain is part of the model

Definition 1.2.3 (Variable types). A decision variable is

- *continuous* if it may take any real value in an interval, e.g. tonnes shipped, $x \geq 0$;
- *integer* if it must lie in $\{0, 1, 2, \dots\}$ (or $\mathbb{Z}$), e.g. number of trucks;
- *binary* if it must lie in $\{0, 1\}$, e.g. “open a warehouse?”.

Interpretation & discussion

Is 2.4 tonnes meaningful? Yes. Is 2.4 trucks meaningful? No. The domain of a variable is not a technicality: it encodes a physical fact, and getting it wrong produces answers that cannot be implemented. A useful check on any finished model: *does every mathematically allowed choice make sense in reality?*

Three possible outcomes

Definition 1.2.4 (Outcomes of a mathematical program). A minimisation problem $\min_{x \in X} f(x)$ is

- *infeasible* if $X = \emptyset$;
- *unbounded* if $X \neq \emptyset$ and for every $M \in \mathbb{R}$ there is $x \in X$ with $f(x) < M$ (the objective can be made arbitrarily good);
- *solvable* if an optimal solution exists.

For a maximisation problem, “unbounded” means f can be made arbitrarily large.

Example 1.2.5.

- *Infeasible*: requiring $x \geq 5$ and $x \leq 3$ allows no choice at all.
- *Unbounded*: maximising x subject only to $x \geq 0$ has no finite best value.
- *Solvable*: the production model of Section 1.3 has best profit 100 MAD.

Remark 1.2.6 (The three outcomes are not exhaustive). A problem can be feasible, bounded, and still have no optimal solution: $\min x$ over $X = \{x \in \mathbb{R} : x > 0\}$ has infimum 0 but no feasible point achieves it. For linear programs this pathology never happens (Chapter 2 proves it), which is one reason linear programs are so pleasant.

Interpretation & discussion

An unexpected outcome is a *message about the model*, not a failure of the mathematics. Infeasibility usually means inconsistent data or a forgotten degree of freedom; unboundedness almost always means a missing limit (nobody can really sell infinitely many litres).

Recognising the structure

Definition 1.2.7 (Classes of mathematical programs).

- A *linear program* (LP) has continuous variables and an objective and constraints that are all *linear* expressions (weighted sums of the variables with constant coefficients).
- An *integer program* (IP) or *mixed-integer program* (MIP) has linear expressions, with some or all variables restricted to integers.
- A *nonlinear program* (NLP) has at least one nonlinear term such as x^2 or xy in the objective or the constraints.

The structure decides the solution method: linear programs have fast, exact algorithms (Chapter 2 begins this story); integer programs are much harder in general; nonlinear programs range from easy (convex, see Section 1.7) to hopeless.

Technique: Spotting the class in five seconds

- Any product of two variables, any square, any $\sqrt{\cdot}$, $\log$, $|\cdot|$ of a variable: **nonlinear**.

- A coefficient that itself depends on a variable (“price drops as quantity rises”): **nonlinear**, even if it looks like a product of constant and variable.
- Everything linear but some variable in $\{0, 1\}$ or $\mathbb{Z}$: **integer/mixed-integer**.
- Everything linear, all variables continuous: **LP**.

The liquid-production model below has linear expressions and continuous litres, so it is an LP.

Summary: Mathematical programs

- $\min_{x \in X} f(x)$: decision x , feasible set X , objective f . Optimal means feasible and best.
- Maximising p is minimising $-p$ (Proposition 1.2.2); the optimal value flips sign.
- The domain (continuous, integer, binary) is part of the model.
- Outcomes: an optimum, infeasible, or unbounded. Interpret unexpected outcomes as messages about the model.
- Class (LP, IP/MIP, NLP) is decided by linearity and by variable domains, and it decides the algorithm.

1.3 Linear Programs

Motivation

Linear programs are the workhorse of OR: they are expressive enough to model production, transport, blending, scheduling and flows, and they can be solved exactly and quickly even with millions of variables. To talk about “the” LP algorithm we need a single agreed shape for every LP. This section fixes a standard form and proves that *every* LP can be rewritten in it without changing the decisions or their costs.

Definition 1.3.1 (Linear program, standard (canonical) form). A *linear program in standard form* (this course’s convention) is

$$\begin{aligned} &\text{maximise} && c_1x_1 + \cdots + c_nx_n \\ &\text{subject to} && a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i \quad (i = 1, \dots, m), \\ &&& x_1, \dots, x_n \geq 0. \end{aligned}$$

The numbers a_{ij}, b_i, c_j are given data; only the x_j are chosen. The convention is: *maximise*, all constraints of type $\leq$, all variables nonnegative (zero allowed).

Remark 1.3.2 (Conventions are not definitions). The directions “max”, “ $\leq$ ”, “ ≥ 0 ” are a convention, not the definition of linearity. An LP is any problem with a linear objective and linear equality or inequality constraints on continuous variables. Other textbooks call $Ax = b, x \geq 0$ the standard form (Chapter 2 calls it the *equational form*). Both conventions represent exactly the same problems.

Real decisions do not always fit the format

Example 1.3.3 (Electricity for a workshop). A workshop needs at least 3 kWh during the next hour. It can generate locally, $g \geq 0$ kWh at 5 MAD/kWh, and it exchanges with the grid a net amount t with $-2 \leq t \leq 2$: $t > 0$ means buying at 2 MAD/kWh, $t < 0$ means selling and earning 2 MAD/kWh. The model is

$$\min 5g + 2t \quad \text{s.t.} \quad g + t \geq 3, \quad -2 \leq t \leq 2, \quad g \geq 0.$$

This is an LP: every expression is linear and the variables are continuous. But it minimises, has a $\geq$ constraint, and has a variable t that may be negative.

Three rewrites

Three elementary rewrites bring any LP into standard form. Each is justified by a small proposition stating that the feasible choices and their objective values correspond exactly.

Nothing is “solved” during a rewrite. We only translate; the decisions and their costs are the same before and after.

Proposition 1.3.4 (Reverse a sign). *For any real numbers, $a^\top x \geq b \iff -a^\top x \leq -b$. Hence $a \geq$ constraint can be replaced by $a \leq$ constraint with the same solution set.*

Proof. Multiplying an inequality by -1 reverses it: $u \geq v \iff -u \leq -v$. Apply with $u = a^\top x$, $v = b$. $\square$

Proposition 1.3.5 (Split a free variable). *Let $t \in \mathbb{R}$ be unrestricted in sign. The map $(u, v) \mapsto u - v$ from $\{(u, v) : u \geq 0, v \geq 0\}$ onto $\mathbb{R}$ is surjective. Consequently, replacing every occurrence of t by $u - v$ and adding $u, v \geq 0$ yields a problem whose feasible $(u, v, \dots)$ correspond to feasible $(t, \dots)$ via $t = u - v$, with equal objective values; the optimal values of the two problems coincide.*

Proof. Surjectivity. Given $t \in \mathbb{R}$, take $(u, v) = (t, 0)$ if $t \geq 0$ and $(u, v) = (0, -t)$ if $t < 0$; both are nonnegative and $u - v = t$.

Correspondence. Every constraint and the objective are functions of t (and the other variables). Substituting $t = u - v$ is an identity, so a point $(u, v, \dots)$ with $u, v \geq 0$ satisfies the new constraints exactly when $(u - v, \dots)$ satisfies the old ones, and the objective values agree. Every old feasible point is reached by surjectivity; every new feasible point maps to an old one. The two problems therefore have the same set of attainable objective values, hence the same optimal value. $\square$

Gotcha

The representation $t = u - v$ is *not unique*: $t = 2$ is $(2, 0)$ but also $(5, 3)$. This is harmless for the optimal value, but do not be surprised when a solver returns both $u > 0$ and $v > 0$; the decision is still $t = u - v$. Also, keep the original bounds on t : for $-2 \leq t \leq 2$ we must add $u - v \leq 2$ and $-u + v \leq 2$.

Proposition 1.3.6 (Min to max). *$\min f(x)$ over X and $\max(-f(x))$ over X have the same optimal solutions, and the minimum equals the negative of the maximum.*

Proof. This is Proposition 1.2.2 applied to $p = -f$. $\square$

Example 1.3.7 (The electricity problem in standard form). Start from Example 1.3.3. Reverse the sign of the supply constraint: $g + t \geq 3 \iff -g - t \leq -3$. Split $t = u - v$ with $u, v \geq 0$, keeping the bounds $u - v \leq 2$ and $-u + v \leq 2$. Flip the objective: $\min(5g + 2t) \iff \max(-5g - 2t)$. Substituting everywhere:

$$\begin{array}{llll} \text{maximise} & -5g - 2u + 2v & & \\ \text{subject to} & -g - u + v \leq -3 & & \text{(minimum supply)} \\ & u - v \leq 2 & & \text{(import limit)} \\ & -u + v \leq 2 & & \text{(export limit)} \\ & g, u, v \geq 0. & & \end{array}$$

Afterwards we recover the real decision as $t = u - v$, and the original minimum cost is the negative of this maximum. Same choices, same costs, after translation; no solving was needed.

For the record, the optimum is easy to see: buying from the grid (2 MAD) is cheaper than generating (5 MAD), so buy the maximum $t = 2$ and generate $g = 1$, for a cost of 9 MAD. Section 1.5 shows how to *prove* that nothing beats 9.

A small production problem, from words to a picture

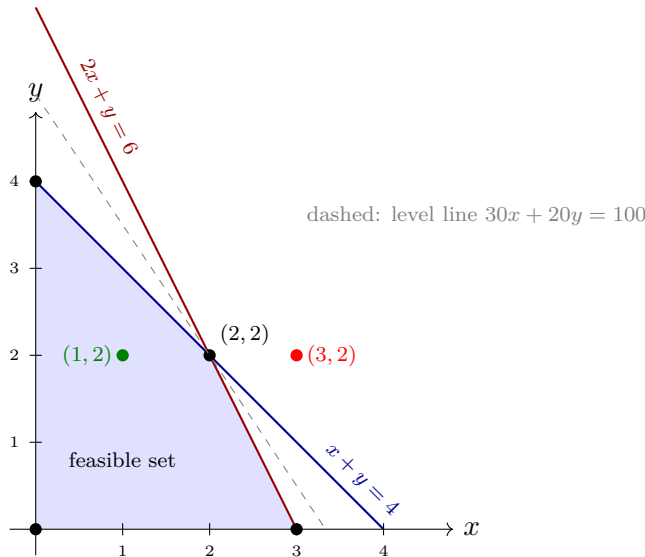
Example 1.3.8 (Liquid products). A workshop makes two liquid products, A and B . Per litre, A needs 1 hour of machine time and 2 kg of raw material and earns 30 MAD; B needs 1 hour and 1 kg and earns 20 MAD. Today 4 hours and 6 kg are available. Assumptions: production is divisible, and everything produced can be sold.

Decisions: x = litres of A , y = litres of B . *Objective:* $\max 30x + 20y$ (MAD).

Constraints:

$$x + y \leq 4 \text{ (machine hours)}, \quad 2x + y \leq 6 \text{ (raw material, kg)}, \quad x, y \geq 0.$$

Check the units: both sides of each constraint are hours, or kilograms. Check the direction: “at most” gives $\leq$.



Feasible means respecting every constraint. The plan $(1, 2)$ uses 3 hours and 4 kg: feasible. The plan $(3, 2)$ would use 5 hours and 8 kg: infeasible, however profitable it looks. A profitable plan is useless if it violates the constraints.

Optimal means best among feasible plans. The dashed line $30x + 20y = 100$ is a *level line*: all plans on it earn the same profit. Sliding it upward increases profit; the last feasible point it touches is the corner $(2, 2)$, with profit $30(2) + 20(2) = 100$ MAD. The picture is convincing, but a picture is not a proof. Here is the proof, and it contains one of the central ideas of linear programming.

Proposition 1.3.9 (A certificate of optimality for Example 1.3.8). *Every feasible plan satisfies $30x + 20y \leq 100$. Hence $(2, 2)$ is optimal.*

Proof. For any feasible (x, y) ,

$$30x + 20y = 10(x + y) + 10(2x + y) \leq 10 \cdot 4 + 10 \cdot 6 = 100,$$

using $x + y \leq 4$ and $2x + y \leq 6$ and the fact that we multiplied both by *nonnegative* numbers, which preserves the inequalities. Since $(2, 2)$ is feasible and achieves 100, it is optimal. $\square$

Key insight

To prove no plan beats 100, we did not examine plans at all. We *combined the constraints* with nonnegative weights (10 and 10) so that the combination reproduces the objective exactly. The weights are a *certificate*: anyone can check the two-line computation. This idea, made general, is the next lemma; made fully general it becomes LP duality later in the course.

Lemma 1.3.10 (Weak duality, elementary form). *Consider the standard-form LP $\max\{c^\top x : Ax \leq b, x \geq 0\}$ with $A \in \mathbb{R}^{m \times n}$. If $y \in \mathbb{R}^m$ satisfies $y \geq 0$ and $A^\top y \geq c$ (componentwise), then $c^\top x \leq b^\top y$ for every feasible x . In particular, if some feasible x^* has $c^\top x^* = b^\top y$, then x^* is optimal.*

Proof. Let x be feasible. Since $x \geq 0$ and $A^\top y \geq c$, multiplying componentwise and summing gives $c^\top x \leq (A^\top y)^\top x = y^\top (Ax)$. Since $y \geq 0$ and $Ax \leq b$, the same argument gives $y^\top (Ax) \leq y^\top b$. Chain the two: $c^\top x \leq b^\top y$. If $c^\top x^* = b^\top y$

for a feasible x^* , then $c^\top x \leq c^\top x^*$ for all feasible x , which is the definition of optimality. $\square$

Interpretation & discussion

In Example 1.3.8, $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, $b = (4, 6)^\top$, $c = (30, 20)^\top$ and $y = (10, 10)^\top$ gives $A^\top y = (30, 20)^\top = c$ and $b^\top y = 100$. The two places where signs matter are exactly the two places where the proof used nonnegativity: we need $x \geq 0$ to pass from $c \leq A^\top y$ to $c^\top x \leq y^\top Ax$, and $y \geq 0$ to pass from $Ax \leq b$ to $y^\top Ax \leq y^\top b$. Forgetting either is the classic error.

Technique: Solving a two-variable LP by hand

1. Draw each constraint as a line; shade the half-plane it allows. The feasible set is the intersection (a polygon, possibly unbounded or empty).
2. Draw one level line of the objective, $c_1x + c_2y = k$; the direction of increase is the vector (c_1, c_2) .
3. Slide the level line in the direction of increase until it is about to leave the feasible set. The last contact is at a corner (or along a whole edge if the level line is parallel to it).
4. Compute the objective at the candidate corners and pick the best. *Shortcut:* corners are intersections of two constraint lines, so solve 2×2 systems.
5. To *prove* optimality on an exam, exhibit nonnegative multipliers of the tight constraints that reproduce the objective (Lemma 1.3.10).

Classic problems

- *Corners of Example 1.3.8.* $(0, 0), (3, 0), (2, 2), (0, 4)$ with profits 0, 90, 100, 80. Best: $(2, 2)$.
- Write $\min 2x - y$ s.t. $x - y \geq 1$, $x \in \mathbb{R}$, $y \geq 0$ in standard form. $\max -2u + 2v + y$ s.t. $-u + v + y \leq -1$, $u, v, y \geq 0$, with $x = u - v$.
- Is $\{x : |x| \leq 1\}$ an LP constraint? Yes: $|x| \leq 1 \iff x \leq 1$ and $-x \leq 1$. But $|x| \geq 1$ is *not* (the set is two disjoint pieces; see convexity, Section 1.7).
- *Certificate for the electricity problem.* From $g + t \geq 3$ (weight 5) and $t \leq 2$ (weight 3): $5g + 5t \geq 15$ and $-3t \geq -6$; adding, $5g + 2t \geq 9$. The plan $g = 1, t = 2$ costs 9, so it is optimal.

Gotcha

- Reading “at least” as $\leq$. Always re-read each constraint in words after writing it.
- Forgetting $x, y \geq 0$. Without them the production LP is unbounded (produce -1000 litres of B !).
- Treating a level line as a constraint. Level lines are where the objective

is constant; they do not restrict anything.

- Concluding optimality from the picture alone. On an exam, give the certificate or evaluate all corners with justification.

Summary: Linear programs

- Standard form: maximise a linear objective, all constraints $\leq$, all variables ≥ 0 .
- Any LP reaches standard form by three rewrites: reverse signs of $\geq$ constraints, split free variables as $u - v$, and turn min into max of the negative. Each is an exact translation (Propositions 1.3.4–1.3.6).
- Feasible = respects every constraint; optimal = best among feasible.
- Weak duality (Lemma 1.3.10): a nonnegative combination of constraints that dominates the objective is a checkable proof of optimality.

1.4 Integer and Nonlinear Models

Motivation

Not every decision is a divisible quantity. Some are yes/no choices (take an item, fund a project, open a warehouse); some interact multiplicatively (area is length times width). These produce integer and nonlinear programs. They are usually harder to solve than LPs, but writing them correctly is the same discipline: decisions, objective, constraints, domain.

Integer programming: pack a backpack

Example 1.4.1 (Knapsack). Your essentials are packed and 5 kg of capacity remain for optional gear. Each item is available once: take it whole or leave it.

Item	Weight (kg)	Usefulness
Camera kit	3	9
Binoculars	2	6
Warm blanket	4	7
Book	1	2

Define $z_i = 1$ if item i is taken and $z_i = 0$ otherwise, in the order camera, binoculars, blanket, book. The model is

$$\max 9z_1 + 6z_2 + 7z_3 + 2z_4 \quad \text{s.t.} \quad 3z_1 + 2z_2 + 4z_3 + z_4 \leq 5, \quad z_i \in \{0, 1\}.$$

All expressions are linear; the decisions are binary integers. This is the *knapsack problem*, a binary integer program.

Proposition 1.4.2 (Solution of Example 1.4.1). *The optimal selection is camera and binoculars, with usefulness 15.*

Proof. There are $2^4 = 16$ selections; we may organise the check. The blanket (4

kg) can be combined only with the book: usefulness 9, or alone: 7. Without the blanket, the items weigh 3, 2, 1: all three weigh $6 > 5$, so at most two. Pairs: camera+binoculars (5 kg, 15), camera+book (4 kg, 11), binoculars+book (3 kg, 8). Singles give at most 9. The maximum over all feasible selections is 15. $\square$

Remark 1.4.3 (Why the domain matters). Replacing $z_i \in \{0, 1\}$ by $0 \leq z_i \leq 1$ gives an LP, the *linear relaxation*, whose solutions may take a fraction of an item. Here the relaxation happens to have the integer optimum 15 as well (fill by best usefulness-per-kilogram: camera 3, binoculars 3, then the bag is full), but in general the relaxation can be strictly better than any integer plan, and its fractional answer is not implementable. Section 1.5 explains how such relaxations are nevertheless useful as *bounds*.

Example 1.4.4 (Your turn: fund projects). A team can spend at most 7 budget units; each of projects A, B, C is funded in full or not at all.

	A	B	C
Cost	4	3	2
Benefit	6	5	4

Model: $a, b, c \in \{0, 1\}$ ($1 = \text{funded}$), $\max 6a + 5b + 4c$ s.t. $4a + 3b + 2c \leq 7$.

Solution: all three cost $9 > 7$. Pairs: $A+B$ costs 7, benefit 11; $A+C$ costs 6, benefit 10; $B+C$ costs 5, benefit 9. Singles give at most 6. Fund A and B : $(a, b, c) = (1, 1, 0)$, benefit 11.

Nonlinear programming: design a garden

Example 1.4.5 (Garden). A school has 40 m of fencing for a rectangular garden; all four sides must be fenced. Choose length $x \geq 0$ and width $y \geq 0$ (fractional metres allowed) to enclose the largest area using at most 40 m of fence:

$$\max xy \quad \text{s.t.} \quad 2x + 2y \leq 40, \quad x, y \geq 0.$$

The constraint $2x + 2y$ is linear but the objective xy multiplies two decision variables: the model is an NLP because its objective is nonlinear.

Proposition 1.4.6 (Solution of Example 1.4.5). *The maximum area is 100 m^2 , attained only by the square $x = y = 10$.*

Proof. For $x, y \geq 0$, $0 \leq (x-y)^2 = x^2 - 2xy + y^2$, so $4xy \leq x^2 + 2xy + y^2 = (x+y)^2$, i.e. $xy \leq \left(\frac{x+y}{2}\right)^2$, with equality iff $x = y$. The constraint gives $x + y \leq 20$, so $xy \leq 10^2 = 100$. The point $(10, 10)$ is feasible ($2 \cdot 10 + 2 \cdot 10 = 40$) and achieves 100. Equality in the chain forces $x = y$ and $x + y = 20$, so the square is the unique maximiser. $\square$

Key insight

The proof is again a *bound plus a plan that attains it*: the inequality $xy \leq ((x+y)/2)^2$ (the arithmetic-geometric mean inequality) is a lower-effort certificate than any calculus. The pattern “bound, then attain” is the subject of the next section.

Technique: Writing integer and nonlinear models

- A yes/no decision is a binary variable $z \in \{0, 1\}$. “Take the whole item or nothing” means *do not* relax to $[0, 1]$.
- Count things with integers, measure things with reals. Ask: would 2.4 of this make sense?
- Products of *data* and variables are linear ($3z_1$); products of two *variables* are not (xy).
- For tiny integer programs, enumerate cleverly: fix the heaviest item in or out, then treat the rest.
- For a symmetric NLP with a sum constraint, try AM–GM before calculus.

Gotcha

Rounding an LP solution is not a way to solve an integer program: the rounded point may be infeasible, and even when feasible it may be far from optimal. Keep the domain $\{0, 1\}$ or $\mathbb{Z}$ in the model and solve the integer problem as such.

Summary: Integer and nonlinear models

- Binary variables model select/assign/activate decisions; linear expressions with integer variables give an IP or MIP (knapsack, project funding).
- A single product of variables makes a model nonlinear (garden design).
- Small instances are solved by structured enumeration or by an inequality that bounds the objective; large instances need the algorithms of later chapters.

1.5 Bounds: How Good Is Our Solution?

Motivation

Finding the best solution can take a long time. Suppose we already have a feasible plan. Should we keep searching? The honest answer requires two numbers. For a cost-minimisation problem:

- **What cost can we actually achieve?** Any feasible plan gives an *upper bound* U on the optimal cost.
- **What cost can we prove is impossible to beat?** A *lower bound* L .

Then

$$L \leq \text{best possible cost} \leq U,$$

and the *gap* $U - L$ tells us how much improvement might remain. If $L = U$, we have proved optimality without examining every plan.

Two questions: what can we achieve? What can we prove impossible? The answers bracket the optimum.

Upper bounds from feasible plans

Proposition 1.5.1 (Feasible plans give upper bounds). *Let $z^* = \min_{x \in X} f(x)$ be the optimal value of a minimisation problem (assumed to exist). For every feasible $\bar{x} \in X$, $z^* \leq f(\bar{x})$.*

Proof. z^* is the minimum of f over X and $\bar{x} \in X$, so $z^* \leq f(\bar{x})$ by definition of a minimum. $\square$

Lower bounds from relaxations

Definition 1.5.2 (Relaxation). Let (P) be $\min_{x \in X} f(x)$. A problem (R) : $\min_{x \in X'} f'(x)$ is a *relaxation* of (P) if

$$X \subseteq X' \quad \text{and} \quad f'(x) \leq f(x) \quad \text{for all } x \in X.$$

That is, (R) allows at least the same choices and never charges more for them. The most common relaxations *drop constraints* ($X' \supseteq X$, $f' = f$) or *drop integrality* ($z \in \{0, 1\}$ becomes $0 \leq z \leq 1$).

Theorem 1.5.3 (Relaxations give lower bounds). *If (R) is a relaxation of (P) , then $\inf_{x \in X'} f'(x) \leq \inf_{x \in X} f(x)$. In words: the optimal value of the relaxation is a lower bound on the optimal value of the original problem. For maximisation problems, a relaxation gives an upper bound.*

Proof. Let $x \in X$ be any feasible point of (P) . Then $x \in X'$, so $\inf_{x \in X'} f' \leq f'(x) \leq f(x)$. The number $\inf_{x \in X'} f'$ is therefore a lower bound for the set $\{f(x) : x \in X\}$, and hence is at most its greatest lower bound $\inf_X f$. (If X is empty the right side is $+\infty$ and the inequality is trivial.) For maximisation, apply the same argument with $\geq$ throughout. $\square$

Key insight

Removing a constraint can only make the minimum smaller or equal: you are minimising over a bigger set. Making the objective cheaper can only make the minimum smaller or equal. Relaxations exploit exactly these two monotonicities.

Matching bounds prove optimality

Theorem 1.5.4 (Optimality by matching bounds). *Let (P) be $\min_{x \in X} f(x)$. Suppose $L \leq f(x)$ for every $x \in X$ (a valid lower bound) and $\bar{x} \in X$ satisfies $f(\bar{x}) = L$. Then $\bar{x}$ is optimal and the optimal value is L .*

Proof. For every $x \in X$, $f(\bar{x}) = L \leq f(x)$. This is precisely the definition of $\bar{x}$ being an optimal solution, and its value is L . $\square$

Interpretation & discussion

The gap $U - L$ is a guarantee, not a promise: our plan is at most $U - L$ worse than optimal, but it does *not* follow that all of $U - L$ can be saved. Either the plan may be better than we can prove (the lower bound is weak), or the lower bound may be tight and the plan poor. Closing the gap can happen

from either side: find a better plan (lower U) or a stronger argument (raise L).

Worked example: the fertiliser delivery problem

Example 1.5.5 (Upper bound: build a plan we can actually use). Return to Example 1.1.5. Propose: $A \rightarrow 1$: 500 t, $A \rightarrow 2$: 0, $B \rightarrow 1$: 100 t, $B \rightarrow 2$: 400 t. *Feasibility*: customer 1 receives $500 + 100 = 600$, customer 2 receives 400; plant A ships $500 \leq 700$, plant B ships $100 + 400 = 500 \leq 500$. *Cost*:

$$U = 500(100) + 0(180) + 100(140) + 400(80) = 96,000 \text{ MAD.}$$

By Proposition 1.5.1, the optimal cost is at most 96,000.

Example 1.5.6 (Lower bound: make the problem easier). Temporarily ignore the plant capacities but keep the demands. Each customer may now use its cheapest route, since nothing limits the plant. Customer 1 pays at least 100 MAD/t for 600 t, customer 2 at least 80 MAD/t for 400 t:

$$L = 600(100) + 400(80) = 92,000 \text{ MAD.}$$

Justify the direction. Dropping the two supply constraints enlarges the feasible set with the same objective, so this is a relaxation, and by Theorem 1.5.3 its optimal value 92,000 is a lower bound. (To be precise, we also need that 92,000 is the optimum of the relaxed problem: for any feasible x of the relaxation, $100x_{A1} + 140x_{B1} \geq 100(x_{A1} + x_{B1}) = 60,000$ and $180x_{A2} + 80x_{B2} \geq 80(x_{A2} + x_{B2}) = 32,000$, using nonnegativity of the variables.)

So $92,000 \leq \text{optimal cost} \leq 96,000$: the gap is 4,000 MAD. Our plan is at most 4,000 MAD more expensive than optimal.

Example 1.5.7 (Matching bounds). Can we achieve 92,000? Send 600 t from A to customer 1 and 400 t from B to customer 2, nothing on the other routes. Check the *original* constraints: both demands are met, and $600 \leq 700$, $400 \leq 500$. Cost = $600(100) + 400(80) = 92,000$. Now $U = 92,000 = L$, and by Theorem 1.5.4 this plan is optimal: no plan can beat it.

Key insight

We never enumerated plans, and we never ran an algorithm. A relaxation gave a number no plan can beat; a single plan reached it. This “sandwich” is how every modern optimisation solver reports progress, and it is how you should justify optimality on an exam whenever you can.

Technique: Producing bounds quickly

- *Upper bound (minimisation)*: any feasible plan. A greedy or “obvious” plan is fine; just *verify feasibility* explicitly.
- *Lower bound (minimisation)*: drop the constraints that make the problem hard, solve what is left. Typical drops: capacities, integrality, coupling constraints between subproblems.

- *Per-unit argument*: “each unit of demand costs at least $c_{\min}$ ” is a relaxation in disguise (it ignores everything except demand).
- *Combine constraints*: nonnegative combinations of constraints that dominate the objective give bounds directly (Lemma 1.3.10); for maximisation this yields upper bounds.
- Always *state the direction*: relaxations bound minima from below and maxima from above.

Classic problems

- *Knapsack bound*. In Example 1.4.1 the LP relaxation gives usefulness ≤ 15 ; the integer plan camera+binoculars achieves 15. Optimal by matching bounds.
- *Transport with a single cheapest route*. If every customer’s cheapest plant has enough capacity to serve it alone, then “everyone uses the cheapest route” is feasible and equals the relaxation bound, hence optimal. (That is exactly what happened above.)
- *Gap interpretation*. “ $L = 90$, $U = 96$: is the plan within 10% of optimal?” Yes. We know $U - z^* \leq U - L = 6$ and $z^* \geq L = 90$, so the relative excess is $(U - z^*)/z^* \leq 6/90 \approx 6.7\% < 10\%$.

Gotcha

- A relaxed solution is usually *infeasible* for the original problem. Its *value* is a bound; its *plan* may be nonsense (fractional items, exceeded capacities).
- Adding constraints does not give a lower bound for a minimisation; it gives an upper bound (you restrict the choices).
- For maximisation problems every direction flips: feasible plans give lower bounds, relaxations give upper bounds.

Summary: Bounds

- Minimisation: feasible plan $\Rightarrow$ upper bound; relaxation (drop constraints or cheapen the objective) $\Rightarrow$ lower bound (Theorem 1.5.3).
- $L \leq z^* \leq U$; the gap $U - L$ bounds the possible improvement.
- $L = U$ proves optimality (Theorem 1.5.4) with no enumeration.

1.6 Modelling with Graphs

Motivation

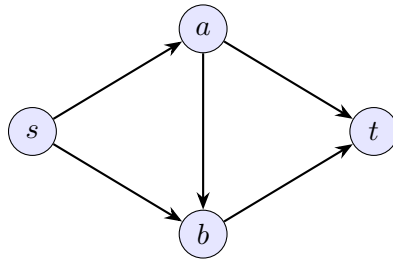
Many decisions depend on which objects are connected to which: cities and roads, tasks and precedence relations, people and the jobs they can do.

Setting	Objects	Connections
Transport	Cities	Roads
Project planning	Tasks	Precedence relations
Staff allocation	People and jobs	Possible assignments

A *graph* keeps exactly the relationships that matter. Different real situations share the same mathematical structure, so one model and one algorithm serve many fields. We will illustrate everything with a small computer network: a monitoring system must send security logs from a sensor s to an analysis server t through relay machines a and b . Machines become nodes; allowed transfers become arcs. Which allowed route gets a log to the server fastest? This is the same question as finding a fast route between cities.

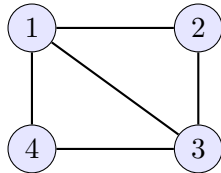
Directed and undirected graphs

Definition 1.6.1 (Directed graph). A *directed graph* (digraph) is a pair $D = (N, A)$ where N is a finite set of *nodes* and $A \subseteq N \times N$ is a set of *arcs*. An arc (i, j) goes *from* i *to* j ; its reverse (j, i) need not exist. We write $n = |N|$ and $m = |A|$. In routing examples, s is the *source* and t the *sink*.



Here $N = \{s, a, b, t\}$, $A = \{(s, a), (s, b), (a, b), (a, t), (b, t)\}$, $n = 4$, $m = 5$.

Definition 1.6.2 (Undirected graph). An *undirected graph* is a pair $G = (V, E)$ where V is a finite set of *vertices* and E is a set of *edges*, each edge being an unordered pair $e = \{i, j\}$ of distinct vertices. Note $\{i, j\} = \{j, i\}$: there is no orientation. (When multiple edges between the same pair are needed, as in Königsberg, E is taken to be a multiset.)



Here $V = \{1, 2, 3, 4\}$ and $E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}\}$: candidate physical links between sites, where we care only whether two sites are connected.

What leaves, what enters, what touches

Definition 1.6.3 (Incidence sets). In a digraph $D = (N, A)$, for a node i :

$\delta^+(i) = \{(i, j) \in A\}$ (outgoing arcs), $\delta^-(i) = \{(j, i) \in A\}$ (incoming arcs).

In an undirected graph $G = (V, E)$, for a vertex v : $\delta(v) = \{e \in E : v \in e\}$ is

the set of edges *incident* to v , and $\deg v = |\delta(v)|$ is the *degree* of v . For a set of vertices $S \subseteq V$:

$$\begin{aligned}\delta(S) &= \{\{i, j\} \in E : \text{exactly one of } i, j \text{ is in } S\} \quad (\text{boundary edges}), \\ E(S) &= \{\{i, j\} \in E : i, j \in S\} \quad (\text{internal edges}).\end{aligned}$$

In a digraph, $\delta^+(S)$ is the set of arcs leaving S (tail in S , head outside) and $\delta^-(S)$ the set entering S .

These are sets of *arcs or edges*, not sets of neighbours. In the digraph above, $\delta^+(a) = \{(a, b), (a, t)\}$ and $\delta^-(a) = \{(s, a)\}$. In the undirected graph, $\delta(1) = \{\{1, 2\}, \{1, 3\}, \{1, 4\}\}$, so $\deg 1 = 3$; for $S = \{1, 2\}$, $\delta(S) = \{\{1, 4\}, \{1, 3\}, \{2, 3\}\}$ and $E(S) = \{\{1, 2\}\}$. If no boundary edge of S is selected in a design, the group S is isolated from the rest.

Lemma 1.6.4 (Handshake lemma). *In any undirected graph, $\sum_{v \in V} \deg v = 2|E|$. Consequently the number of odd-degree vertices is even. In any digraph, $\sum_{i \in N} |\delta^+(i)| = \sum_{i \in N} |\delta^-(i)| = |A|$.*

Proof. Count pairs (v, e) with $v \in e$. Summing over vertices counts $\deg v$ for each v ; summing over edges counts 2 for each edge (an edge has two endpoints). Hence $\sum_v \deg v = 2|E|$. The right side is even, so the number of odd terms on the left is even. In a digraph, each arc has exactly one tail and one head, so it is counted once in $\sum_i |\delta^+(i)|$ and once in $\sum_i |\delta^-(i)|$. $\square$

Key insight

Double counting: count the same set of pairs two ways. It is the single most useful proof technique in elementary graph theory, and it reappears in the flow arguments below as “sum the conservation equations.”

Paths, cycles, connectivity, weights

Definition 1.6.5 (Walks, paths, cycles, connectivity). A *path* is a sequence of vertices $v_0, v_1, \dots, v_k$ in which consecutive vertices are joined by an edge (or, in a digraph, by an arc (v_{i-1}, v_i) pointing the way we travel) and no vertex repeats. A *cycle* is such a sequence with $v_k = v_0$, $k \geq 3$, and no other repetition. A graph is *connected* if every pair of vertices is joined by a path.

For instance, in a triangle 1, 2, 3 with an extra vertex 4 attached to 3: 1, 2, 3 is a path; 1, 2, 3, 1 is a cycle; the graph is connected.

Definition 1.6.6 (Weights). A *weight* is a number attached to each arc or edge, e.g. w_{ij} on arc (i, j) . The weight of a path is the sum of the weights of its arcs. Always state the meaning and the units of the weights (time in ms, cost in MAD, capacity in MB/s, ...); the same graph can carry different weights for different questions.

In the network with latencies $w_{sa} = 2$, $w_{at} = 4$, $w_{sb} = 5$, $w_{bt} = 2$ (ms), the path $s \rightarrow a \rightarrow t$ takes $2 + 4 = 6$ ms.

Variables on links and a summation shorthand

The graph and its weights are *data*. The *decisions* live on the links:

- choose whether to use an arc: $x_{ij} \in \{0, 1\}$, e.g. $x_{sa} = 1$ selects (s, a) ;
- choose an amount on an arc: $x_{ij} \geq 0$, e.g. $x_{sa} = 2.5$ sends 2.5 units along (s, a) .

The meaning and domain of x depend on the problem.

Definition 1.6.7 (Sum over a set of links). For a set F of arcs or edges, $x(F) = \sum_{e \in F} x_e$.

For example $x(\delta^+(a)) = x_{ab} + x_{at}$ is the total leaving a ; if $x_{ab} = 1$ and $x_{at} = 2$ the total is 3. For a group S , $x(\delta(S))$ is the total use of boundary edges. First identify the set of links; then add their variables.

Four questions, four graph models

Shortest path	Which route uses the fewest links (or least total weight)?
Maximum flow	How much traffic can reach the sink?
Minimum-cost flow	How cheaply can we send a required amount?
Assignment	Which analyst should handle each incident?

For each we write decisions, objective and constraints, and we prove the basic structural facts.

Shortest path

Definition 1.6.8 (Shortest path as an integer program). Send one unit from s to t using the fewest links. Decision $x_{ij} \in \{0, 1\}$ for each arc ($1 =$ use it). For the example network:

$$\begin{array}{ll}
 \min & x_{sa} + x_{sb} + x_{ab} + x_{at} + x_{bt} \\
 \text{s.t.} & x_{sa} + x_{sb} = 1 && \text{leave the source,} \\
 & x_{ab} + x_{at} - x_{sa} = 0 && \text{at } a : \text{out} = \text{in,} \\
 & x_{bt} - x_{sb} - x_{ab} = 0 && \text{at } b : \text{out} = \text{in,} \\
 & x_{at} + x_{bt} = 1 && \text{reach the sink.}
 \end{array}$$

In general the constraints read $x(\delta^+(i)) - x(\delta^-(i)) = 1$ for $i = s$, $= -1$ for $i = t$, $= 0$ otherwise, and the objective is $\sum_{(i,j)} w_{ij}x_{ij}$ for weights w (all equal to 1 when counting links).

Proposition 1.6.9. For every directed s - t path P , the vector x^P with $x_{ij}^P = 1$ on the arcs of P and 0 elsewhere is feasible for the shortest-path program, with objective value equal to the number of arcs of P (or its weight). Hence the optimal value is at most the length of any s - t path.

Proof. On a path $s = v_0, v_1, \dots, v_k = t$ with no repeated vertex, s has exactly one outgoing path arc and no incoming one, t has one incoming and no outgoing, and every intermediate v_i has exactly one incoming arc (v_{i-1}, v_i) and one outgoing arc

(v_i, v_{i+1}) . So $x^P(\delta^+(i)) - x^P(\delta^-(i))$ equals 1, -1 , 0 as required, and the objective sums the arcs of P . $\square$

In the example, $x_{sa} = x_{at} = 1$, all others 0, is the route $s \rightarrow a \rightarrow t$ with value 2, which beats $s \rightarrow a \rightarrow b \rightarrow t$ (value 3). Since any s - t path has at least 2 arcs (there is no arc (s, t)), the value 2 is optimal, by matching bounds again. All expressions are linear, so this is an integer linear program.

Maximum flow

Definition 1.6.10 (Maximum flow). Each arc (i, j) has a *capacity* $u_{ij} \geq 0$ (e.g. MB/s). A *flow* is a vector x with $0 \leq x_{ij} \leq u_{ij}$ satisfying *conservation* $x(\delta^+(i)) = x(\delta^-(i))$ at every node $i \notin \{s, t\}$: every relay forwards all the traffic it receives. Its *value* is $x(\delta^+(s)) - x(\delta^-(s))$, the net amount leaving the source. The *maximum flow problem* asks for a flow of maximum value:

$$\max x(\delta^+(s)) - x(\delta^-(s)) \quad \text{s.t. conservation at } i \notin \{s, t\}, \quad 0 \leq x_{ij} \leq u_{ij}.$$

Flow may split across several routes; we choose amounts, not a single route. All expressions are linear: this is an LP.

Lemma 1.6.11 (What leaves s arrives at t). *For every flow x , $x(\delta^+(s)) - x(\delta^-(s)) = x(\delta^-(t)) - x(\delta^+(t))$.*

Proof. Sum $x(\delta^+(i)) - x(\delta^-(i))$ over all nodes $i \in N$. Each arc (i, j) contributes $+x_{ij}$ to the term of i and $-x_{ij}$ to the term of j , so the total is 0. Conservation makes every term with $i \notin \{s, t\}$ vanish, leaving $(x(\delta^+(s)) - x(\delta^-(s))) + (x(\delta^-(t)) - x(\delta^+(t))) = 0$, which rearranges to the claim. $\square$

Theorem 1.6.12 (Cuts bound flows). *Let $S \subseteq N$ with $s \in S$ and $t \notin S$ (an s - t cut). For every flow x ,*

$$\text{value}(x) = x(\delta^+(S)) - x(\delta^-(S)) \leq u(\delta^+(S)) := \sum_{(i,j) \in \delta^+(S)} u_{ij}.$$

Hence the maximum flow value is at most the capacity of every cut.

Proof. Sum $x(\delta^+(i)) - x(\delta^-(i))$ over $i \in S$. An arc with both ends in S contributes $+x_{ij}$ (at its tail) and $-x_{ij}$ (at its head), cancelling. An arc leaving S contributes $+x_{ij}$ only; an arc entering S contributes $-x_{ij}$ only. So the sum equals $x(\delta^+(S)) - x(\delta^-(S))$. On the other hand, conservation kills every term except $i = s$, so the sum equals $\text{value}(x)$. Finally $x(\delta^+(S)) \leq u(\delta^+(S))$ since $x_{ij} \leq u_{ij}$, and $x(\delta^-(S)) \geq 0$. $\square$

Key insight

Summing the conservation equations over a set S telescopes: internal arcs cancel and only boundary arcs survive. This is the graph version of “combine the constraints with weights 1 on S , 0 elsewhere,” i.e. weak duality once more.

Example 1.6.13 (Maximum flow in the log network). Capacities (MB/s): $u_{sa} = 4$, $u_{sb} = 3$, $u_{ab} = 2$, $u_{at} = 2$, $u_{bt} = 4$. *Upper bound.* The cut $S = \{s, a, b\}$

has $\delta^+(S) = \{(a, t), (b, t)\}$ with capacity $2 + 4 = 6$, so no flow exceeds 6. A flow of value 6. Send $x_{sa} = 4$, split at a into $x_{at} = 2$, $x_{ab} = 2$; send $x_{sb} = 2$; then $x_{bt} = x_{sb} + x_{ab} = 4$. All capacities respected, conservation holds at a ($4 = 2 + 2$) and at b ($2 + 2 = 4$). Value $= x_{sa} + x_{sb} = 6$. By matching bounds, 6 MB/s is the maximum. (The full max-flow/min-cut theorem, proved later in the course, says a matching cut *always* exists.)

Minimum-cost flow

Definition 1.6.14 (Minimum-cost flow). Each arc has a cost c_{ij} per unit and a capacity u_{ij} . Each node has a balance b_i : $b_i > 0$ is a supply, $b_i < 0$ a demand, $b_i = 0$ a relay. Find the cheapest flow meeting all balances:

$$\begin{aligned} \min \quad & \sum_{(i,j) \in A} c_{ij} x_{ij} \\ \text{s.t.} \quad & x(\delta^+(i)) - x(\delta^-(i)) = b_i & (i \in N), \\ & 0 \leq x_{ij} \leq u_{ij} & ((i,j) \in A). \end{aligned}$$

Maximum flow *chooses* the amount; here the amount is *given*. For a required 4 MB/s from s to t : $b_s = 4$, $b_t = -4$, $b_a = b_b = 0$.

Proposition 1.6.15 (Balances must sum to zero). *If the minimum-cost flow problem is feasible, then $\sum_{i \in N} b_i = 0$.*

Proof. Sum the balance equations over all nodes: the left side is $\sum_i (x(\delta^+(i)) - x(\delta^-(i))) = 0$ (each arc appears once with $+$ and once with $-$, as in Lemma 1.6.11); the right side is $\sum_i b_i$. $\square$

Example 1.6.16 (Minimum-cost flow in the log network). Labels (cost; capacity): (s, a) : 1; 4, (s, b) : 3; 3, (a, b) : 1; 2, (a, t) : 4; 2, (b, t) : 1; 4. Send 4 units from s to t . The three s - t paths cost $s \rightarrow a \rightarrow b \rightarrow t$: 3, $s \rightarrow b \rightarrow t$: 4, $s \rightarrow a \rightarrow t$: 5 per unit. The cheapest path is limited by $u_{ab} = 2$. *Plan*: 2 units on $s \rightarrow a \rightarrow b \rightarrow t$ and 2 units on $s \rightarrow b \rightarrow t$: $x_{sa} = 2$, $x_{ab} = 2$, $x_{sb} = 2$, $x_{bt} = 4$, $x_{at} = 0$. Capacities hold ($x_{bt} = 4 \leq 4$, $x_{sb} = 2 \leq 3$), balances hold. Cost $= 2 \cdot 3 + 2 \cdot 4 = 14$. *Lower bound*. Use the balance equations to eliminate variables. Conservation gives $x_{sa} = x_{ab} + x_{at}$ and $x_{bt} = x_{sb} + x_{ab}$, and the demand at t gives $x_{at} + x_{bt} = 4$, so $x_{sb} = 4 - x_{at} - x_{ab}$. Substituting into the cost,

$$x_{sa} + 3x_{sb} + x_{ab} + 4x_{at} + x_{bt} = 3x_{ab} + 5x_{at} + 4x_{sb} = 16 + x_{at} - x_{ab} \geq 16 + 0 - 2 = 14,$$

using $x_{at} \geq 0$ and the capacity $x_{ab} \leq 2$. Every feasible flow costs at least 14, so the plan is optimal by matching bounds. (Intuitively: the cheap path $s \rightarrow a \rightarrow b \rightarrow t$ at 3 per unit can carry only 2 units because of $u_{ab} = 2$; the rest costs at least 4 per unit.)

Assignment

Definition 1.6.17 (Assignment). A *bipartite graph* has vertex set $L \cup R$ with every edge joining L to R . With cost c_{ij} on edge $\{i, j\}$ (e.g. estimated handling time), the *assignment problem* chooses a set of edges such that every vertex

on both sides belongs to exactly one chosen edge, at minimum total cost:

$$\begin{aligned}
 \min \quad & \sum_{\{i,j\} \in E} c_{ij} x_{ij} \\
 \text{s.t.} \quad & \sum_{j: \{i,j\} \in E} x_{ij} = 1 & (i \in L), \\
 & \sum_{i: \{i,j\} \in E} x_{ij} = 1 & (j \in R), \\
 & x_{ij} \in \{0, 1\} & (\{i, j\} \in E).
 \end{aligned}$$

Example 1.6.18 (Analysts and incidents). Analysts A, B and incidents 1, 2 with handling times $c_{A1} = 2, c_{A2} = 5, c_{B1} = 4, c_{B2} = 3$ hours. Constraints: $x_{A1} + x_{A2} = 1, x_{B1} + x_{B2} = 1$ (each analyst handles one incident); $x_{A1} + x_{B1} = 1, x_{A2} + x_{B2} = 1$ (each incident handled once). There are only two complete assignments: $\{A1, B2\}$ costing $2 + 3 = 5$, and $\{A2, B1\}$ costing $5 + 4 = 9$. Optimal: $A \rightarrow 1, B \rightarrow 2$, total 5 hours.

Proposition 1.6.19 (Assignment requires equal sides). *If the assignment problem is feasible then $|L| = |R|$, and the number of chosen edges is $|L|$.*

Proof. Sum the L -constraints to get $\sum_{i \in L} \sum_j x_{ij} = |L|$, and sum the R -constraints to get $\sum_{j \in R} \sum_i x_{ij} = |R|$. Both sums count every chosen edge exactly once, so they are equal: $|L| = |R|$, and this common value is the number of chosen edges. $\square$

Technique: Writing graph models

- *Data vs. decision.* Nodes, arcs, weights, capacities, balances are data. Variables live on arcs: $x_{ij} \in \{0, 1\}$ to select, $x_{ij} \geq 0$ to send an amount.
- *One constraint per node.* Balance/conservation constraints have the shape $x(\delta^+(i)) - x(\delta^-(i)) = b_i$. Write $b_s = +1$ (or the supply), $b_t = -1$ (or the demand), 0 at relays. Check $\sum_i b_i = 0$.
- *One constraint per arc.* Capacity: $0 \leq x_{ij} \leq u_{ij}$.
- *Objective.* $\sum_{(i,j)} c_{ij} x_{ij}$: count links ($c \equiv 1$), total time, total cost.
- *Bounds from cuts.* For max flow, any cut's capacity is an upper bound; for shortest path, any path's length is an upper bound; for min-cost flow, "cheapest path times demand" is a lower bound.

Classic problems

- *Is there a graph with degrees 3, 3, 3, 3, 3?* No: the degree sum 15 is odd (Lemma 1.6.4).
- *Max flow with capacities $u_{sa} = 4, u_{sb} = 3, u_{ab} = 2, u_{at} = 2, u_{bt} = 4$.* Cut $\{s, a, b\}$ has capacity 6; the flow in Example 1.6.13 has value 6. Answer: 6.
- *Shortest path with unit weights from s to t when no arc (s, t) exists.* At least 2 arcs; any 2-arc path is optimal.

- *Assignment* 2×2 . Compare the two diagonals of the cost matrix. Optimal is the cheaper diagonal.

Gotcha

- $\delta^+(i)$ is a set of *arcs*, not of neighbours. Count arcs, then sum their variables.
- In a directed path every arc must point the way you travel. $s \rightarrow b$, then “ $b \rightarrow a$ ” is illegal if only (a, b) exists.
- Conservation is *not* imposed at s and t in max flow; imposing it would force value 0.
- In min-cost flow the objective is $\sum c_{ij}x_{ij}$ with cost per unit; do not multiply by capacities.

Summary: Graph models

- Digraph $D = (N, A)$ with ordered arcs; undirected graph $G = (V, E)$ with unordered edges. Incidence sets $\delta^+(i), \delta^-(i), \delta(v), \delta(S), E(S)$ are sets of links; $x(F) = \sum_{e \in F} x_e$.
- Handshake: $\sum \deg v = 2|E|$; summing conservation over a node set telescopes to the boundary.
- Four models: shortest path (ILP, select arcs), maximum flow (LP, choose amounts, cuts give upper bounds), minimum-cost flow (LP, given balances summing to 0), assignment (binary, exactly one on each side).
- Graphs and equations describe the same problem; the graph guides the equations.

1.7 Convexity

Motivation

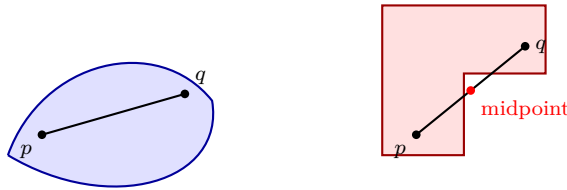
Why are linear programs easy while integer programs are hard? The deepest answer is *convexity*. Two properties of LPs are responsible: any mixture of two feasible plans is feasible, and a linear objective never rewards “hiding” a good solution behind a worse one nearby. Together they imply that a solution which cannot be improved by small changes cannot be improved at all. This section makes that precise.

Convex: if two plans are allowed, every blend of them is allowed. Convex objective: blending never costs more than the blended cost.

Definition 1.7.1 (Convex set). A set $X \subseteq \mathbb{R}^n$ is *convex* if for all $p, q \in X$ and all $\lambda \in [0, 1]$,

$$\lambda p + (1 - \lambda)q \in X.$$

Geometrically: the whole segment between any two points of X stays inside X .



convex: the whole segment stays inside nonconvex: even the midpoint is outside

Proposition 1.7.2 (Feasible sets of linear programs are convex). *Any set of the form $X = \{x \in \mathbb{R}^n : Ax \leq b\}$ (with equality constraints and sign constraints allowed, since $a^\top x = \beta$ is the pair $a^\top x \leq \beta, -a^\top x \leq -\beta$, and $x_j \geq 0$ is $-x_j \leq 0$) is convex.*

Proof. Let $p, q \in X$ and $\lambda \in [0, 1]$. Then $A(\lambda p + (1 - \lambda)q) = \lambda Ap + (1 - \lambda)Aq \leq \lambda b + (1 - \lambda)b = b$, where the inequality uses $Ap \leq b, Aq \leq b$ and $\lambda, 1 - \lambda \geq 0$. So $\lambda p + (1 - \lambda)q \in X$. $\square$

In Example 1.3.8, half of each of two feasible production plans gives their average, again feasible.

Definition 1.7.3 (Convex function). Let X be convex. A function $f : X \rightarrow \mathbb{R}$ is *convex* if for all $p, q \in X$ and $\lambda \in [0, 1]$,

$$f(\lambda p + (1 - \lambda)q) \leq \lambda f(p) + (1 - \lambda)f(q).$$

Geometrically: between any two points, the graph of f lies on or below the chord joining them.

Example 1.7.4 ($f(x) = x^2$). Choose $x = 0$ or $x = 2$: costs 0 and 4, average cost 2. The cost at the average choice is $f(1) = 1 \leq 2$: the curve lies below the joining line. In general, for $\lambda \in [0, 1]$,

$$\lambda p^2 + (1 - \lambda)q^2 - (\lambda p + (1 - \lambda)q)^2 = \lambda(1 - \lambda)(p - q)^2 \geq 0,$$

(expand and simplify) so x^2 is convex on $\mathbb{R}$.

Proposition 1.7.5 (Linear functions are convex). $f(x) = c^\top x + d$ is convex (and so is $-f$; such functions are called affine).

Proof. $f(\lambda p + (1 - \lambda)q) = \lambda c^\top p + (1 - \lambda)c^\top q + d = \lambda f(p) + (1 - \lambda)f(q)$: equality holds, so in particular $\leq$ holds. The same computation applies to $-f$. $\square$

Definition 1.7.6 (Local and global minima). Let $f : X \rightarrow \mathbb{R}$. A point $x^* \in X$ is a *global minimum* if $f(x^*) \leq f(x)$ for all $x \in X$. It is a *local minimum* if there is $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for all $x \in X$ with $\|x - x^*\| < \varepsilon$: no better feasible choice nearby.

Theorem 1.7.7 (In convex minimisation, local is global). *Let X be a convex set and $f : X \rightarrow \mathbb{R}$ a convex function. Every local minimum of f over X is a global minimum.*

Proof. Step 1 (set-up). Let x^* be a local minimum with radius ε , and suppose for contradiction that some $y \in X$ has $f(y) < f(x^*)$.

Step 2 (walk toward y). For $\lambda \in (0, 1]$ put $x_\lambda = x^* + \lambda(y - x^*) = (1 - \lambda)x^* + \lambda y$. By convexity of X , $x_\lambda \in X$. By convexity of f ,

$$f(x_\lambda) \leq (1 - \lambda)f(x^*) + \lambda f(y) < (1 - \lambda)f(x^*) + \lambda f(x^*) = f(x^*),$$

where the strict inequality uses $f(y) < f(x^*)$ and $\lambda > 0$.

Step 3 (contradiction). $\|x_\lambda - x^*\| = \lambda\|y - x^*\|$, which is smaller than ε for λ small enough. So for such λ , x_λ is a feasible point within distance ε of x^* with a strictly smaller objective, contradicting local minimality. Hence no such y exists and x^* is a global minimum. $\square$

Key insight

Convexity turns a *far-away* better point y into *nearby* better points: the segment from x^* to y stays feasible, and the objective along it stays below the chord, so it decreases immediately. A convex landscape has no “valleys” in which to get trapped.

Corollary 1.7.8 (Linear programming is convex minimisation). *For an LP, the feasible set is convex (Proposition 1.7.2) and the objective is convex (Proposition 1.7.5; for maximisation, minimise $-c^\top x$, also convex). Hence every local optimum of an LP is a global optimum.*

Proof. Combine the two propositions with Theorem 1.7.7. $\square$

Proposition 1.7.9 (The set of optimal solutions of an LP is convex). *Let X be convex, f linear, and $X^* = \{x \in X : f(x) = z^*\}$ the set of optimal solutions of $\min_X f$. Then X^* is convex.*

Proof. For $p, q \in X^*$ and $\lambda \in [0, 1]$, $\lambda p + (1 - \lambda)q \in X$ by convexity of X , and $f(\lambda p + (1 - \lambda)q) = \lambda z^* + (1 - \lambda)z^* = z^*$ by linearity. So the point is optimal. $\square$

Interpretation & discussion

Two consequences shape everything that follows. First, an algorithm that only makes *local improvements* (like the simplex method) can be trusted to find a *global* optimum of an LP. Second, if an LP has two optimal solutions it has infinitely many (the whole segment between them). Integer programs break both: the feasible set $\{0, 1\}^n$ is not convex (the midpoint of two integer points is usually fractional), so local search can get stuck and the theory above gives no guarantee.

Technique: Checking convexity

- *Sets:* intersections of convex sets are convex; half-spaces $\{a^\top x \leq \beta\}$ and hyperplanes are convex; so any system of linear (in)equalities defines a convex set. A set described with “or” ($x \leq -1$ or $x \geq 1$) is usually not convex: test the midpoint.
- *Functions:* linear/affine functions, x^2 , $|x|$, e^x are convex; sums of convex functions with nonnegative weights are convex; the maximum of convex functions is convex. A twice-differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex iff $f'' \geq 0$.

- *Quick disproof:* find two points where the midpoint fails. For $f(x) = -x^2$: $f(0) = 0 > \frac{f(-1)+f(1)}{2} = -1$, not convex.
- *Maximising a convex function* is *not* convex minimisation: $\max xy$ in the garden example needed a separate argument.

Classic problems

- Is $\{(x, y) : xy \geq 1, x, y > 0\}$ convex? Yes: it is the region on or above the graph of $g(x) = 1/x$ on $x > 0$, and g is convex there ($g''(x) = 2/x^3 > 0$). The region above the graph of a convex function is convex: if $y_1 \geq g(x_1)$ and $y_2 \geq g(x_2)$ then $\lambda y_1 + (1-\lambda)y_2 \geq \lambda g(x_1) + (1-\lambda)g(x_2) \geq g(\lambda x_1 + (1-\lambda)x_2)$.
- Is $\{x \in \mathbb{R} : |x| \geq 1\}$ convex? No: -1 and 1 are in the set, their midpoint 0 is not.
- An LP has optimal solutions $(0, 4)$ and $(2, 2)$. Give a third. $(1, 3)$, by Proposition 1.7.9.

Gotcha

- “Convex” is a property of the *pair* (set, objective) for minimisation. A convex objective over a nonconvex set (integer points) or a concave objective to be minimised both lose the local-is-global guarantee.
- x^2 is convex, but $\max x^2$ over $[-1, 2]$ is attained at an endpoint, not where the derivative vanishes; the theorem is about *minimising* convex functions.

Summary: Convexity

- Convex set: segments stay inside. Convex function: graph below chords.
- Feasible sets of linear systems are convex; linear objectives are convex.
- In convex minimisation every local minimum is global (Theorem 1.7.7); LPs are a special case, and their optimal sets are convex.

Notes on the original slides

Every number in the Lecture 1 slides was recomputed while writing this chapter, and all of them check out (the transport bounds 96,000 and 92,000, the certificate $30x + 20y = 10(x+y) + 10(2x+y)$, the knapsack and project tables, the incidence sets, the flow constraints, the assignment costs). A few statements are correct but easy to misread, and are worth a warning.

- **“A model can have different outcomes” (slide 26).** The three outcomes listed (an optimum, infeasible, unbounded) are exhaustive for *linear* programs (Chapter 2) but not for general mathematical programs: $\min x$ over $x > 0$ is feasible and bounded yet has no optimum. See the remark after Definition 1.2.4.
- **“Path: a sequence of linked vertices” (slide 47).** As stated this also covers walks that revisit vertices. The standard convention, used in

this chapter, is that a path has no repeated vertex; the slides' examples all satisfy it.

- **Shortest-path constraints (slide 53).** The sink equation $x_{at} + x_{bt} = 1$ is implied by the other three (add them). Keeping it is harmless and mirrors the general form $x(\delta^+(i)) - x(\delta^-(i)) = b_i$ at every node, but do not be surprised if a solver reports a redundant row.
- **Lower bound by “cheapest route” (slide 35).** The slide states $L = 92,000$ as if it were obvious; strictly, one must also argue that 92,000 is the *optimum* of the relaxed problem, not just the cost of one relaxed plan. The one-line justification is given in Section 1.5.
- **Knapsack relaxation (slide 29).** Here the LP relaxation happens to give the integer optimum 15; this is a coincidence of the data, not a general fact.
- **Maximum flow objective (slide 55).** $\max x_{sa} + x_{sb}$ is the net outflow of s only because no arc enters s ; in general the objective is $x(\delta^+(s)) - x(\delta^-(s))$.

Chapter takeaways

Summary: From a decision to a model

- **Build the model.** Variables: what can we choose (continuous, integer, binary)? Constraints: what must we respect? Objective: what do we minimise or maximise?
- **Judge the solution.** Feasible: satisfies every constraint. Optimal: best objective value among feasible choices. Bounds: for minimisation, lower bound $\leq$ optimal cost $\leq$ feasible cost; matching bounds prove optimality.
- **Choose a useful representation.** LPs have continuous variables and linear expressions, and any LP can be put in standard form. Convexity makes local optima global. Binary variables select, assign or activate. Graphs (vertices, links, weights with stated units) organise shortest-path, flow and assignment models; the graph and the equations describe the same problem.
- **Check the model.** Does every mathematically allowed choice make sense in reality? Start simple, state assumptions, and check the result against reality.
- **Next.** Solving linear programs (Chapter 2); graph algorithms later.

Chapter 2

The Geometry and Algebra of Linear Programs

2.1 From Scalars to Matrices

Motivation

Chapter 1 ended with a promise: linear programs can be solved exactly and efficiently. Delivering on that promise requires a change of viewpoint. Instead of a list of inequalities in $x_1, \dots, x_n$, we regard an LP as a statement about a matrix A , a right-hand side b and a cost vector c . Then we can ask linear-algebra questions (are these columns independent? is this system consistent? what is the rank?) whose answers turn out to locate the optimal solutions.

Matrix notation is not decoration: it lets us apply the whole toolbox of linear algebra (rank, independence, inverses) to a decision problem.

Recall from Chapter 1 that many OR problems have the shape

$$\max_{x \in X} f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \quad (i = 1, \dots, m), \quad h_k(x) = 0 \quad (k = 1, \dots, p),$$

where $x = (x_1, \dots, x_n)$ collects the decisions and $X \subseteq \mathbb{R}^n$ specifies their domain. A linear program is the continuous case in which f , g_i and h_k are all linear.

Definition 2.1.1 (Canonical form, matrix notation). Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. The *canonical form* of an LP is

$$\max_{x \in \mathbb{R}^n} c^\top x \quad \text{subject to} \quad Ax \leq b, \quad x \geq 0,$$

where vector inequalities are understood *componentwise*: $Ax \leq b$ means $\sum_{j=1}^n a_{ij}x_j \leq b_i$ for every $i = 1, \dots, m$, and $x \geq 0$ means $x_j \geq 0$ for every j . Row i of A holds the coefficients of inequality i ; the numbers a_{ij}, b_i, c_j are data.

A dimension check keeps everyone honest: A is $m \times n$, x is $n \times 1$, so Ax is $m \times 1$, matching b . The objective $c^\top x$ is $(1 \times n)(n \times 1) = 1 \times 1$: a scalar, not a vector.

Proposition 2.1.2 (Componentwise inequalities behave well). For $u, v, w \in \mathbb{R}^m$ and $\lambda \geq 0$: if $u \leq v$ then $u + w \leq v + w$ and $\lambda u \leq \lambda v$; if $u \leq v$ and $y \geq 0$ then $y^\top u \leq y^\top v$.

Proof. Each statement holds coordinate by coordinate for real numbers; the last is $\sum_i y_i u_i \leq \sum_i y_i v_i$ because each term satisfies $y_i u_i \leq y_i v_i$ for $y_i \geq 0$. $\square$

Example 2.1.3 (Paint workshop, canonical form). A workshop makes paints A and B . Per litre, A needs 1 litre of base liquid and 2 g of pigment, earning 3 MAD; B needs 1 litre and 1 g, earning 2 MAD. Available: 4 litres of base, 6 g of pigment. Assumptions: fractional litres possible, constant recipes and profits, all production sold.

$$\max 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4, \quad 2x_1 + x_2 \leq 6, \quad x_1, x_2 \geq 0.$$

In matrix notation, with

$$G = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad h = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad p = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

the LP is $\max p^\top x$ s.t. $Gx \leq h, x \geq 0$: two decisions, two resource inequalities.

Summary: Matrix notation

- Canonical form: $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$, inequalities componentwise.
- Check dimensions: $(m \times n)(n \times 1) = m \times 1$; $c^\top x$ is a scalar.
- Nonnegative multipliers preserve componentwise inequalities (Proposition 2.1.2); this is the algebraic heart of weak duality.

2.2 Equational Form

Motivation

Inequalities are awkward to manipulate algebraically: you cannot “solve” $x_1 + x_2 \leq 4$ for x_1 . Equations, by contrast, are the natural habitat of linear algebra. The equational form converts every inequality into an equation by giving the unused part of each resource its own name. The price is a few extra variables; the reward is that we can choose independent columns, solve a square system, and then simply *check* whether the result is nonnegative.

*Equalities let us solve;
nonnegativity lets us check.
The equational form
separates the two jobs.*

Definition 2.2.1 (Equational form). An LP is in *equational form* if it reads

$$\max_{z \in \mathbb{R}^n} c^\top z \quad \text{subject to} \quad Az = b, \quad z \geq 0,$$

with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$: m equations and n variables, where n counts *all* variables including any that were added during the conversion. Its *feasible set* is $P = \{z \in \mathbb{R}^n : Az = b, z \geq 0\}$.

Slack: turn unused resources into variables

Example 2.2.2 (Slack in the paint workshop). For the plan $(x_1, x_2) = (1, 2)$, base liquid used is $x_1 + x_2 = 3$ of 4 litres: 1 litre is unused. Pigment used is $2x_1 + x_2 = 4$ of 6 g: 2 g unused. Call the unused amounts $s_1 = 1$ and $s_2 = 2$. In general

$$x_1 + x_2 + s_1 = 4, \quad s_1 \geq 0, \quad 2x_1 + x_2 + s_2 = 6, \quad s_2 \geq 0 :$$

used stock + unused stock = available stock. The equational form of Example 2.1.3 is

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + 0s_1 + 0s_2 \\ \text{s.t.} \quad & x_1 + x_2 + s_1 = 4, \quad 2x_1 + x_2 + s_2 = 6, \quad x_1, x_2, s_1, s_2 \geq 0, \end{aligned}$$

i.e. $\max c^\top z$ s.t. $Az = b, z \geq 0$ with

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 4}, \quad z = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

The slack variables have objective coefficient 0 because profit depends only on production.

Definition 2.2.3 (Slack and surplus).

- *At most:* $a^\top x \leq b$ becomes $a^\top x + s = b, s \geq 0$. The *slack* $s = b - a^\top x$ is what is missing to reach capacity (a tank holding $q \leq 10$ litres: $s = 10 - q$).
- *At least:* $a^\top x \geq b$ becomes $a^\top x - s = b, s \geq 0$. The *surplus* $s = a^\top x - b$ is the amount beyond the minimum (an order of $q \geq 10$: $s = q - 10$).
- *Exactly:* $a^\top x = b$ stays as it is. No extra variable.

Gotcha

Never add a nonnegative variable to an *equality*. Requiring exactly $q = 10$ and writing $q + s = 10, s \geq 0$ would allow $q = 7, s = 3$; writing $q - s = 10$ would allow $q = 13$. Both change the requirement. Keep $a^\top x = b$ unchanged.

Theorem 2.2.4 (Canonical and equational forms are equivalent). *Let (C) be $\max\{c^\top x : Ax \leq b, x \geq 0\}$ and (E) be $\max\{c^\top x + 0^\top s : Ax + s = b, x \geq 0, s \geq 0\}$. The map $\phi(x) = (x, b - Ax)$ is a bijection from the feasible set of (C) onto the feasible set of (E) , with $c^\top x$ preserved. Consequently (C) and (E) have the same optimal value, x is optimal for (C) iff $\phi(x)$ is optimal for (E) , and (x, s) is optimal for (E) iff x is optimal for (C) .*

Proof. Step 1 (ϕ lands in the feasible set of (E)). If $Ax \leq b$ and $x \geq 0$, then $s := b - Ax \geq 0$ and $Ax + s = b$, so (x, s) is feasible for (E) .

Step 2 (injective). $\phi(x) = \phi(x')$ forces $x = x'$ (first component).

Step 3 (surjective). If (x, s) is feasible for (E) , then $s = b - Ax$ (from $Ax + s = b$) and $s \geq 0$ gives $Ax \leq b$; with $x \geq 0$, x is feasible for (C) and $\phi(x) = (x, s)$. So every feasible point of (E) has unique slacks determined by x , and dropping the slacks recovers a feasible x .

Step 4 (objective). $c^\top x + 0^\top s = c^\top x$. A bijection between feasible sets that preserves the objective maps optimal solutions to optimal solutions and preserves the optimal value. $\square$

Interpretation & discussion

Nothing about the decision has changed. Every feasible production plan has *unique* slacks, and the slacks carry useful information: a slack of 0 means the resource is fully used (the constraint is *tight* or *active*); a positive slack means spare capacity.

Make every variable nonnegative

Proposition 2.2.5 (Substitutions for bounded and free variables). *Each substitution below is a bijection (or, in the free case, a surjection) between the original domain of t and the new domain, so replacing t by the expression in u (and v) everywhere yields an LP with the same attainable objective values and the same optimal value.*

Original domain	Substitution	New domain
$t \geq L$	$t = L + u$	$u \geq 0$
$t \leq U$	$t = U - u$	$u \geq 0$
$L \leq t \leq U$	$t = L + u$	$0 \leq u \leq U - L$
$t \in \mathbb{R}$	$t = u - v$	$u, v \geq 0$

Proof. $u \mapsto L+u$ maps $[0, \infty)$ bijectively onto $[L, \infty)$ (inverse $u = t-L$); $u \mapsto U-u$ maps $[0, \infty)$ bijectively onto $(-\infty, U]$ (inverse $u = U-t$); $u \mapsto L+u$ maps $[0, U-L]$ bijectively onto $[L, U]$. The free case is Proposition 1.3.5. In each case the substitution is an identity in the constraints and objective, so feasible points correspond with equal objective values, and the sets of attainable values (hence the optimal values) coincide. $\square$

Example 2.2.6. For $-2 \leq t \leq 3$, use $t = u - 2$ with $0 \leq u \leq 5$. One shifted variable is enough; the upper bound $u \leq 5$ then becomes an ordinary $\leq$ constraint with its own slack. Use $u - v$ only for a genuinely free variable: it adds a variable and destroys uniqueness for nothing.

Technique: Converting to equational form

1. Make the objective a maximisation ($\min f \rightarrow \max -f$).
2. For every $\leq$ constraint add a slack $+s_i$; for every $\geq$ constraint subtract a surplus $-s_i$; leave equalities alone. Each new variable is ≥ 0 with objective coefficient 0.
3. For every variable not already ≥ 0 : shift ($t = L + u$), flip ($t = U - u$), or split ($t = u - v$) as in Proposition 2.2.5.
4. Recount: n is the total number of variables now; m is the number of equations. Write A with one column per variable and one row per equation, and c with a 0 for each added variable.
5. *Sanity checks:* slack columns of $\leq$ constraints form (part of) an identity matrix; every original feasible point must still be reachable; objective coefficients of slacks are 0.

Classic problems

- *Convert* $\min x_1 - x_2$ s.t. $x_1 + x_2 \geq 2$, $x_1 \leq 5$, $x_1 \geq 0$, $x_2 \in \mathbb{R}$. Set $x_2 = u - v$: $\max -x_1 + u - v$ s.t. $x_1 + u - v - s_1 = 2$, $x_1 + s_2 = 5$, $x_1, u, v, s_1, s_2 \geq 0$.
- *Read off slacks*. In the paint LP, at $(x_1, x_2) = (2, 2)$: $s_1 = 4 - 4 = 0$, $s_2 = 6 - 6 = 0$; both resources fully used.
- *How many variables?* A canonical LP with n variables and m constraints has $n + m$ variables in equational form.

Gotcha

- Adding a slack to an equality (see above).
- Forgetting to keep the bounds on a split variable: $-2 \leq t \leq 2$ with $t = u - v$ needs $u - v \leq 2$ and $-u + v \leq 2$ (each then gets its own slack).
- Giving the slack a nonzero objective coefficient. Unused stock earns nothing.
- Mixing up the sign: surplus is *subtracted*, slack is *added*.

Summary: Equational form

- $\max c^\top z$ s.t. $Az = b$, $z \geq 0$; feasible set $P = \{z : Az = b, z \geq 0\}$.
- Slack (+ s) for $\leq$, surplus ($-s$) for $\geq$, nothing for $=$; shift/flip/split variables to make them nonnegative.
- The conversion is an exact translation (Theorem 2.2.4, Proposition 2.2.5): same decisions, same optimal value.

2.3 A Linear Algebra Interlude: Solving $Az = b$

Motivation

Now that constraints are equations, the natural question is: what does the solution set of $Az = b$ look like, and how do we compute with it? Three facts are needed: row operations do not change the solution set; consistency can be read from the rank; and zero rows can be discarded. We prove them and then apply them to an LP with three variables and four constraints.

Definition 2.3.1 (Elementary row operations). On a matrix (typically the augmented matrix $[A \mid b]$): (i) swap two rows; (ii) multiply a row by a nonzero scalar; (iii) add a multiple of one row to another row. *Row reduction* (Gaussian elimination) applies these operations to reach a form in which each nonzero row starts with a *pivot* (its first nonzero entry) strictly to the right of the pivot of the row above, and all zero rows are at the bottom. The number of pivots is the *rank*.

Proposition 2.3.2 (Row operations preserve solution sets). If $[A' \mid b']$ is obtained from $[A \mid b]$ by elementary row operations, then $\{z : Az = b\} = \{z : A'z = b'\}$.

Proof. Each row of $[A | b]$ encodes one equation. Swapping equations, multiplying an equation by $\lambda \neq 0$, or adding a multiple of one equation to another produces equations that are consequences of the old ones, so every old solution satisfies the new system. Each operation is invertible by an operation of the same kind (swap back; multiply by $1/\lambda$; subtract the multiple), so every new solution also satisfies the old system. $\square$

Key insight

Invertibility. Each step can be undone, so nothing is lost or gained. The single most common error is to apply an operation to A but not to b ; the equation is then no longer the same equation.

Theorem 2.3.3 (Consistency). *The system $Az = b$ has a solution if and only if b lies in the column space $\text{col}(A) = \text{span}(a_1, \dots, a_n)$, if and only if $\text{rank}(A) = \text{rank}([A | b])$. Equivalently, after row reduction of $[A | b]$, there is no row of the form $[0 \cdots 0 | d]$ with $d \neq 0$.*

Proof. First equivalence. $Az = z_1 a_1 + \cdots + z_n a_n$ is a linear combination of the columns, so a solution exists iff b is such a combination, i.e. $b \in \text{col}(A)$.

Second equivalence. We have $\text{rank}([A | b]) = \dim \text{span}(a_1, \dots, a_n, b)$. If $b \in \text{span}(a_1, \dots, a_n)$, adding b does not enlarge the span, so the ranks are equal. If $b \notin \text{span}(a_1, \dots, a_n)$, adding b increases the dimension by one, so $\text{rank}([A | b]) = \text{rank}(A) + 1$.

Row form. Row reduction preserves the rank of A and of $[A | b]$ (row operations do not change row spaces). A reduced row $[0 \cdots 0 | d]$ with $d \neq 0$ is a pivot in the last column, which happens exactly when $\text{rank}([A | b]) > \text{rank}(A)$; and such a row reads $0 = d$, which no z satisfies. $\square$

Proposition 2.3.4 (Zero rows may be deleted). *If the reduced augmented matrix contains a row $[0 \cdots 0 | 0]$, deleting it does not change the solution set. After deleting all such rows, the remaining matrix $\tilde{A} \in \mathbb{R}^{r \times n}$ has $r = \text{rank}(A)$ rows, which are linearly independent, and $\{z : \tilde{A}z = \tilde{b}\} = \{z : Az = b\}$.*

Proof. The row $0 = 0$ holds for every z , so removing it changes nothing. The remaining rows are the nonzero rows of a row-reduced matrix; each has a pivot in a distinct column, so they are independent, and their number equals the number of pivots, i.e. the rank. The equality of solution sets is Proposition 2.3.2 plus the deletion just justified. $\square$

Redundant inequalities

Before adding slacks, it is wise to discard inequalities that are implied by others: they only add variables and equations without changing the feasible set.

Proposition 2.3.5 (Nonnegative combinations are redundant). *Suppose inequality k satisfies $a_k = \sum_{i \neq k} \lambda_i a_i$ and $b_k \geq \sum_{i \neq k} \lambda_i b_i$ for some $\lambda_i \geq 0$. Then every x satisfying the other inequalities $a_i^\top x \leq b_i$ ($i \neq k$) also satisfies $a_k^\top x \leq b_k$; the feasible set is unchanged when inequality k is deleted.*

Proof. $a_k^\top x = \sum_{i \neq k} \lambda_i a_i^\top x \leq \sum_{i \neq k} \lambda_i b_i \leq b_k$, using $\lambda_i \geq 0$ for the first inequality. $\square$

Worked example: build a basic feasible solution by hand**Example 2.3.6** (A three-variable LP).

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + x_3 \quad \text{s.t.} \quad \begin{aligned} & x_1 + x_2 + x_3 \leq 4, \\ & 2x_1 + 3x_2 + x_3 \leq 10, \\ & 3x_1 + 4x_2 + 2x_3 \leq 14, \\ & 3x_1 + 5x_2 + x_3 \leq 18, \end{aligned} \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

Our goal: construct a basic feasible solution. We simplify, add slacks, eliminate, select variables, and check.

Step 1: remove a redundant inequality. Adding the first two inequalities gives $3x_1 + 4x_2 + 2x_3 \leq 14$, which is exactly the third. By Proposition 2.3.5 (with $\lambda_1 = \lambda_2 = 1$), the third inequality is implied and we remove it before adding slacks. (The fourth is also redundant: $\frac{1}{2}(\text{first}) + \frac{3}{2}(\text{second})$ gives $3.5x_1 + 5x_2 + 2x_3 \leq 17$, which implies the fourth for $x \geq 0$; we keep it anyway, to practise with three equations.)

Step 2: add three slacks.

$$x_1 + x_2 + x_3 + s_1 = 4, \quad 2x_1 + 3x_2 + x_3 + s_2 = 10, \quad 3x_1 + 5x_2 + x_3 + s_3 = 18,$$

with $z = (x_1, x_2, x_3, s_1, s_2, s_3)^\top \geq 0$ and $c = (3, 2, 1, 0, 0, 0)^\top$; $A \in \mathbb{R}^{3 \times 6}$.

Step 3: row-reduce $[A \mid b]$, columns in the order $x_1, x_2, x_3, s_1, s_2, s_3 \mid b$:

$$\begin{aligned} [A \mid b] &= \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 2 & 3 & 1 & 0 & 1 & 0 & 10 \\ 3 & 5 & 1 & 0 & 0 & 1 & 18 \end{array} \right] \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 3R_1}} \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 0 & 1 & -1 & -2 & 1 & 0 & 2 \\ 0 & 2 & -2 & -3 & 0 & 1 & 6 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 4 \\ 0 & 1 & -1 & -2 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 & 1 & 2 \end{array} \right]. \end{aligned}$$

Three pivots (columns 1, 2, 4): $\text{rank}(A) = \text{rank}([A \mid b]) = 3$. The system is consistent with three independent rows, and there is no zero row to delete: each retained inequality has its own slack, and the three slack columns form an identity matrix in the original A , which already guarantees full row rank. The equivalent LP is

$$\begin{aligned} & \max \quad 3x_1 + 2x_2 + x_3 \quad \text{s.t.} \quad \begin{aligned} & x_1 + x_2 + x_3 + s_1 = 4, \\ & x_2 - x_3 - 2s_1 + s_2 = 2, \quad z \geq 0. \\ & s_1 - 2s_2 + s_3 = 2, \end{aligned} \end{aligned}$$

Step 4: choose three of the six variables, whose columns must be independent; set the other three to zero; solve; check signs.

First choice: x_1, x_2, x_3 , with $s_1 = s_2 = s_3 = 0$. The reduced system restricted to these columns is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 2 \end{array} \right].$$

Only two pivots: the three selected columns are dependent, and the last row reads $0 = 2$. This choice gives no solution at all. Reject it; the LP itself may still be feasible.

Second choice: x_1, x_2, s_3 , with $x_3 = s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \implies s_3 = 2, x_2 = 2, x_1 = 4 - x_2 = 2.$$

Three pivots: independent columns. The solution $z = (2, 2, 0, 0, 0, 2)^\top \geq 0$ is a *basic feasible solution*.

Check in the canonical LP. For $x = (2, 2, 0)$ all four original inequalities hold: $4 \leq 4$, $10 \leq 10$, $14 \leq 14$, $16 \leq 18$. The slacks of the three retained constraints are $s_1 = 0$, $s_2 = 0$, $s_3 = 2$, matching the computation; the removed inequality is automatically satisfied (with equality, as it must be, being the sum of two tight ones). Objective: $3(2) + 2(2) + 0 = 10$.

Third choice: x_1, x_3, s_3 , with $x_2 = s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \implies s_3 = 2, x_3 = -2, x_1 = 6.$$

Three pivots again: independent columns, and the system has a unique solution. But $x_3 = -2 < 0$: a basic solution that is *not feasible*. Nonnegativity is a separate check *after* solving.

Interpretation & discussion

Three lessons from one example. (1) Independence of the chosen columns is necessary to get a solution at all. (2) Independence is not sufficient for feasibility: the sign check is separate. (3) A basic feasible solution is one *candidate*, not the optimum. Indeed $x = (4, 0, 0)$ is feasible ($4 \leq 4$, $8 \leq 10$, $12 \leq 18$) with objective $12 > 10$; it corresponds to the choice x_1, s_2, s_3 giving $z = (4, 0, 0, 0, 2, 6)$. And 12 is in fact optimal: by Lemma 1.3.10 with $y = (3, 0, 0)$ on the three retained constraints, $A^\top y = (3, 3, 3) \geq (3, 2, 1) = c$ and $b^\top y = 12$, so no feasible point exceeds 12.

Technique: The general recipe: reduce, then build a candidate

Given $\max c^\top z$ s.t. $Az = b$, $z \geq 0$ with $A \in \mathbb{R}^{m \times n}$:

1. **Row-reduce** $[A | b]$. Apply every operation to the right-hand side too.
2. **Check consistency.** A row $[0 \cdots 0 | d]$ with $d \neq 0$ means no solution: the LP is infeasible (Theorem 2.3.3).
3. **Delete zero rows.** With $r = \text{rank}(A)$, the equivalent system $\tilde{A}z = \tilde{b}$ has $\tilde{A} \in \mathbb{R}^{r \times n}$ of rank r (Proposition 2.3.4). Feasible set and objective are unchanged.
4. **Choose r independent columns.** $B \subseteq \{1, \dots, n\}$, $|B| = r$, with $\tilde{A}_B$ invertible; $N = \{1, \dots, n\} \setminus B$.
5. **Set nonbasic variables to zero; solve for the rest.** $z_N = 0$, $\tilde{A}_B z_B = \tilde{b}$.
6. **Check nonnegativity.** If $z_B \geq 0$ the result is a BFS; otherwise this choice gives an infeasible basic solution. One choice gives one candidate,

not necessarily an optimum.

Gotcha

- Forgetting to transform b along with A .
- Confusing “this choice of columns has no solution” with “the LP is infeasible.” Only a row $[0 \cdots 0 \mid d \neq 0]$ in the reduction of the *full* $[A \mid b]$ proves infeasibility.
- Skipping the sign check. Solving the square system is half the job.
- Removing an inequality that is *not* implied (e.g. one obtained by subtracting others). Only nonnegative combinations are safe.

Summary: Solving $Az = b$

- Row operations are invertible and preserve the solution set; do them on $[A \mid b]$.
- Consistent iff $b \in \text{col}(A)$ iff $\text{rank } A = \text{rank}[A \mid b]$ iff no row $[0 \cdots 0 \mid d \neq 0]$.
- Delete zero rows to reach full row rank r ; then choose r independent columns, set the others to zero, solve, check signs.
- An inequality that is a nonnegative combination of others is redundant; drop it before adding slacks.

2.4 Bases and Basic Feasible Solutions

Motivation

The feasible set of an LP usually contains infinitely many points, so “compare them all” is not an algorithm. The worked example suggested a way out: certain feasible points are obtained by fixing all but a few variables to zero and solving a square system. There are only finitely many ways to choose the square system. If these special points suffice to find the optimum, we have a finite algorithm. This section defines them algebraically, connects them to the geometric idea of a corner, and works through every basis of a small example.

Infinitely many feasible points, but only finitely many corners. If we can prove the optimum sits at a corner, the search becomes finite.

Example 2.4.1 (Sharing network resources). A server handles traffic for two services with throughputs $x_1, x_2 \geq 0$ (Gbit/s). Per Gbit/s, service A uses 1 CPU unit and 2 bandwidth units; service B uses 1 and 1. Capacities: 6 CPU units, 8 bandwidth units:

$$x_1 + x_2 \leq 6, \quad 2x_1 + x_2 \leq 8.$$

With slacks $s_1 = 6 - x_1 - x_2$ (unused CPU) and $s_2 = 8 - 2x_1 - x_2$ (unused bandwidth), the equational feasible set is

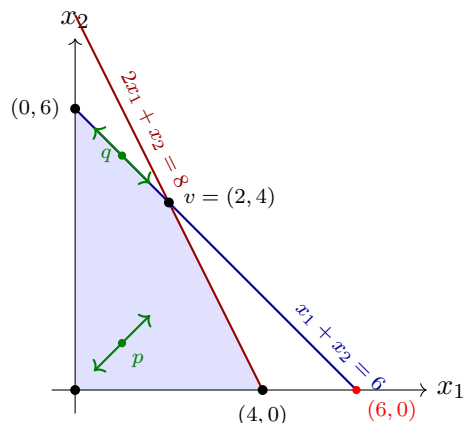
$$P = \{z = (x_1, x_2, s_1, s_2) \geq 0 : x_1 + x_2 + s_1 = 6, 2x_1 + x_2 + s_2 = 8\},$$

that is, $Az = b, z \geq 0$ with

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 8 \end{bmatrix}.$$

A feasible solution is any point of P ; it need not be optimal. For instance $z = (1, 1, 4, 5)$ satisfies both equations and is nonnegative. We could maximise total throughput $x_1 + x_2$; for now we study feasibility and bases.

Geometric intuition



At p (interior) or q (on an edge) we can move a little in *both* directions along some straight line and remain feasible. At a vertex such as $v = (2, 4)$, no such two-sided movement is possible: every line through v leaves the region on at least one side. The picture uses (x_1, x_2) ; the complete vector also contains the slacks.

Definition 2.4.2 (Extreme point). A point z of a convex set P is an *extreme point* (or *vertex*) of P if it is not the midpoint of two distinct points of P : there do not exist $u \neq w$ in P with $z = \frac{1}{2}(u + w)$. Equivalently, z cannot be written as $\lambda u + (1 - \lambda)w$ with $u \neq w$ in P and $0 < \lambda < 1$.

(The two formulations agree: if $z = \lambda u + (1 - \lambda)w$ with $0 < \lambda < 1$, $u \neq w$, then replacing u, w by suitable points on the segment $[u, w]$ symmetric about z writes z as a midpoint; the converse is the case $\lambda = \frac{1}{2}$.)

The algebraic definition

Definition 2.4.3 (Basis, basic solution, basic feasible solution). Let $A \in \mathbb{R}^{m \times n}$ have $\text{rank}(A) = m$ (full row rank), and write $A = (a_1 \cdots a_n)$ by columns. A *basis* is an index set $B \subseteq \{1, \dots, n\}$ with $|B| = m$ such that the square matrix $A_B = (a_j)_{j \in B}$ is invertible. Let $N = \{1, \dots, n\} \setminus B$ (the *nonbasic* indices) and write $z_B = (z_j)_{j \in B}$, $z_N = (z_j)_{j \in N}$. The *basic solution* associated with B is the unique vector z with

$$z_N = 0, \quad z_B = A_B^{-1}b.$$

It is a *basic feasible solution* (BFS) if in addition $z_B \geq 0$. A feasible vector $z \in P$ is called *basic* if it is the basic solution of at least one basis. The variables z_j , $j \in B$, are the *basic variables*.

Proposition 2.4.4 (Basic solutions are well defined and satisfy $Az = b$). For a basis B , the basic solution exists, is unique, and satisfies $Az = b$. It has at most m nonzero entries.

Proof. The equations split as $Az = A_B z_B + A_N z_N = b$. Imposing $z_N = 0$ leaves $A_B z_B = b$, a square system with invertible matrix, so $z_B = A_B^{-1}b$ is its unique solution. Substituting back, $Az = A_B A_B^{-1}b + A_N 0 = b$. Only the m entries in z_B can be nonzero. $\square$

Remark 2.4.5 (Degeneracy). Basic variables may themselves equal zero: a BFS has *at most* m positive entries, not necessarily exactly m . Such a BFS is called *degenerate*, and it typically arises from several different bases. For example, with constraints $x_1 \leq 1$, $x_1 + x_2 \leq 1$ (so $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$, $b = (1, 1)^\top$), the bases $\{x_1, x_2\}$, $\{x_1, s_1\}$ and $\{x_1, s_2\}$ all produce the same point $z = (1, 0, 0, 0)$.

Every basis of the network example

Example 2.4.6 (Basis $\{1, 2\}$: both throughputs). Choose $B = \{1, 2\}$, $N = \{3, 4\}$. $A_B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ with $\det A_B = -1 \neq 0$: a basis. Set $s_1 = s_2 = 0$ and solve $x_1 + x_2 = 6$, $2x_1 + x_2 = 8$: $x_1 = 2$, $x_2 = 4$. So $z = (2, 4, 0, 0)^\top \geq 0$ is a BFS. It is the corner where both resources are fully used.

Example 2.4.7 (Basis $\{1, 3\}$: a different corner). $B = \{1, 3\}$ (solve for x_1 and s_1), $A_B = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$, $\det A_B = -2 \neq 0$. Set $x_2 = s_2 = 0$: $2x_1 = 8$ gives $x_1 = 4$, then $s_1 = 6 - 4 = 2$. So $z = (4, 0, 2, 0)^\top \geq 0$: another BFS. No service- B traffic; bandwidth fully used; two CPU units remain. This is the corner $(4, 0)$.

Example 2.4.8 (Basis $\{1, 4\}$: a basis that produces an infeasible point). $B = \{1, 4\}$ selects x_1, s_2 ; set $x_2 = s_1 = 0$. Then $x_1 = 6$ and $2x_1 + s_2 = 8$ gives $s_2 = -4$. $A_B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ is nonsingular, so B is a basis, but $s_2 = -4 < 0$: not a feasible basis. Geometrically, $(6, 0)$ is the intersection of $x_2 = 0$ and $x_1 + x_2 = 6$ but violates $2x_1 + x_2 \leq 8$ (the red point in the figure).

Example 2.4.9 (All bases: finite candidates). There are $\binom{4}{2} = 6$ two-column sets.

Basis B	Basic solution (x_1, x_2, s_1, s_2)	Status
$\{3, 4\}$	$(0, 0, 6, 8)$	feasible
$\{1, 3\}$	$(4, 0, 2, 0)$	feasible
$\{1, 2\}$	$(2, 4, 0, 0)$	feasible
$\{2, 4\}$	$(0, 6, 0, 2)$	feasible
$\{1, 4\}$	$(6, 0, 0, -4)$	infeasible
$\{2, 3\}$	$(0, 8, -2, 0)$	infeasible

(Every pair of columns here is independent, so all six are bases.) Four feasible bases, four vertices of the original polygon: $(0, 0)$, $(4, 0)$, $(2, 4)$, $(0, 6)$. The two infeasible basic solutions are intersections of two constraint lines that fall outside the polygon.

Vertices are exactly the basic feasible solutions

Theorem 2.4.10 (BFS = extreme point). *Let $A \in \mathbb{R}^{m \times n}$ have rank m and $P = \{z : Az = b, z \geq 0\}$. A point $z \in P$ is a basic feasible solution if and only if it is an extreme point of P .*

Proof. ($\Rightarrow$) **A BFS is extreme.** Let z be the BFS of a basis B , and suppose $z = \lambda u + (1 - \lambda)w$ with $u, w \in P$ and $0 < \lambda < 1$. For $j \in N$: $0 = z_j = \lambda u_j + (1 - \lambda)w_j$ with $u_j, w_j \geq 0$ and both coefficients positive, so $u_j = w_j = 0$. Thus $u_N = w_N = 0$, and $Au = b$ reduces to $A_B u_B = b$; likewise $A_B w_B = b$. Since A_B is invertible, $u_B = w_B = A_B^{-1}b = z_B$. Hence $u = w = z$, so z is not a proper combination of two distinct points: it is extreme.

($\Leftarrow$) **An extreme point is a BFS.** Let $z \in P$ be extreme and let $S = \{j : z_j > 0\}$ be its support.

Step 1: the columns $\{a_j : j \in S\}$ are linearly independent. Suppose not: there is $d \neq 0$ with $d_j = 0$ for $j \notin S$ and $Ad = \sum_{j \in S} d_j a_j = 0$. For $\varepsilon > 0$, the points $z \pm \varepsilon d$ satisfy $A(z \pm \varepsilon d) = b$. Their entries outside S are 0, and inside S they are $z_j \pm \varepsilon d_j$, which are positive for ε small because $z_j > 0$. So $z + \varepsilon d$ and $z - \varepsilon d$ are two distinct points of P with midpoint z , contradicting extremality.

Step 2: extend to a basis. The independent set $\{a_j : j \in S\}$ has at most m elements. Since $\text{rank } A = m$, the n columns of A span $\mathbb{R}^m$, so any independent subset of columns can be extended, using further columns of A , to a set of m independent columns, i.e. to a basis $B \supseteq S$. (Repeatedly add a column not in the span of the current set; this is possible until the span is all of $\mathbb{R}^m$, which happens exactly when m independent columns have been collected.)

Step 3: z is the basic solution of B . For $j \notin B$ we have $j \notin S$, so $z_j = 0$: $z_N = 0$. Then $Az = b$ reads $A_B z_B = b$, so $z_B = A_B^{-1}b$. Thus z is the basic solution of B , and $z \geq 0$ makes it a BFS. $\square$

Key insight

The direction d in Step 1 is a *feasible direction of two-sided movement*: because $Ad = 0$, moving along $\pm d$ keeps the equations, and because d lives only on coordinates where z is strictly positive, small moves keep nonnegativity. A point admits such a d exactly when the columns under its support are dependent. So “no two-sided wiggle room” (geometry) is the same as “independent support columns” (algebra).

Interpretation & discussion

The theorem is why we may say “vertex” and “basic feasible solution” interchangeably. It also explains the table above: the four feasible bases produced exactly the four corners of the polygon, and no corner was missed. Note the role of the hypothesis $\text{rank } A = m$: without it, an independent support might not extend to m columns; this is why Section 2.3 insisted on deleting redundant equations first.

Corollary 2.4.11 (Finitely many vertices). *P has at most $\binom{n}{m}$ extreme points.*

Proof. Each extreme point is the basic solution of some basis (Theorem 2.4.10), each basis is a subset of size m of $\{1, \dots, n\}$, and each basis determines exactly one basic solution (Proposition 2.4.4). There are $\binom{n}{m}$ such subsets. $\square$

Technique: Working with bases by hand

- *Two-by-two invertibility*: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \neq 0$. Larger: row-reduce A_B and count pivots (m pivots means invertible).
- *Identity columns are free*: if the slack columns form an identity matrix, then $B = \{\text{all slacks}\}$ is always a basis and its basic solution is $x = 0$, $s = b$; it is feasible iff $b \geq 0$ (the origin of the polygon).
- *Read the corner*: a basis selects which constraints are *tight*. Nonbasic slack $s_i = 0$ means constraint i is tight; nonbasic $x_j = 0$ means the axis $x_j = 0$ is tight. Two tight lines in the plane meet at the corner.
- *Sanity check a BFS*: plug x back into the *original* inequalities. If any fails, the corresponding slack is negative and you mis-solved.
- *Count*: at most $\binom{n}{m}$ bases; in a 2-constraint, 4-variable problem that is 6, and you can list them all.

Classic problems

- Which point does basis $\{2, 4\}$ of Example 2.4.1 represent? $x_1 = s_1 = 0$, $x_2 = 6$, $s_2 = 8 - 6 = 2$: the corner $(0, 6)$; CPU fully used, 2 bandwidth units spare.
- Is $(1, 1, 4, 5)$ a BFS of Example 2.4.1? No: it has four positive entries but $m = 2$.
- Is $(3, 3, 0, -1)$ basic? It is the basic solution of $B = \{1, 2\}$ only if it solves $A_B z_B = b$; but $B = \{1, 2\}$ gives $(2, 4)$. In fact $(3, 3, 0, -1)$ is not even feasible ($s_2 < 0$), so it is not a BFS.
- A point with m positive entries whose columns are independent is a BFS. Yes: its support is a basis and it solves $A_B z_B = b$ with $z_B > 0$.

Gotcha

- *Basis $\neq$ feasible basis*. Invertibility of A_B says nothing about signs.
- *Not every m -subset is a basis*. Columns can be dependent (x_1, x_2, x_3 in Example 2.3.6).
- *Degenerate BFS*. One point, several bases. Do not conclude “different basis, different corner.”
- *Full row rank is required*. If $\text{rank } A < m$, no $m \times m$ submatrix is invertible and Definition 2.4.3 is vacuous; reduce first.

Summary: Bases and BFSs

- For A of full row rank m : a basis is m independent columns; set the other variables to 0 and solve $A_B z_B = b$; a BFS additionally has $z_B \geq 0$.
- BFSs are exactly the extreme points (vertices) of P (Theorem 2.4.10); there are at most $\binom{n}{m}$ of them.

- A BFS has at most m positive entries; fewer means degenerate.

2.5 The Fundamental Theorem of Linear Programming

Motivation

We now have a finite list of candidates: the basic feasible solutions. The question is whether the optimum is guaranteed to be among them. The answer is yes, under the two hypotheses one would expect (the problem is feasible, and the objective cannot grow without bound), and the proof is constructive: it shows how to walk from any feasible point to a vertex that is at least as good.

If a feasible point is not a vertex, slide along a feasible direction without losing objective value until a coordinate hits zero. Repeat. You arrive at a vertex at least as good.

Lemma 2.5.1 (Purification). *Let $A \in \mathbb{R}^{m \times n}$ have rank m , let $P = \{z : Az = b, z \geq 0\}$, and suppose $c^\top z$ is bounded above on P . Then for every $z \in P$ there is a basic feasible solution z' with $c^\top z' \geq c^\top z$.*

Proof. We argue by induction on the number $k = |\text{supp}(z)|$ of positive entries of z , where $\text{supp}(z) = \{j : z_j > 0\}$.

Base and “independent” case. If the columns $\{a_j : j \in \text{supp}(z)\}$ are linearly independent (this includes $k = 0$, the empty set being independent), then exactly as in Steps 2–3 of the proof of Theorem 2.4.10, $\text{supp}(z)$ extends to a basis B and z is the BFS of B . Take $z' = z$.

Dependent case. Otherwise there is $d \neq 0$ with $\text{supp}(d) \subseteq \text{supp}(z)$ and $Ad = 0$. Replacing d by $-d$ if necessary, we may assume $c^\top d \geq 0$. We claim we may also assume d has at least one negative entry:

- if $c^\top d > 0$ and $d \geq 0$, then $z + \theta d \in P$ for all $\theta \geq 0$ (equations hold since $Ad = 0$; nonnegativity since $d \geq 0$) and $c^\top(z + \theta d) = c^\top z + \theta c^\top d \rightarrow \infty$, contradicting boundedness; so some $d_j < 0$;
- if $c^\top d = 0$ and $d \geq 0$, replace d by $-d$: still $Ad = 0$, $c^\top d = 0$, and now some entry is negative since $d \neq 0$.

Now define

$$\theta^* = \min \left\{ \frac{z_j}{-d_j} : d_j < 0 \right\} > 0,$$

positive because every j with $d_j < 0$ lies in $\text{supp}(z)$, where $z_j > 0$. Set $z' = z + \theta^* d$. Then:

- $Az' = b$, since $Ad = 0$;
- $z' \geq 0$: for $d_j \geq 0$, $z'_j \geq z_j \geq 0$; for $d_j < 0$, $z'_j = z_j + \theta^* d_j \geq z_j - \frac{z_j}{-d_j}(-d_j) = 0$ by the choice of θ^* ;
- $c^\top z' = c^\top z + \theta^* c^\top d \geq c^\top z$;
- $|\text{supp}(z')| < |\text{supp}(z)|$: no new positive entries appear (outside $\text{supp}(z)$ both z and d vanish), and for the index j attaining the minimum, $z'_j = 0$.

By the induction hypothesis applied to z' there is a BFS z'' with $c^\top z'' \geq c^\top z' \geq c^\top z$. $\square$

Key insight

Two ingredients. *Dependence* gives a direction d with $Ad = 0$ supported where z is positive, so moving along it keeps the equations and, for a while, the signs. *Boundedness forbids* an improving direction that never hits a bound; therefore some coordinate decreases, and the *ratio test* $\theta^* = \min z_j/(-d_j)$ tells us exactly how far to go before the first coordinate reaches zero. The ratio test will reappear as the heart of the simplex method.

Theorem 2.5.2 (Existence of an optimal BFS). *Let $A \in \mathbb{R}^{m \times n}$ have full row rank (after removing redundant equations) and consider*

$$\max c^\top z \quad \text{subject to} \quad Az = b, \quad z \geq 0.$$

If the feasible set P is nonempty and the objective is bounded above on P , then the problem has an optimal solution, and at least one optimal solution is a basic feasible solution.

Proof. Step 1 (there is at least one BFS). $P \neq \emptyset$; pick any $z \in P$ and apply Lemma 2.5.1 to obtain a BFS. So the set $\mathcal{B}$ of basic feasible solutions is nonempty, and it is finite by Corollary 2.4.11.

Step 2 (the best BFS beats every feasible point). Let $z^* \in \mathcal{B}$ maximise $c^\top z$ over the finite set $\mathcal{B}$ (a maximum over a nonempty finite set exists). For any $z \in P$, Lemma 2.5.1 gives a BFS z' with $c^\top z \leq c^\top z' \leq c^\top z^*$.

Step 3 (conclusion). $z^* \in P$ and $c^\top z^* \geq c^\top z$ for all $z \in P$: z^* is an optimal solution, and it is a BFS. $\square$

Interpretation & discussion

- *Finding the best BFS gives the optimal value.* The infinite search over P collapses to a finite comparison. This is the licence for every vertex-based algorithm, the simplex method first among them.
- *Other optimal solutions may exist that are not basic.* If two vertices are optimal, the whole edge between them is optimal (Proposition 1.7.9), and its interior points are not BFSs. The theorem promises *at least one* optimal vertex, not that all optima are vertices.
- *The hypotheses are exactly the two bad outcomes.* Infeasible: no BFS exists at all. Unbounded: the purification argument breaks precisely because an improving ray $z + \theta d$ exists. So for LPs, “feasible and bounded” already implies “solvable,” closing the gap noted after Definition 1.2.4.

Example 2.5.3 (The network example, optimised). Maximise total throughput $x_1 + x_2$ over Example 2.4.1. The four BFSs have values 0 at $(0,0)$, 4 at $(4,0)$, 6 at $(2,4)$ and 6 at $(0,6)$. By Theorem 2.5.2 the optimal value is 6. Both $(2,4)$ and $(0,6)$ are optimal, and so is every point of the segment between them, e.g. $(1,5)$ with slacks $(0,2)$: an optimal solution with three positive entries, hence not basic. Certificate: $x_1 + x_2 \leq 6$ is itself a constraint, so $y = (1,0)$ proves the bound.

A first algorithm: compare the bases

Technique: Complete enumeration of bases

Assume A has m independent rows and the LP is feasible with bounded objective.

1. For each set B of m column indices:
2. if A_B is invertible, solve $A_B z_B = b$ and set $z_N = 0$;
3. keep the solutions with $z_B \geq 0$;
4. compare $c^\top z$ over the kept solutions and return the best.

Theorem 2.5.2 guarantees the answer is optimal. The method is practical for small examples: $\binom{4}{2} = 6$ systems above. It is hopeless at scale: an LP with $m = 50$ equations and $n = 100$ variables has $\binom{100}{50} \approx 10^{29}$ candidate bases. The simplex method, next in the course, walks from BFS to neighbouring BFS improving the objective at each step, visiting only a tiny fraction of them.

Classic problems

- “An LP is feasible and bounded. Must it have an optimal BFS?” Yes, after reducing to full row rank (Theorem 2.5.2).
- “An LP has an optimal solution that is not a vertex. Contradiction?” No: it then has infinitely many optimal solutions, at least one of which is a vertex.
- “The feasible set is $\{x \geq 0 : x_1 - x_2 = 0\}$ and we maximise x_1 .” Unbounded: $d = (1, 1)$ has $Ad = 0$, $d \geq 0$, $c^\top d = 1 > 0$. This is exactly the ray the purification lemma detects.
- Purify $(1, 1, 4, 5)$ in Example 2.4.1 for $c = (1, 1, 0, 0)$. Support is everything; four columns in $\mathbb{R}^2$ are dependent. Take $d = (1, 0, -1, -2)$ ($Ad = 0$, $c^\top d = 1 > 0$). Ratio test: $\min\{4/1, 5/2\} = 2.5$, giving $(3.5, 1, 1.5, 0)$, value 4.5. Support is now $\{1, 2, 3\}$, still dependent: $d = (-1, 2, -1, 0)$ has $Ad = 0$, $c^\top d = 1 > 0$; ratio test $\min\{3.5/1, 1.5/1\} = 1.5$ gives $(2, 4, 0, 0)$, the BFS of $\{1, 2\}$ with value 6, which is optimal.

Gotcha

- The theorem needs *both* feasibility and boundedness. Do not invoke it before checking them (or before an algorithm has established them).
- “Optimal solutions are vertices” is false as a universal statement; “some optimal solution is a vertex” is the theorem.
- Enumerating bases of the *canonical* form makes no sense; bases live in the *equational* form with slacks included.

Summary: The fundamental theorem

- Purification (Lemma 2.5.1): from any feasible point, walk along a null direction of A with the ratio test to a BFS at least as good; boundedness guarantees the walk stops.

- Fundamental theorem (Theorem 2.5.2): feasible and bounded above $\Rightarrow$ an optimal BFS exists.
- Consequence: comparing the finitely many BFSs solves the LP; large instances need a smarter walk (the simplex method).

Notes on the original slides

All row reductions, determinants and basic solutions in the Lecture 2 slides were recomputed and are correct (the three pivots, the six bases of the network example with their signs, $\det A_B = -1$ and -2 , the point $(6, 0, 0, -4)$). The following points deserve a clarification.

- **“The slack columns form an identity matrix” (slide 22).** This is true of the *original* matrix A , where the slack columns are e_1, e_2, e_3 , and it is what guarantees $\text{rank } A = 3$ in advance. It is *not* true of the row-reduced matrix shown on that slide, whose slack columns are $(1, -2, 1)$, $(0, 1, -2)$, $(0, 0, 1)$.
- **“The fourth inequality is also redundant” (slide 19).** True, but not by simply adding constraints: one needs $x \geq 0$ as well. Indeed $\frac{1}{2}(\text{first}) + \frac{3}{2}(\text{second})$ gives $3.5x_1 + 5x_2 + 2x_3 \leq 17$, which dominates $3x_1 + 5x_2 + x_3 \leq 18$ only because $x_1, x_3 \geq 0$.
- **“For equational form, vertices are exactly the BFSs” (slide 32).** This needs A to have full row rank (redundant equations removed), as the slides assume from slide 33 on; Theorem 2.4.10 states the hypothesis explicitly.
- **Theorem on slide 41.** The statement “feasible set nonempty and objective bounded above $\Rightarrow$ an optimal BFS exists” silently includes the claim that an optimum exists at all; the slides do not prove it. Lemma 2.5.1 and Theorem 2.5.2 supply the missing argument.
- **Basic solution of the first choice (slide 24).** The slide says “this choice gives no solution,” which is right; note that this does not mean the LP is infeasible (it is not), nor that x_1, x_2, x_3 can never all be positive in a feasible point (e.g. $(1, 1, 1)$ is feasible). It only means these three columns cannot form a basis.
- **Bases versus corners (slide 39).** “Four feasible bases, four vertices” holds here because no BFS is degenerate. In general several bases can give the same vertex (see the degeneracy remark after Proposition 2.4.4).

Chapter takeaways

Summary: The geometry and algebra of linear programs

- **Canonical form** $\max c^\top x$, $Ax \leq b$, $x \geq 0$ is the modelling format; **equational form** $\max c^\top z$, $Az = b$, $z \geq 0$ is the computing format. Slacks (for $\leq$), surpluses (for $\geq$) and variable shifts translate one into the other exactly.
- **Linear algebra** organises the equalities: row-reduce $[A \mid b]$, detect in-

consistency, delete zero rows, reach full row rank.

- **A basis** is m independent columns; its basic solution sets the other variables to 0; it is a **BFS** if nonnegative. BFSs are exactly the **vertices** of the feasible set, and there are finitely many.
- **The fundamental theorem:** a feasible, bounded LP has an optimal BFS. Hence comparing vertices suffices, and the stage is set for the simplex method.

Chapter 3

Solving Linear Programs: The Simplex Method

3.1 Preparing an LP: Redundancy and Two Routes to Equational Form

Motivation

Chapter 2 established the pipeline: model in canonical form, convert to equational form, and search among basic feasible solutions. Before the conversion there is an optional but useful step. Some inequalities in a model are *implied* by the others: they exclude nothing that was not already excluded. Removing them changes neither the feasible set nor the optimum, but it reduces the number of slack variables and therefore the number of column selections to examine. This section makes the test precise, then solves one LP both with and without the removal to see exactly what changes and what does not.

Recall the running example: a workshop chooses quantities x_1, x_2 of two products,

$$\max 2x_1 + x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 4, \quad x_1 \leq 3, \quad 2x_1 + x_2 \leq 7, \quad x_1, x_2 \geq 0.$$

Adding the first two inequalities gives $2x_1 + x_2 \leq 7$, which is exactly the third. The third inequality is therefore redundant: every point satisfying the first two automatically satisfies it.

Proposition 3.1.1 (Redundancy test). *Let $a_i^\top x \leq b_i$ ($i = 1, \dots, m$) be a system of inequalities and let $a^\top x \leq b$ be a candidate inequality. If there exist $\lambda_1, \dots, \lambda_m \geq 0$ with*

$$\sum_i \lambda_i a_i = a \quad \text{and} \quad \sum_i \lambda_i b_i \leq b,$$

then every x satisfying the system satisfies the candidate; the candidate is redundant and may be removed without changing the feasible set. Variable bounds such as $x_j \geq 0$ may be used as inequalities $-x_j \leq 0$ in the combination.

Proof. For any x satisfying the system, $a^\top x = \sum_i \lambda_i a_i^\top x \leq \sum_i \lambda_i b_i \leq b$, where the first inequality uses $\lambda_i \geq 0$ (multiplying an inequality by a nonnegative number preserves it) and the second is the hypothesis. So the candidate holds automatically. Removing it leaves the feasible set unchanged because the set of x satisfying “system and candidate” equals the set satisfying “system”. $\square$

Every inequality you keep becomes a slack variable, a column, and more candidate bases. Dropping an inequality that says nothing new is free.

Key insight

Redundancy is a *proof*: a nonnegative combination of the other inequalities that dominates the candidate. It is the same certificate mechanism as weak duality (Lemma 1.3.10), pointed at a constraint instead of at the objective.

Example 3.1.2 (Three variables, five inequalities). Which inequality can be removed from

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \quad x_2 + x_3 \leq 5, \quad x_1 + 2x_3 \leq 6, \\ & x_3 \leq 2, \quad 2x_1 + 3x_2 + x_3 \leq 14, \quad x \geq 0 ? \end{aligned}$$

Take $\lambda = (2, 1, 0, 0)$ on the first four: $2(x_1 + x_2) + (x_2 + x_3) = 2x_1 + 3x_2 + x_3$, and $2 \cdot 4 + 5 = 13 \leq 14$. The fifth inequality is redundant; the others imply the even tighter bound 13. Removing it leaves four inequalities and hence four slacks instead of five.

Technique: Spotting redundancy fast

- Try to *match the left-hand side* first: which combination of rows produces the candidate's coefficients? Only then compare right-hand sides.
- Small integer multipliers are the norm in exercises: $1+1$, $2 \cdot (\text{row}) + (\text{row})$.
- If the candidate has a coefficient that no other row can supply (e.g. x_3 appears only in the candidate), it cannot be redundant.
- Do not remove an inequality that is merely *loose at the optimum*; redundancy is about the whole feasible set, not about one point.

Route A: remove the redundant inequality, then add slacks

Example 3.1.3 (Route A). Keeping $x_1 + x_2 \leq 4$ and $x_1 \leq 3$ only, add slacks $s_1, s_2 \geq 0$:

$$x_1 + x_2 + s_1 = 4, \quad x_1 + s_2 = 3,$$

so that, with $z = (x_1, x_2, s_1, s_2)^\top$,

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

The slack columns form I_2 , so $\text{rank } A = 2$. There are $\binom{4}{2} = 6$ column selections. One in detail: select x_1, x_2 and set $s_1 = s_2 = 0$:

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & 0 & 3 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -1 & -1 \end{array} \right],$$

so $x_2 = 1$, $x_1 = 3$, $z = (3, 1, 0, 0) \geq 0$, profit $2 \cdot 3 + 1 = 7$. Another: select x_2, s_1 .

Their columns are $(1, 0)^\top$ and $(1, 0)^\top$, so $M = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ has rank $1 < 2$: not a basis; skip it (the columns stay in the LP and may belong to other bases). All six:

Selected	(x_1, x_2, s_1, s_2)	Feasible?	Profit
x_1, x_2	$(3, 1, 0, 0)$	yes	7
x_1, s_1	$(3, 0, 1, 0)$	yes	6
x_1, s_2	$(4, 0, 0, -1)$	no	
x_2, s_1	dependent columns		
x_2, s_2	$(0, 4, 0, 3)$	yes	4
s_1, s_2	$(0, 0, 4, 3)$	yes	0

Six selections, five bases, four distinct BFSs. The best BFS is $(x_1, x_2) = (3, 1)$ with $v^* = 7$. Comparing BFSs suffices because the LP is feasible and bounded (Theorem 2.5.2); we enumerated all BFSs, not all feasible points.

Route B: keep every inequality

Example 3.1.4 (Route B). With one slack per inequality,

$$\tilde{A} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}, \quad z = (x_1, x_2, s_1, s_2, s_3)^\top.$$

I_3 in the slack columns guarantees rank 3; nothing to delete. Now $\binom{5}{3} = 10$ selections. One in detail: select x_1, x_2, s_3 , set $s_1 = s_2 = 0$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 1 & 0 & 0 & 3 \\ 2 & 1 & 1 & 7 \end{array} \right] \xrightarrow[R_3 \leftarrow R_3 - R_1 - R_2]{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right],$$

so $x_2 = 1$, $x_1 = 3$, $s_3 = 0$. Three pivots, so the columns are independent; the BFS is $(3, 1, 0, 0, 0)$. Note that s_3 is basic yet equals zero: a *degenerate* BFS. All ten selections:

Selected	$(x_1, x_2, s_1, s_2, s_3)$	Feasible?	Profit
x_1, x_2, s_1	$(3, 1, 0, 0, 0)$	yes	7
x_1, x_2, s_2	$(3, 1, 0, 0, 0)$	yes	7
x_1, x_2, s_3	$(3, 1, 0, 0, 0)$	yes	7
x_1, s_1, s_2	$(\frac{7}{2}, 0, \frac{1}{2}, -\frac{1}{2}, 0)$	no	
x_1, s_1, s_3	$(3, 0, 1, 0, 1)$	yes	6
x_1, s_2, s_3	$(4, 0, 0, -1, -1)$	no	
x_2, s_1, s_2	$(0, 7, -3, 3, 0)$	no	
x_2, s_1, s_3	dependent columns		
x_2, s_2, s_3	$(0, 4, 0, 3, 3)$	yes	4
s_1, s_2, s_3	$(0, 0, 4, 3, 7)$	yes	0

Ten selections, nine bases, six feasible bases, four distinct BFSs. Three different bases produce the same degenerate vertex $(3, 1)$.

Proposition 3.1.5 (The two routes are equivalent). *In Examples 3.1.3 and 3.1.4, the map $(x_1, x_2, s_1, s_2) \mapsto (x_1, x_2, s_1, s_2, s_1 + s_2)$ is a bijection between the feasible sets of Route A and Route B that preserves the objective. Hence both routes have the same optimal decisions and value.*

Proof. For any feasible point of Route B, $s_3 = 7 - 2x_1 - x_2 = (4 - x_1 - x_2) + (3 - x_1) = s_1 + s_2$, so s_3 is determined by the other variables and is automatically ≥ 0 when $s_1, s_2 \geq 0$. Thus dropping s_3 gives a feasible point of Route A, and adding $s_3 = s_1 + s_2$ recovers it; the two operations are inverse. The objective involves only x_1, x_2 . $\square$

	Route A: remove	Route B: keep
Slack variables	2	3
Column selections	6	10
Bases	5	9
Feasible bases	4	6
Distinct BFSs	4	4

Interpretation & discussion

Same optimal decision $(3, 1)$, same profit 7, same four corners of the polygon. Route B paid for the redundant inequality with more variables and more selections, and it introduced degeneracy: the vertex $(3, 1)$ lies on three constraint lines ($x_1 + x_2 = 4$, $x_1 = 3$, $2x_1 + x_2 = 7$) in a plane where two suffice, so three bases describe it. Removal is optional, but here it reduces the work and removes the degeneracy.

Summary: Redundancy and the two routes

- An inequality that is a nonnegative combination of others (with a no-larger right-hand side) is redundant (Proposition 3.1.1).
- Recipe: optionally remove redundant inequalities; add one slack per retained inequality; $A = [G \ I_m]$ has rank m ; choose m independent columns, solve, check signs; compare BFSs.
- Keeping a redundant inequality changes nothing about the optimum but adds variables, selections and degeneracy.

3.2 Enumeration: Correct but Expensive

Motivation

The fundamental theorem (Theorem 2.5.2) turns an infinite search into a finite one. This section records what that finite search costs, checks the two hypotheses of the theorem on examples, and shows what happens when boundedness fails. The conclusion motivates the simplex method: the theorem narrows the search, but does not make enumeration efficient.

Theorem 3.2.1 (Fundamental theorem, restated). *Let $A \in \mathbb{R}^{m \times n}$ have full row rank and consider $\max c^\top z$ s.t. $Az = b$, $z \geq 0$. If the feasible set is nonempty and the objective is bounded above, there is an optimal basic feasible solution. (Proof: Theorem 2.5.2.)*

Technique: Enumeration algorithm

1. Select m columns of A . Skip dependent selections.
2. Set $z_N = 0$ and solve $A_B z_B = b$. Keep the solution only if $z_B \geq 0$.
3. Compare the objective values of all distinct BFSs; the best is optimal.

The number of candidates satisfies

$$\#\{\text{distinct BFSs}\} \leq \#\{\text{bases}\} \leq \binom{n}{m}.$$

Example 3.2.2 (The count explodes). $\binom{4}{2} = 6$, $\binom{5}{3} = 10$, $\binom{20}{10} = 184,756$. A model with 20 decisions and 20 inequalities has $n = 40$ variables after slacks and $\binom{40}{20} = 137,846,528,820$ column selections. Each requires solving a 20×20 system. Exhaustive enumeration is impractical at that scale, and industrial LPs have millions of variables.

Checking the hypotheses

Example 3.2.3 (A bounded objective: the running example). For $\max 2x_1 + x_2$ with $x_1 + x_2 \leq 4$, $x_1 \leq 3$, $x \geq 0$: feasibility holds since $(0,0)$ satisfies everything. Boundedness: adding the two resource inequalities, $2x_1 + x_2 = (x_1 + x_2) + x_1 \leq 4 + 3 = 7$, so $U = 7$ is a finite upper bound and the theorem applies. The feasible point $(3,1)$ attains 7, so it is optimal by matching bounds (Theorem 1.5.4).

Proposition 3.2.4 (A feasible improving ray proves unboundedness). *Let x^0 be feasible for $\max c^\top x$ over a feasible set X , and let d be a direction such that $x^0 + \theta d \in X$ for all $\theta \geq 0$ and $c^\top d > 0$. Then the LP is unbounded.*

Proof. $c^\top(x^0 + \theta d) = c^\top x^0 + \theta c^\top d \rightarrow +\infty$ as $\theta \rightarrow \infty$, and every point of the ray is feasible, so no finite M bounds the objective over X . $\square$

Example 3.2.5 (When boundedness fails). $\max x_1 + x_2$ s.t. $x_1 \leq 3$, $x \geq 0$. The ray $(x_1, x_2) = (3, t)$, $t \geq 0$, is feasible and $x_1 + x_2 = 3 + t \rightarrow \infty$. There is no finite optimum. The best BFS is $(3,0)$ (in equational form $A = [1 \ 0 \ 1]$, the only bases are $\{x_1\}$ and $\{x_3\}$, since the x_2 column is zero), yet feasible points achieve arbitrarily larger values: the theorem's conclusion fails because its hypothesis fails. Same region, different objective: $\max x_1$ is bounded by 3.

Gotcha

- An *unbounded feasible region* does not imply an *unbounded objective*. The region above is unbounded for both objectives; only one objective is unbounded.
- “Best BFS” is meaningful only when the theorem applies. Without boundedness, the best BFS can be beaten by non-basic points.

Summary: Enumeration

- Under feasibility and boundedness, comparing all BFSs is a correct algorithm.
- It costs up to $\binom{n}{m}$ linear solves: fine for $\binom{4}{2}$, hopeless for $\binom{40}{20}$.
- Check boundedness with a certificate (a nonnegative combination bounding the objective) or refute it with a feasible improving ray (Proposition 3.2.4).

3.3 The Simplex Method

Motivation

Can we reach an optimum without examining every column selection? The simplex method (Dantzig, 1947) answers yes: start at a BFS, use the objective to pick a direction in which the objective increases, move along it until a nonnegativity constraint would be violated, and thereby arrive at a neighbouring BFS with a better value. Repeat until the algebra certifies optimality. Each step updates one basis, exchanging a single variable; no list of all bases is ever formed.

Instead of visiting every corner, walk from corner to corner, always uphill, and let the algebra tell you when no uphill direction remains.

A worked run

Example 3.3.1 (Setting up).

$$\max x_1 + x_2 \quad \text{s.t.} \quad -x_1 + x_2 \leq 1, \quad x_1 \leq 3, \quad x_2 \leq 2, \quad x_1, x_2 \geq 0.$$

The origin is feasible and $x_1 + x_2 \leq 3 + 2 = 5$ bounds the objective, so an optimal BFS exists. Slacks $x_3, x_4, x_5 \geq 0$ give

$$-x_1 + x_2 + x_3 = 1, \quad x_1 + x_4 = 3, \quad x_2 + x_5 = 2,$$

with $x_3 = 1 + x_1 - x_2$, $x_4 = 3 - x_1$, $x_5 = 2 - x_2$: each slack measures the gap between the two sides of its inequality.

Definition 3.3.2 (Tableau). A *tableau* (also called a *dictionary*) for a basis B is the system $Az = b$ rewritten so that each basic variable is expressed in terms of the nonbasic variables only, together with the objective expressed in the nonbasic variables:

$$x_i = p_i + \sum_{j \in N} q_{ij} x_j \quad (i \in B), \quad v = v_0 + \sum_{j \in N} r_j x_j.$$

The tableau is *feasible* if $p_i \geq 0$ for all $i \in B$. Setting every nonbasic variable to zero reads off the current BFS: $x_j = 0$ ($j \in N$), $x_i = p_i$ ($i \in B$), $v = v_0$. The numbers r_j are the *reduced costs*.

Example 3.3.3 (Tableau 0). Basic x_3, x_4, x_5 ; nonbasic x_1, x_2 :

$$\begin{aligned}x_3 &= 1 + x_1 - x_2 \\x_4 &= 3 - x_1 \\x_5 &= 2 \quad - x_2 \\v &= 0 + x_1 + x_2\end{aligned}$$

BFS $x = (0, 0, 1, 3, 2)$, $v = 0$. Both reduced costs are $+1$. Choose x_2 to enter (either choice is allowed); keep $x_1 = 0$. The rows give $x_3 = 1 - x_2 \geq 0$ and $x_5 = 2 - x_2 \geq 0$, and x_4 does not depend on x_2 . So x_2 can increase to 1, at which point x_3 reaches zero first: x_2 enters, x_3 leaves.

Example 3.3.4 (Pivot 1). Solve the leaving row for the entering variable: $x_3 = 1 + x_1 - x_2 \iff x_2 = 1 + x_1 - x_3$. Substitute into every other row, including the objective:

$$\begin{aligned}x_2 &= 1 + x_1 - x_3 \\x_4 &= 3 - x_1 \\x_5 &= 2 - (1 + x_1 - x_3) = 1 - x_1 + x_3 \\v &= x_1 + (1 + x_1 - x_3) = 1 + 2x_1 - x_3\end{aligned}$$

New nonbasic variables x_1, x_3 ; set them to zero: $x = (0, 1, 0, 3, 1)$, $v = 1$.

Example 3.3.5 (Pivot 2). Only x_1 has a positive reduced cost ($+2$). Keep $x_3 = 0$. Limits: $x_4 = 3 - x_1$ allows $x_1 \leq 3$; $x_5 = 1 - x_1$ allows only $x_1 \leq 1$; $x_2 = 1 + x_1$ does not limit. x_1 enters, x_5 leaves. From $x_5 = 1 - x_1 + x_3$: $x_1 = 1 + x_3 - x_5$.

$$\begin{aligned}x_1 &= 1 + x_3 - x_5 \\x_2 &= 1 + (1 + x_3 - x_5) - x_3 = 2 - x_5 \\x_4 &= 3 - (1 + x_3 - x_5) = 2 - x_3 + x_5 \\v &= 1 + 2(1 + x_3 - x_5) - x_3 = 3 + x_3 - 2x_5\end{aligned}$$

BFS $x = (1, 2, 0, 2, 0)$, $v = 3$.

Example 3.3.6 (Pivot 3: even a slack can enter). The reduced cost of x_3 is $+1$. Keep $x_5 = 0$. Only $x_4 = 2 - x_3$ limits: $x_3 \leq 2$; meanwhile $x_1 = 1 + x_3$ increases and $x_2 = 2$ is fixed. x_3 enters, x_4 leaves. From $x_4 = 2 - x_3 + x_5$: $x_3 = 2 - x_4 + x_5$.

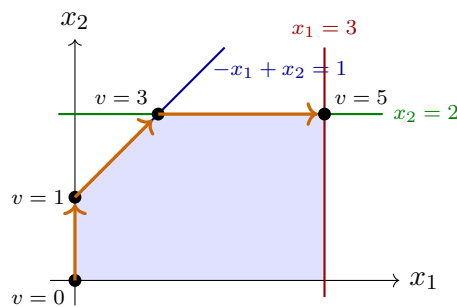
$$\begin{aligned}x_1 &= 1 + (2 - x_4 + x_5) - x_5 = 3 - x_4 \\x_2 &= 2 - x_5 \\x_3 &= 2 - x_4 + x_5 \\v &= 3 + (2 - x_4 + x_5) - 2x_5 = 5 - x_4 - x_5\end{aligned}$$

BFS $x = (3, 2, 2, 0, 0)$, $v = 5$.

Proposition 3.3.7 (Why we can stop: the objective row is a certificate). *In the final tableau, $v = 5 - x_4 - x_5$ holds for every feasible solution, and $x_4, x_5 \geq 0$, so $v \leq 5$ for all feasible points. The BFS $(x_1, x_2) = (3, 2)$ attains 5; hence it is optimal. Moreover $v = 5$ forces $x_4 = x_5 = 0$, which fixes every other variable through the tableau rows, so the optimum is unique.*

Proof. Each pivot rewrote the equations $Az = b$ by solving one equation for one variable and substituting; these are invertible operations, so the final system has exactly the same solution set as the original (Proposition 2.3.2). The objective row is an identity valid on that solution set. The rest is reading off signs. $\square$

Geometrically the run visited $(0, 0) \rightarrow (0, 1) \rightarrow (1, 2) \rightarrow (3, 2)$: four corners of the polygon, each adjacent to the previous one, with the objective climbing $0, 1, 3, 5$.



The general step

Consider a feasible tableau as in Definition 3.3.2.

Theorem 3.3.8 (Optimality test). *If every reduced cost satisfies $r_j \leq 0$, the current BFS is optimal. If moreover $r_j < 0$ for all $j \in N$, it is the unique optimal solution.*

Proof. For every feasible x , the objective row holds and $x_j \geq 0$, so $v = v_0 + \sum_{j \in N} r_j x_j \leq v_0$. The current BFS has value v_0 , hence it is optimal. If all $r_j < 0$, then $v = v_0$ forces $x_j = 0$ for every $j \in N$, and the basic rows then give $x_i = p_i$: the only feasible point with value v_0 is the current BFS. $\square$

Theorem 3.3.9 (Ratio test and unboundedness). *Suppose $r_j > 0$ for some $j \in N$, and increase $x_j = \theta \geq 0$ while keeping the other nonbasic variables at zero. Then each basic variable becomes $x_i = p_i + q_{ij}\theta$.*

1. *If $q_{ij} \geq 0$ for all $i \in B$, the LP is unbounded: the ray $\theta \mapsto x(\theta)$ is feasible for all $\theta \geq 0$ and $v = v_0 + r_j\theta \rightarrow \infty$.*
2. *Otherwise the largest feasible step is*

$$\theta^* = \min_{i \in B: q_{ij} < 0} \frac{p_i}{-q_{ij}} \geq 0,$$

and any row ℓ attaining the minimum has $x_\ell(\theta^) = 0$. The point $x(\theta^*)$ is feasible with objective $v_0 + r_j\theta^* \geq v_0$.*

Proof. The equations of the tableau hold along the whole line $x(\theta)$ by construction, so feasibility reduces to nonnegativity. Rows with $q_{ij} \geq 0$ satisfy $x_i(\theta) \geq p_i \geq 0$ for every $\theta \geq 0$; rows with $q_{ij} < 0$ require $\theta \leq p_i/(-q_{ij})$. If no row is of the second kind, all $\theta \geq 0$ are feasible and Proposition 3.2.4 applies with $c^\top d = r_j > 0$. Otherwise the binding constraint is the smallest ratio, which is ≥ 0 because $p_i \geq 0$; at θ^* the minimising row reaches exactly zero and no row is negative. $\square$

Key insight

Each basic row is a one-variable inequality $p_i + q_{ij}\theta \geq 0$. The ratio test is nothing more than solving all of them and taking the tightest. Zero ratios count: if some $p_i = 0$ with $q_{ij} < 0$, the step is $\theta^* = 0$.

Proposition 3.3.10 (Pivoting produces a new feasible tableau). *Let x_j enter and let x_ℓ be a row attaining θ^* (so $q_{\ell j} < 0$). Solving the row of x_ℓ for x_j and substituting into the other rows and the objective yields a tableau for the basis $B' = (B \setminus \{\ell\}) \cup \{j\}$, $N' = (N \setminus \{j\}) \cup \{\ell\}$. It is feasible, and its BFS is $x(\theta^*)$ with objective $v_0 + r_j\theta^*$.*

Proof. The row $x_\ell = p_\ell + q_{\ell j}x_j + \sum_{k \neq j} q_{\ell k}x_k$ can be solved for x_j because $q_{\ell j} \neq 0$: $x_j = -p_\ell/q_{\ell j} + x_\ell/q_{\ell j} - \sum_{k \neq j} (q_{\ell k}/q_{\ell j})x_k$. Substituting eliminates x_j from every other row, so each variable of B' is expressed in the variables of N' only, and the same equations still hold (Proposition 2.3.2). Setting N' to zero gives $x_j = -p_\ell/q_{\ell j} = p_\ell/(-q_{\ell j}) = \theta^*$ and, for $i \in B \setminus \{\ell\}$, $x_i = p_i + q_{ij}\theta^* \geq 0$ by the choice of θ^* ; these are exactly the coordinates of $x(\theta^*)$, hence nonnegative. The objective value at that point is $v_0 + r_j\theta^*$. $\square$

Technique: The simplex algorithm

1. Start with a feasible tableau (with all $b_i \geq 0$ in canonical form, the slack basis is one).
2. Read the objective row. If every reduced cost is ≤ 0 , stop: the current BFS is optimal (Theorem 3.3.8).
3. Choose an entering variable with $r_j > 0$.
4. Ratio test (Theorem 3.3.9). If no row limits the increase, stop: unbounded, return the ray. Otherwise the minimising row's variable leaves.
5. Pivot (Proposition 3.3.10) and return to step 2.

The LP never changes; only its description does.

Theorem 3.3.11 (Termination without degeneracy). *If every tableau visited is nondegenerate (all $p_i > 0$), then $\theta^* > 0$ at every pivot, the objective strictly increases, no basis is visited twice, and the algorithm terminates after finitely many pivots with an optimal BFS or a proof of unboundedness.*

Proof. With all $p_i > 0$ every ratio $p_i/(-q_{ij})$ is positive, so $\theta^* > 0$ and the objective increases by $r_j\theta^* > 0$. A basis determines its BFS and hence its objective value; a strictly increasing sequence of values cannot repeat a basis. There are at most $\binom{n}{m}$ bases, so the sequence is finite, and the only ways to stop are the two certificates. $\square$

Interpretation & discussion

Simplex is enumeration made lazy: it examines bases in an order chosen by the objective and typically visits a tiny fraction of them. Each visited basis costs a pivot, roughly the cost of one elimination step, rather than a full linear solve. The theorem also shows what *could* go wrong: degenerate pivots with $\theta^* = 0$ do not increase the objective, so the finiteness argument breaks. The next section treats the two exceptional cases.

Exceptional cases

Example 3.3.12 (Case 1: unboundedness). $\max x_1$ s.t. $x_1 - x_2 \leq 1$, $-x_1 + x_2 \leq 2$, $x \geq 0$. Slacks x_3, x_4 ; tableau 0:

$$x_3 = 1 - x_1 + x_2, \quad x_4 = 2 + x_1 - x_2, \quad v = x_1.$$

x_1 enters; only x_3 decreases, so $\theta^* = 1$ and x_3 leaves: $x_1 = 1 + x_2 - x_3$. After the pivot,

$$x_1 = 1 + x_2 - x_3, \quad x_4 = 2 + (1 + x_2 - x_3) - x_2 = 3 - x_3, \quad v = 1 + x_2 - x_3.$$

Now x_2 has reduced cost $+1$. Let $x_2 = \theta$ with $x_3 = 0$: $x_1 = 1 + \theta$ and $x_4 = 3$; no basic variable decreases. There is no leaving variable: the objective $v = 1 + \theta$ is unbounded. The certificate is the feasible ray $(x_1, x_2) = (1, 0) + \theta(1, 1)$, $\theta \geq 0$: both inequalities hold along it ($x_1 - x_2 = 1$, $-x_1 + x_2 = -1 \leq 2$).

Example 3.3.13 (Case 2: degeneracy, a zero step). $\max x_2$ s.t. $-x_1 + x_2 \leq 0$, $x_1 \leq 2$, $x \geq 0$. Tableau 0:

$$x_3 = x_1 - x_2, \quad x_4 = 2 - x_1, \quad v = x_2.$$

The initial BFS is degenerate: $x_3 = 0$. x_2 can enter, but with $x_1 = 0$ the row $x_3 = -x_2 \geq 0$ forces $\theta^* = 0$. Pivot anyway: $x_2 = x_1 - x_3$, giving

$$x_2 = x_1 - x_3, \quad x_4 = 2 - x_1, \quad v = x_1 - x_3.$$

The basis changed from $\{3, 4\}$ to $\{2, 4\}$, but the point $(0, 0, 0, 2)$ and the value 0 did not. A zero step does not prove optimality; it only re-describes the same vertex. Now x_1 has reduced cost $+1$: with $x_3 = 0$, $x_1 = \theta$, $x_2 = \theta$, $x_4 = 2 - \theta$, so $\theta^* = 2$ and x_4 leaves: $x_1 = 2 - x_4$,

$$x_1 = 2 - x_4, \quad x_2 = 2 - x_3 - x_4, \quad v = 2 - x_3 - x_4 \leq 2.$$

Optimum $(2, 2)$, $v^* = 2$. The first pivot changed the description; the second improved the solution.

Remark 3.3.14 (Cycling and how it is prevented). With degenerate pivots the objective can stall, and in rare, carefully constructed examples the algorithm can return to a basis it has already visited and loop forever. Simple entering/leaving rules prevent this. *Bland's rule*: among candidates, always pick the variable with the smallest index, both for entering and (among ties in the ratio test) for leaving. With Bland's rule the simplex method never cycles, so it always terminates. This is beyond the lecture; in exam-sized examples cycling does not occur, but zero

steps do, and you should recognise them.

A production decision: three juice blends

Example 3.3.15 (Model). A cooperative produces apple, berry and tropical juice. Per 100 litres: blending 1, 1, 1 hours (available 12); heat treatment 1, 2, 1 hours (available 18); imported concentrate 0, 1, 1 units of 10 L (available 8); profit 200, 400, 300 MAD. With x_1, x_2, x_3 in units of 100 L and v in hundreds of MAD,

$$\begin{aligned} \max \quad & v = 2x_1 + 4x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 12, \quad x_1 + 2x_2 + x_3 \leq 18, \quad x_2 + x_3 \leq 8, \quad x \geq 0. \end{aligned}$$

The zero plan is feasible; $v \leq 4(x_1 + x_2 + x_3) \leq 48$ bounds the objective; an optimal BFS exists. Enumeration would test $\binom{6}{3} = 20$ selections. Slacks x_4, x_5, x_6 give the starting BFS $x = (0, 0, 0, 12, 18, 8)$, $v = 0$.

Example 3.3.16 (Three pivots). *Tableau 0.*

$$\begin{aligned} x_4 &= 12 - x_1 - x_2 - x_3 \\ x_5 &= 18 - x_1 - 2x_2 - x_3 \\ x_6 &= 8 - x_2 - x_3 \\ v &= 2x_1 + 4x_2 + 3x_3 \end{aligned}$$

Choose x_2 (largest reduced cost). With $x_1 = x_3 = 0$: $x_2 \leq 12, 9, 8$; x_6 leaves at 8. $x_2 = 8 - x_3 - x_6$.

Tableau 1.

$$\begin{aligned} x_2 &= 8 - x_3 - x_6 \\ x_4 &= 12 - x_1 - (8 - x_3 - x_6) - x_3 = 4 - x_1 + x_6 \\ x_5 &= 18 - x_1 - 2(8 - x_3 - x_6) - x_3 = 2 - x_1 + x_3 + 2x_6 \\ v &= 2x_1 + 4(8 - x_3 - x_6) + 3x_3 = 32 + 2x_1 - x_3 - 4x_6 \end{aligned}$$

BFS $(0, 8, 0, 4, 2, 0)$, $v = 32$. Only x_1 has a positive reduced cost. With $x_3 = x_6 = 0$: $x_4 = 4 - x_1$, $x_5 = 2 - x_1$; x_5 leaves at 2. $x_1 = 2 + x_3 - x_5 + 2x_6$.

Tableau 2.

$$\begin{aligned} x_1 &= 2 + x_3 - x_5 + 2x_6 \\ x_2 &= 8 - x_3 - x_6 \\ x_4 &= 4 - (2 + x_3 - x_5 + 2x_6) + x_6 = 2 - x_3 + x_5 - x_6 \\ v &= 32 + 2(2 + x_3 - x_5 + 2x_6) - x_3 - 4x_6 = 36 + x_3 - 2x_5 \end{aligned}$$

BFS $(2, 8, 0, 2, 0, 0)$, $v = 36$. x_3 enters. With $x_5 = x_6 = 0$: $x_1 = 2 + x_3$ increases, $x_2 = 8 - x_3$ allows 8, $x_4 = 2 - x_3$ allows only 2; x_4 leaves. $x_3 = 2 - x_4 + x_5 - x_6$.

Tableau 3.

$$\begin{aligned} x_1 &= 2 + (2 - x_4 + x_5 - x_6) - x_5 + 2x_6 = 4 - x_4 + x_6 \\ x_2 &= 8 - (2 - x_4 + x_5 - x_6) - x_6 = 6 + x_4 - x_5 \\ x_3 &= 2 - x_4 + x_5 - x_6 \\ v &= 36 + (2 - x_4 + x_5 - x_6) - 2x_5 = 38 - x_4 - x_5 - x_6 \end{aligned}$$

BFS $(4, 6, 2, 0, 0, 0)$, $v = 38$. All reduced costs are negative: optimal and unique (Theorem 3.3.8).

Example 3.3.17 (Interpretation). Produce 400 L apple, 600 L berry, 200 L tropical, for a daily profit of $100 \times 38 = 3,800$ MAD. All three slacks are zero: the plan uses all 12 blending hours, all 18 heat-treatment hours and all 80 L of concentrate. Check: $4 + 6 + 2 = 12$, $4 + 12 + 2 = 18$, $6 + 2 = 8$. The final objective row $v = 38 - x_4 - x_5 - x_6$ is also a certificate anyone can verify without redoing the pivots: it is the identity $2x_1 + 4x_2 + 3x_3 = 38 - (12 - x_1 - x_2 - x_3) - (18 - x_1 - 2x_2 - x_3) - (8 - x_2 - x_3)$, i.e. the weak-duality multipliers $y = (1, 1, 1)$ of Lemma 1.3.10: $A^\top y = (2, 4, 3) = c$ and $b^\top y = 38$.

Key insight

The negatives of the final reduced costs of the slack variables are exactly the weak-duality multipliers y . Simplex does not only find the optimal plan; it finds the certificate that proves it optimal. That is the seed of LP duality, developed later in the course.

Technique: Running simplex by hand without errors

- Keep every row in the form *basic* = *constant* + (*nonbasic terms*). A row with a basic variable on the right is a sign you forgot to substitute.
- *Ratio test*: only rows where the entering variable has a *negative* coefficient limit the step. Write the ratio next to each such row; include zero ratios.
- After a pivot, set the new nonbasic variables to zero and check every basic value is ≥ 0 . If not, you chose the wrong leaving row.
- Check the objective: the new v_0 must equal old $v_0 + r_j\theta^*$. In the juice example: $0 \rightarrow 32$ ($4 \cdot 8$), $32 \rightarrow 36$ ($2 \cdot 2$), $36 \rightarrow 38$ ($1 \cdot 2$).
- Verify the final answer against the *original* inequalities and read the slacks as unused resources.
- With several positive reduced costs, any choice is valid. The largest coefficient (Dantzig's rule) is the usual default; in exams, state your choice.

Classic problems

- The objective row reads $v = 10 + 3x_2 - x_4$ and the column of x_2 has no negative entry. Conclusion? Unbounded; the ray increasing x_2 is feasible with $v \rightarrow \infty$.
- The objective row reads $v = 10 - 3x_2 - x_4$. Conclusion? Optimal and unique, $v^* = 10$.
- The objective row reads $v = 10 + 0 \cdot x_2 - x_4$. Conclusion? Optimal, $v^* = 10$, but possibly not unique: increasing x_2 (if some row limits it) reaches another optimal solution, and the whole segment is optimal.

- Rows $x_3 = 0 - 2x_1 + x_2$ and $x_4 = 5 - x_1$; x_1 enters. Ratios $0/2 = 0$ and $5/1 = 5$; $\theta^* = 0$, x_3 leaves; a degenerate pivot.
- How many pivots can simplex take? At most the number of bases, $\binom{n}{m}$, with an anti-cycling rule; in practice far fewer.

Gotcha

- A positive coefficient of the entering variable in a basic row is *not* a limit: that variable increases. Only negative coefficients limit.
- Forgetting to substitute into the objective row leaves a stale certificate: the reduced costs must be recomputed at every pivot.
- “ $\theta^* = 0$ ” does not mean “optimal.” Check the reduced costs, not the step size.
- Simplex needs a *feasible* starting tableau. When some $b_i < 0$ the slack basis is infeasible and a preliminary phase is needed (not covered in this lecture).

Summary: The simplex method

- A tableau writes basic variables and the objective in terms of nonbasic variables; setting nonbasics to zero reads the BFS.
- All reduced costs ≤ 0 : optimal (strictly negative: unique). A positive reduced cost with no negative column entry: unbounded, with a ray certificate.
- Otherwise the ratio test picks the leaving variable and a pivot moves to an adjacent BFS with objective $\geq$ before; $>$ unless degenerate.
- Without degeneracy the method terminates; with it, an anti-cycling rule guarantees termination.
- The final reduced costs of the slacks are the weak-duality multipliers: simplex delivers an optimal plan and its proof.

Notes on the original slides

Every pivot in the Lecture 3 slides was recomputed (all four tableaux of the small example, the three of the juice example, the unbounded and degenerate examples), together with both enumeration tables and the binomial counts. All are correct. Points worth a reader’s attention:

- **Redundancy rule (slide 4).** The rule requires $\sum_i \lambda_i a_i = a$ exactly. When variables are nonnegative, a combination whose coefficients *dominate* a componentwise also works, because the bounds $-x_j \leq 0$ can be added with their own multipliers; the slide’s remark “you may also use variable bounds” covers this.
- **Tableau versus dictionary (slides 27–33).** What the slides call a tableau is the *dictionary* form (basic variables written in terms of nonbasic ones). Many textbooks reserve “tableau” for a matrix layout with the same information; the arithmetic is identical.

- **“The slack columns form I_3 : this is a BFS” (slide 27).** The identity columns make the slack set a *basis*; it is a *feasible* basis because every right-hand side is nonnegative. When some $b_i < 0$, the slack basis is not feasible and simplex cannot start from it.
- **Choice of entering variable (slide 28).** Both reduced costs equal 1; the slides choose x_2 . Choosing x_1 is equally valid and gives a different path to the same optimum. The theorems require only $r_j > 0$.
- **Termination (slide 40).** The slides state the algorithm without discussing whether it must stop. Theorem 3.3.11 proves termination when no pivot is degenerate; the remark after Example 3.3.13 explains why degenerate pivots need an anti-cycling rule.
- **Uniqueness (slide 34).** The argument “ $v = 5$ forces $x_4 = x_5 = 0$ ” is correct and general: strictly negative reduced costs imply a unique optimum (Theorem 3.3.8). A zero reduced cost does not.
- **Slide 21.** The best BFS of $\max x_1 + x_2$ s.t. $x_1 \leq 3$ is $(3, 0)$ because the x_2 column in equational form is the zero vector, so no basis contains x_2 .

Chapter takeaways

Summary: Solving linear programs

- **Prepare:** define decisions, units, constraints and objective; optionally prove and remove redundant inequalities; add one slack per inequality.
- **Enumerate** only for tiny problems: a feasible, bounded LP has an optimal BFS, but $\binom{n}{m}$ grows explosively.
- **Simplex:** start from a feasible tableau; enter a variable with positive reduced cost; leave by the ratio test; pivot; stop when the objective row certifies optimality or reveals an improving ray.
- **Exceptions:** no limiting row means unbounded; a zero step means degeneracy, which changes the basis but not the point.
- **Read the answer** in the original units, and read the final objective row as a proof.

Chapter 4

Duality of Linear Programs

4.1 From a Feasible Plan to a Certificate: Bounding the Optimum

Motivation

Chapter 3 showed how to *find* an optimal plan. This chapter asks a different question: how do you *know* a plan is optimal, or how far from optimal it is, without finding the optimum first? A workshop that already runs a feasible production plan does not necessarily want to re-solve its whole problem; it wants to know how much profit it might be leaving on the table. Answering that requires an *upper bound* on the best possible profit. The surprising fact is that the search for the best such bound is itself a linear program, the *dual*, and that its optimal value equals the original optimum exactly. Everything in this chapter grows out of one running example.

A feasible plan tells you what you can get. A certificate tells you what you cannot beat. Optimisation is finished when the two numbers agree.

Example 4.1.1 (The workshop). A workshop chooses two divisible activity levels $x_1, x_2 \geq 0$. Three limited resources constrain production:

$$\begin{aligned} & \max \quad 2x_1 + 3x_2 \\ \text{s.t.} \quad & 4x_1 + 8x_2 \leq 12, \quad 2x_1 + x_2 \leq 3, \quad 3x_1 + 2x_2 \leq 4, \quad x_1, x_2 \geq 0. \end{aligned}$$

In the matrix notation of Definition 2.1.1,

$$A = \begin{bmatrix} 4 & 8 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 12 \\ 3 \\ 4 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Consider the plan $\hat{x} = (0, 3/2)$. Its resource use is $4(0) + 8(3/2) = 12 \leq 12$, $2(0) + 3/2 = 3/2 \leq 3$ and $3(0) + 2(3/2) = 3 \leq 4$; both coordinates are nonnegative, so the plan is feasible, with profit $2(0) + 3(3/2) = 9/2$.

Feasibility says nothing about quality. Write p^* for the (unknown) maximum profit. Because $\hat{x}$ is feasible, $9/2 \leq p^*$, and the *improvement still possible* is $p^* - 9/2 \geq 0$: zero if $\hat{x}$ is optimal, positive otherwise. We would like to bound this quantity before finding an optimal solution.

Upper bounds from the constraints

Suppose we can prove that every feasible plan has profit at most U , i.e. $2x_1 + 3x_2 \leq U$ for every feasible x . Then

$$\frac{9}{2} \leq p^* \leq U \quad \implies \quad 0 \leq p^* - \frac{9}{2} \leq U - \frac{9}{2}.$$

If $U = 9/2$ the plan is optimal; if U is close to $9/2$ the plan is close to optimal. The bound limits the possible loss; it need not give the exact loss.

Example 4.1.2 (A first bound, by eye). Divide the first resource inequality by 2: $4x_1 + 8x_2 \leq 12$ gives $2x_1 + 4x_2 \leq 6$. Since $x_2 \geq 0$, every feasible point satisfies

$$\underbrace{2x_1 + 3x_2}_{\text{profit}} \leq 2x_1 + 4x_2 \leq 6.$$

Hence $9/2 \leq p^* \leq 6$ and $0 \leq p^* - 9/2 \leq 3/2$: the plan is at most 1.5 profit units below the optimum.

Example 4.1.3 (A better bound: combine two constraints). Take one third of each of the first two inequalities and add:

$$\frac{1}{3}(4x_1 + 8x_2) + \frac{1}{3}(2x_1 + x_2) \leq \frac{1}{3} \cdot 12 + \frac{1}{3} \cdot 3 \quad \implies \quad 2x_1 + 3x_2 \leq 5.$$

The left-hand side is *exactly* the profit, so $p^* \leq 5$ and $0 \leq p^* - 9/2 \leq 1/2$. The plan is now guaranteed to be within 0.5 of optimal.

The two bounds came from different choices of multipliers:

Constraint 1	Constraint 2	Constraint 3	Bound
1/2	0	0	6
1/3	1/3	0	5
y_1	y_2	y_3	?

The natural next step is to let the multipliers themselves become decision variables and ask: *which nonnegative weights give the smallest upper bound obtainable this way?*

Multiply, add, dominate

Proposition 4.1.4 (Nonnegative combinations of constraints are valid bounds). *In Example 4.1.1, let $y_1, y_2, y_3 \geq 0$ satisfy*

$$4y_1 + 2y_2 + 3y_3 \geq 2 \quad \text{and} \quad 8y_1 + y_2 + 2y_3 \geq 3.$$

Then every feasible x satisfies $2x_1 + 3x_2 \leq 12y_1 + 3y_2 + 4y_3$.

Proof. **Step 1 (multiply).** Nonnegative weights preserve $\leq$ (Proposition 2.1.2):

$$y_1(4x_1 + 8x_2) \leq 12y_1, \quad y_2(2x_1 + x_2) \leq 3y_2, \quad y_3(3x_1 + 2x_2) \leq 4y_3.$$

Step 2 (add and regroup by variable). Summing,

$$(4y_1 + 2y_2 + 3y_3)x_1 + (8y_1 + y_2 + 2y_3)x_2 \leq 12y_1 + 3y_2 + 4y_3.$$

Step 3 (dominate the profit). Because $x_1, x_2 \geq 0$, a coefficient that is at least the profit coefficient can only make the left-hand side larger: $2x_1 \leq (4y_1 + 2y_2 + 3y_3)x_1$ and $3x_2 \leq (8y_1 + y_2 + 2y_3)x_2$. Adding these two inequalities to the one in Step 2 gives the claim. $\square$

Key insight

Two sign conditions make the argument work, and they will reappear in every proof of this chapter: multipliers $y_i \geq 0$ so that the resource inequalities keep their direction, and variables $x_j \geq 0$ so that a larger coefficient means a larger term. Each condition is used exactly once.

The search for the best bound is an LP

Among all weights that make the argument valid, we want the smallest bound:

$$\begin{aligned} \min \quad & 12y_1 + 3y_2 + 4y_3 \\ \text{s.t.} \quad & 4y_1 + 2y_2 + 3y_3 \geq 2, \quad 8y_1 + y_2 + 2y_3 \geq 3, \quad y_1, y_2, y_3 \geq 0. \end{aligned}$$

This is a new linear program, the *dual problem*. Its feasible solutions are certificates of upper bounds for the original LP. Side by side:

Primal: choose activity levels	Dual: choose constraint weights
$\max \quad 2x_1 + 3x_2$ $\text{s.t.} \quad 4x_1 + 8x_2 \leq 12$ $2x_1 + x_2 \leq 3$ $3x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$	$\min \quad 12y_1 + 3y_2 + 4y_3$ $\text{s.t.} \quad 4y_1 + 2y_2 + 3y_3 \geq 2$ $8y_1 + y_2 + 2y_3 \geq 3$ $y_1, y_2, y_3 \geq 0$
A feasible plan gives a <i>lower</i> bound on the maximum profit.	Feasible weights give an <i>upper</i> bound on the maximum profit.

For every feasible pair x, y : $2x_1 + 3x_2 \leq p^* \leq 12y_1 + 3y_2 + 4y_3$.

Definition 4.1.5 (The canonical primal–dual pair). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$. The *primal* and *dual* linear programs are

$$\begin{aligned} (P) \quad & \max \quad c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0, \\ (D) \quad & \min \quad b^\top y \quad \text{s.t.} \quad A^\top y \geq c, \quad y \geq 0. \end{aligned}$$

(*P*) has n variables and m inequalities; (*D*) has m variables and n inequalities. One primal constraint gives one dual variable; one primal variable gives one dual constraint. We write p^* and d^* for their optimal values when these exist.

Interpretation & discussion

Why does the matrix get transposed? In the workshop,

$$A = \begin{bmatrix} 4 & 8 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad A^\top y = \begin{bmatrix} 4y_1 + 2y_2 + 3y_3 \\ 8y_1 + y_2 + 2y_3 \end{bmatrix}.$$

Primal column j lists the resource use of activity j ; dual row j is the *weighted*

Rows become variables, columns become constraints. The transpose is not a convention: it is what “regroup by variable” does to the coefficients.

resource use of activity j , which must dominate its profit c_j . In indices, the j -th dual constraint reads $\sum_{i=1}^m a_{ij}y_i \geq c_j$, $j = 1, \dots, n$: the same numbers a_{ij} , read down a column instead of along a row. The objective $b^\top y$ is the weighted sum of capacities, i.e. the bound itself.

Technique: Writing the canonical dual in thirty seconds

- Count: m primal rows $\Rightarrow m$ dual variables $y_1, \dots, y_m \geq 0$; n primal variables $\Rightarrow n$ dual constraints.
- Dual objective: the primal right-hand sides b become the cost vector; max becomes min.
- Dual constraint j : read primal *column* j top to bottom, dot it with y , and require $\geq c_j$.
- Dimension check: $A^\top$ is $n \times m$ and y is $m \times 1$, so $A^\top y$ is $n \times 1$, matching c .

Summary: Bounding the optimum

- A feasible plan gives a lower bound on p^* ; a proof that no plan beats U gives an upper bound, and $U - (\text{plan value})$ bounds the possible loss.
- Multiplying constraints by nonnegative weights, adding, and choosing weights whose regrouped coefficients dominate the objective produces a valid upper bound (Proposition 4.1.4).
- Minimising that bound over all valid weights is a linear program: the dual (Definition 4.1.5). Rows become variables, columns become constraints, and A becomes $A^\top$.

4.2 Weak Duality: Every Certificate Is Valid

Motivation

Proposition 4.1.4 was proved for three specific constraints. The general statement, for any primal–dual pair, is the first theorem of duality theory. It is short, but it is the engine behind optimality certificates, the duality gap, and the primal–dual case analysis of Section 4.6. Lemma 1.3.10 already stated it in Chapter 1; here we name it, reprove it, and draw every consequence.

Weak duality is the “always safe” half of the story: whatever dual-feasible weights you pick, the bound they give is never wrong.

Theorem 4.2.1 (Weak duality). *For every primal-feasible x and every dual-feasible y of the canonical pair of Definition 4.1.5,*

$$c^\top x \leq b^\top y.$$

Proof. Two steps, each using one sign condition.

$$\begin{aligned} c^\top x &\leq (A^\top y)^\top x = y^\top Ax && \text{because } A^\top y \geq c \text{ and } x \geq 0, \\ y^\top Ax &= y^\top (Ax) \leq y^\top b = b^\top y && \text{because } Ax \leq b \text{ and } y \geq 0. \end{aligned}$$

Both inequalities are instances of Proposition 2.1.2: $u \leq v$ and $w \geq 0$ imply $w^\top u \leq w^\top v$. Chain them. $\square$

Key insight

The quantity $y^\top Ax$ sits between the two objectives. It can be read in two ways: as $(A^\top y)^\top x$, a dual row applied to x , or as $y^\top (Ax)$, the weights applied to the resource use. Every duality proof in this chapter inserts this middle term.

Corollary 4.2.2 (Bounds and the duality gap). *Let x be primal-feasible and y dual-feasible.*

- (a) $c^\top x \leq p^*$ and $d^* \leq b^\top y$ whenever these optimal values exist; moreover $p^* \leq b^\top y$ and $c^\top x \leq d^*$.
- (b) The duality gap $b^\top y - c^\top x$ is nonnegative, and $p^* - c^\top x \leq b^\top y - c^\top x$: the gap bounds the loss of the plan x .

Proof. (a) $c^\top x \leq p^*$ and $d^* \leq b^\top y$ are the definitions of maximum and minimum. For $p^* \leq b^\top y$: every feasible x' satisfies $c^\top x' \leq b^\top y$ by Theorem 4.2.1, so the maximum does too. Symmetrically for $c^\top x \leq d^*$. (b) Nonnegativity is the theorem; subtract $c^\top x$ from $p^* \leq b^\top y$. $\square$

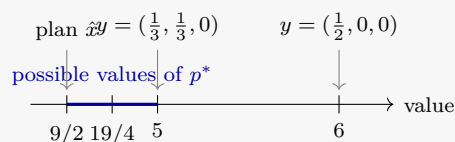
Theorem 4.2.3 (A matching pair proves optimality). *If x is primal-feasible, y is dual-feasible and $c^\top x = b^\top y$, then x is optimal for (P), y is optimal for (D), and $p^* = d^* = c^\top x$.*

Proof. For every primal-feasible x' , $c^\top x' \leq b^\top y = c^\top x$ by Theorem 4.2.1; so x is optimal and $p^* = c^\top x$. For every dual-feasible y' , $b^\top y' \geq c^\top x = b^\top y$; so y is optimal and $d^* = b^\top y$. $\square$

Interpretation & discussion

Theorem 4.2.3 is a *verification* procedure that costs two matrix–vector products and one comparison. It does not require any algorithm, any tableau, or any knowledge of how x and y were found. On an exam, “feasible x , feasible y , equal values” is a complete proof of optimality. The theorem says nothing about whether such a pair *exists*; that is the job of strong duality (Section 4.3).

Example 4.2.4 (Trapping the optimum of the workshop). The plan $\hat{x} = (0, 3/2)$ has value $9/2$. The weights $y = (1/3, 1/3, 0)$ are dual-feasible ($4/3 + 2/3 = 2 \geq 2$, $8/3 + 1/3 = 3 \geq 3$, all ≥ 0) with value $12/3 + 3/3 = 5$. Hence $9/2 \leq p^* \leq 5$.



A better certificate, $y = (5/16, 0, 1/4)$, is also dual-feasible: $4 \cdot \frac{5}{16} + 3 \cdot \frac{1}{4} = \frac{5}{4} + \frac{3}{4} = 2$ and $8 \cdot \frac{5}{16} + 2 \cdot \frac{1}{4} = \frac{5}{2} + \frac{1}{2} = 3$, with value $12 \cdot \frac{5}{16} + 4 \cdot \frac{1}{4} = \frac{15}{4} + 1 = \frac{19}{4}$. So $9/2 \leq p^* \leq 19/4$, and the duality gap of the pair $(\hat{x}, y)$ is $19/4 - 9/2 = 1/4$.

Example 4.2.5 (A matching pair for the workshop). Take $x^* = (1/2, 5/4)$ and $y^* = (5/16, 0, 1/4)$. Primal feasibility:

$$Ax^* = \begin{bmatrix} 4 \cdot \frac{1}{2} + 8 \cdot \frac{5}{4} \\ 2 \cdot \frac{1}{2} + \frac{5}{4} \\ 3 \cdot \frac{1}{2} + 2 \cdot \frac{5}{4} \end{bmatrix} = \begin{bmatrix} 12 \\ 9/4 \\ 4 \end{bmatrix} \leq \begin{bmatrix} 12 \\ 3 \\ 4 \end{bmatrix}, \quad x^* \geq 0.$$

Dual feasibility was checked in Example 4.2.4: $A^\top y^* = (2, 3)^\top = c$ and $y^* \geq 0$. Values:

$$c^\top x^* = 2 \cdot \frac{1}{2} + 3 \cdot \frac{5}{4} = 1 + \frac{15}{4} = \frac{19}{4}, \quad b^\top y^* = \frac{15}{4} + 0 + 1 = \frac{19}{4}.$$

By Theorem 4.2.3 both are optimal and $p^* = d^* = 19/4$. The exact loss of the initial plan $\hat{x}$ is now known: $19/4 - 9/2 = 1/4$.

Technique: Reading a dual solution as a proof

A dual-feasible y is a recipe: “multiply constraint i by y_i and add.” Writing the recipe out makes the bound visible to anyone. For $y^* = (5/16, 0, 1/4)$:

$$\frac{5}{16}(4x_1 + 8x_2) + \frac{1}{4}(3x_1 + 2x_2) = 2x_1 + 3x_2 \leq \frac{5}{16} \cdot 12 + \frac{1}{4} \cdot 4 = \frac{19}{4}.$$

When $A^\top y = c$ exactly (as here), the recipe reproduces the objective and the bound is read off directly; when $A^\top y > c$ in some coordinate, one more line “and $x \geq 0$ ” finishes the argument.

Classic problems

- For $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$, is $x = (2, 2)$ optimal? Add the two constraints: $3x_1 + 2x_2 \leq 10$; $x = (2, 2)$ is feasible with value 10. Yes, by Theorem 4.2.3 with $y = (1, 1)$.
- x has value 7 and some dual-feasible y has value 7.3. What do you know? $7 \leq p^* \leq 7.3$; the loss of x is at most 0.3; nothing more without further work.
- Can a dual-feasible y have $b^\top y < c^\top x$ for a primal-feasible x ? No: that would contradict Theorem 4.2.1. If it “happens” in a calculation, one of the two vectors is infeasible.
- Is $y = (1/2, 0, 0)$ a worse certificate than $y = (1/3, 1/3, 0)$ in the workshop? It gives bound $6 > 5$: valid but weaker. Both are dual-feasible; the dual objective ranks them.

Gotcha

- Weak duality compares a feasible x with a feasible y . A “bound” computed from weights that violate $A^\top y \geq c$ or have a negative component is not a bound at all.
- The gap $b^\top y - c^\top x$ bounds the loss of x but does not equal it: in Example 4.2.4, with $y = (1/3, 1/3, 0)$ the gap was $1/2$ while the true loss of $\hat{x}$ is $1/4$.
- Equal values prove optimality only together with *both* feasibilities.

Checking $c^\top x = b^\top y$ for an infeasible y proves nothing.

Summary: Weak duality

- $c^\top x \leq y^\top Ax \leq b^\top y$ for every primal-feasible x and dual-feasible y (Theorem 4.2.1); for maximisation, x gives a lower bound and y an upper bound.
- The duality gap $b^\top y - c^\top x \geq 0$ bounds how far x is from optimal (Corollary 4.2.2).
- Feasible x , feasible y , equal values $\Rightarrow$ both optimal (Theorem 4.2.3): a two-line proof anyone can check.

4.3 Strong Duality: The Best Bounds Meet

Motivation

In Example 4.2.5 the best plan and the best certificate had the same value, 19/4. Was that luck? Strong duality says no: whenever a canonical LP has a finite optimum, the dual has one too, and the two values coincide. This is the deepest theorem of the chapter and the one the slides state without proof. We prove it from scratch. The route is through a classical alternative theorem, the Farkas lemma, which itself follows from an elimination procedure that is nothing but “combine inequalities with nonnegative weights” used systematically.

Weak duality: every certificate is valid. Strong duality: some certificate is exact. The gap between the best plan and the best bound is zero.

Theorem 4.3.1 (Strong duality). *If the primal (P) of Definition 4.1.5 is feasible and its objective is bounded above on the feasible set, then both (P) and (D) have optimal solutions x^* , y^* , and*

$$\max\{c^\top x : Ax \leq b, x \geq 0\} = \min\{b^\top y : A^\top y \geq c, y \geq 0\}.$$

The feasible region need not be bounded; only the objective value must be. The proof occupies the rest of this section. We first show that an optimal x^* exists, then build y^* .

Step 0: the primal optimum is attained

Lemma 4.3.2 (Existence of a primal optimum). *Under the hypotheses of Theorem 4.3.1, (P) has an optimal solution.*

Proof. Add slacks (Definition 2.2.3) to obtain the equational form $\max c^\top x$ s.t. $Ax + s = b$, $(x, s) \geq 0$, whose constraint matrix $[A \ I_m]$ has rank m . By Theorem 2.2.4 this problem is feasible, and its objective is bounded above by the same bound, since feasible (x, s) correspond to feasible x with equal objective. Theorem 2.5.2 then gives an optimal solution (x^*, s^*) , and x^* is optimal for (P) by the same correspondence. $\square$

Step 1: eliminating a variable from a system of inequalities

Lemma 4.3.3 (Fourier–Motzkin elimination). *Let $Ax \leq b$ be a system of m linear inequalities in the variables $x_1, \dots, x_n$ with $n \geq 1$. There is a system $A'x' \leq b'$ in the variables $x' = (x_2, \dots, x_n)$ such that*

- (a) *every row (a'_k, b'_k) of $[A' \ b']$ is a nonnegative combination of rows of $[A \ b]$ whose x_1 -coefficient is zero, i.e. $a'_k = \sum_i \lambda_{ki} a_{i1}$, $b'_k = \sum_i \lambda_{ki} b_i$ with all $\lambda_{ki} \geq 0$ and $\sum_i \lambda_{ki} a_{i1} = 0$;*
- (b) *$Ax \leq b$ has a solution if and only if $A'x' \leq b'$ has a solution.*

Proof. Step 1 (sort the rows by the sign of the x_1 -coefficient). Let $P = \{i : a_{i1} > 0\}$, $N = \{i : a_{i1} < 0\}$, $Z = \{i : a_{i1} = 0\}$. Write row i as $a_{i1}x_1 + \bar{a}_i^\top x' \leq b_i$ where $\bar{a}_i$ collects the remaining coefficients. For $i \in P$ divide by $a_{i1} > 0$: $x_1 \leq u_i(x') := (b_i - \bar{a}_i^\top x')/a_{i1}$. For $k \in N$ divide by $a_{k1} < 0$, which reverses the inequality (Proposition 1.3.4): $x_1 \geq \ell_k(x') := (b_k - \bar{a}_k^\top x')/a_{k1}$.

Step 2 (define the new system). Take all rows of Z unchanged, and for every pair $(k, i) \in N \times P$ the inequality $\ell_k(x') \leq u_i(x')$, which after clearing denominators reads

$$\left(\frac{1}{a_{i1}} \bar{a}_i - \frac{1}{a_{k1}} \bar{a}_k \right)^\top x' \leq \frac{1}{a_{i1}} b_i - \frac{1}{a_{k1}} b_k.$$

This row is $\frac{1}{a_{i1}}(\text{row } i) + \frac{1}{-a_{k1}}(\text{row } k)$: both multipliers are positive, and the x_1 -coefficient is $\frac{a_{i1}}{a_{i1}} + \frac{a_{k1}}{-a_{k1}} = 1 - 1 = 0$. Rows of Z are combinations with a single multiplier 1. This proves (a). (If P or N is empty, the new system consists of the Z rows only; if Z is also empty, it is the empty system, which every x' satisfies.)

Step 3 (solutions go down). If $x = (x_1, x')$ satisfies $Ax \leq b$, then x' satisfies every Z row, and for each pair, $\ell_k(x') \leq x_1 \leq u_i(x')$, so $\ell_k(x') \leq u_i(x')$. Hence x' solves the new system.

Step 4 (solutions come back up). If x' solves the new system, then $\max_{k \in N} \ell_k(x') \leq \min_{i \in P} u_i(x')$ (every pair was checked). Choose x_1 in this interval; if $N = \emptyset$ take $x_1 = \min_{i \in P} u_i(x')$, if $P = \emptyset$ take $x_1 = \max_{k \in N} \ell_k(x')$, if both are empty take $x_1 = 0$. Then $x_1 \leq u_i(x')$ for all $i \in P$ and $x_1 \geq \ell_k(x')$ for all $k \in N$, which are the P and N rows; the Z rows hold by assumption. So (x_1, x') solves $Ax \leq b$. Together with Step 3 this proves (b). $\square$

Key insight

Eliminating x_1 means: every lower bound on x_1 must be at most every upper bound on x_1 . Writing those comparisons out is exactly “add a positive multiple of a row with $a_{i1} > 0$ to a positive multiple of a row with $a_{k1} < 0$ so that x_1 cancels.” The procedure is a machine that manufactures nonnegative combinations.

Step 2: the Farkas lemma

Theorem 4.3.4 (Farkas lemma, inequality form). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following holds:*

- (i) *there is $x \in \mathbb{R}^n$ with $Ax \leq b$;*
- (ii) *there is $y \in \mathbb{R}^m$ with $y \geq 0$, $A^\top y = 0$ and $b^\top y < 0$.*

Proof. Not both. If x and y satisfied (i) and (ii), then $0 = (A^\top y)^\top x = y^\top (Ax) \leq y^\top b < 0$ (the inequality by $Ax \leq b$, $y \geq 0$ and Proposition 2.1.2), a contradiction.

At least one. Suppose (i) fails. Apply Lemma 4.3.3 to eliminate x_1 , then x_2 from the resulting system, and so on, n times. By part (b) of the lemma, applied n times, the final system has no solution either. But the final system has *no variables*: it is a list of inequalities $0 \leq \beta_k$, which is unsolvable only if some $\beta_k < 0$. By part (a), applied n times, that row $(0, \beta_k)$ is a nonnegative combination of nonnegative combinations of the original rows, hence a nonnegative combination $\sum_i y_i(a_i, b_i)$ with $y \geq 0$ (a product of nonnegative multipliers is nonnegative, and sums of such products are nonnegative). Its coefficient vector is $A^\top y = 0$ and its right-hand side is $b^\top y = \beta_k < 0$. This is (ii). $\square$

Interpretation & discussion

Farkas is a statement about *certificates of infeasibility*: a system of inequalities has no solution exactly when some nonnegative combination of its rows reads $0 \leq$ (negative number). It is the infeasibility twin of weak duality, which produces certificates of optimality. Both say the same thing: for linear inequalities, whatever is true is provable by adding them up with nonnegative weights.

Step 3: the proof of strong duality

Proof of Theorem 4.3.1. By Lemma 4.3.2 let x^* be primal-optimal and set $\gamma = c^\top x^* = p^*$.

Step 1 (what we need). It suffices to find y with $y \geq 0$, $A^\top y \geq c$ and $b^\top y \leq \gamma$. Indeed such a y is dual-feasible, so $b^\top y \geq c^\top x^* = \gamma$ by Theorem 4.2.1; hence $b^\top y = \gamma$, and Theorem 4.2.3 makes y dual-optimal with $d^* = p^*$.

Step 2 (write the requirement as one system). The three conditions form the system $My \leq d$ with

$$M = \begin{bmatrix} -A^\top \\ -I_m \\ b^\top \end{bmatrix} \in \mathbb{R}^{(n+m+1) \times m}, \quad d = \begin{bmatrix} -c \\ 0 \\ \gamma \end{bmatrix}.$$

Step 3 (suppose it is unsolvable and apply Farkas). If the system has no solution, Theorem 4.3.4 gives a vector $(u, v, t) \geq 0$ with $u \in \mathbb{R}^n$, $v \in \mathbb{R}^m$, $t \in \mathbb{R}$ such that $M^\top(u, v, t) = 0$ and $d^\top(u, v, t) < 0$, that is,

$$-Au - v + tb = 0 \quad \text{and} \quad -c^\top u + t\gamma < 0.$$

Rearranged: $Au = tb - v \leq tb$ (since $v \geq 0$) and $c^\top u > t\gamma$.

Step 4 (case $t > 0$: a better plan). Put $\tilde{x} = u/t \geq 0$. Then $A\tilde{x} = b - v/t \leq b$, so $\tilde{x}$ is primal-feasible, and $c^\top \tilde{x} = c^\top u/t > \gamma = p^*$. This contradicts the optimality of x^* .

Step 5 (case $t = 0$: an improving ray). Then $Au = -v \leq 0$, $u \geq 0$ and $c^\top u > 0$. For every $\theta \geq 0$, $x^* + \theta u \geq 0$ and $A(x^* + \theta u) = Ax^* + \theta Au \leq b$, so the whole ray is feasible, and $c^\top u > 0$ makes the objective unbounded along it (Proposition 3.2.4). This contradicts boundedness.

Both cases are impossible, so the system of Step 2 is solvable, and Step 1 finishes the proof. $\square$

Key insight

Farkas turns “no exact certificate exists” into an explicit vector (u, v, t) . Reading that vector back in primal terms gives either a feasible plan better than the optimum ($t > 0$) or a feasible direction of unbounded improvement ($t = 0$). Both contradict the hypotheses, so an exact certificate must exist. The proof is non-constructive; Proposition 4.3.5 below shows where simplex actually finds y^* .

Interpretation & discussion

Strong duality says the bound-finding game is *complete*: the cleverest possible combination of constraints proves the true optimum, never something weaker. It also says the dual is not an auxiliary trick but an equally legitimate description of the same problem; Section 4.4 shows the dual of the dual is the primal. Two readings of the hypotheses matter. First, “bounded objective” is required, not “bounded feasible region”: $\max x_1$ s.t. $x_1 \leq 3, x \geq 0$ in two variables has an unbounded region and a finite optimum, and strong duality applies. Second, if the primal is infeasible or unbounded, the theorem is silent; Section 4.6 covers those cases.

Where simplex finds the dual solution

Strong duality guarantees an exact certificate exists. The simplex method of Chapter 3 hands it to you.

Proposition 4.3.5 (The final objective row is a dual optimal solution). *Run simplex on the equational form $Ax + s = b$, $(x, s) \geq 0$ of (P) , and suppose it stops at a tableau (Definition 3.3.2) in which every reduced cost is ≤ 0 :*

$$v = v_0 + \sum_{j \in N} r_j z_j, \quad z = (x, s), \quad r_j \leq 0.$$

Extend r to all variables by $r_j = 0$ for basic j , and let $r_x \in \mathbb{R}^n$, $r_s \in \mathbb{R}^m$ be the reduced costs of the original and slack variables. Then $y^ := -r_s$ is dual-feasible with $b^\top y^* = v_0$; hence y^* is dual-optimal and the current BFS is primal-optimal.*

Proof. Step 1 (the objective row is an identity). The tableau was obtained from $Ax + s = b$ and $v = c^\top x$ by substitutions that are valid for every (x, s) satisfying the equations, not only for feasible ones. So $c^\top x = v_0 + r_x^\top x + r_s^\top s$ whenever $s = b - Ax$.

Step 2 (substitute the slacks). Replacing s by $b - Ax$, which holds for every $x \in \mathbb{R}^n$ with the corresponding s ,

$$c^\top x = v_0 + r_x^\top x + r_s^\top (b - Ax) = (v_0 + r_s^\top b) + (r_x - A^\top r_s)^\top x \quad \text{for all } x \in \mathbb{R}^n.$$

Step 3 (compare coefficients). Two affine functions of x that agree on all of $\mathbb{R}^n$ have the same constant term and the same linear part: $v_0 + r_s^\top b = 0$ and $r_x - A^\top r_s = c$. With $y^* = -r_s$ these read $b^\top y^* = v_0$ and $A^\top y^* = c - r_x \geq c$, since $r_x \leq 0$. Also $y^* = -r_s \geq 0$. So y^* is dual-feasible with value v_0 , the value of the current BFS, and Theorem 4.2.3 makes both optimal. $\square$

Interpretation & discussion

This is the “seed of duality” noted at the end of Chapter 3: the negatives of the final reduced costs of the slack variables are the optimal dual weights. The juice example there ended with $v = 38 - x_4 - x_5 - x_6$, i.e. $y^* = (1, 1, 1)$. The proposition also explains *why* the optimality test (Theorem 3.3.8) is a proof: a tableau with nonpositive reduced costs *is* a dual-feasible certificate written in a different notation.

Technique: Proof patterns in duality theory

- *Insert the middle term.* To compare $c^\top x$ with $b^\top y$, write $y^\top Ax$ between them and use one sign condition on each side.
- *Equal values $\Rightarrow$ optimal.* Never prove optimality of x directly; exhibit a y and compare values (Theorem 4.2.3).
- *Alternatives.* To prove something *exists*, assume it does not, apply Farkas to get a certificate, and read the certificate as a contradiction (the proof of Theorem 4.3.1).
- *Identities from tableaux.* An objective row is an identity on the whole affine space $\{Az = b\}$; comparing coefficients extracts dual variables (Proposition 4.3.5).

Classic problems

- *The primal has optimal value 19/4. What is the dual optimal value, and why?* 19/4, by strong duality (Theorem 4.3.1); the primal is feasible with a finite maximum.
- *A final tableau reads $v = 38 - x_4 - x_5 - x_6$ where x_4, x_5, x_6 are the slacks. Dual optimal solution? $y^* = (1, 1, 1)$, value 38 (Proposition 4.3.5).*
- *Does strong duality need the feasible region to be bounded?* No; it needs the objective to be bounded above on the region.
- *Show $\{x : x_1 + x_2 \leq 1, -x_1 - x_2 \leq -3\}$ is empty with a Farkas certificate.* $y = (1, 1)$: $A^\top y = 0$, $b^\top y = 1 - 3 = -2 < 0$.

Gotcha

- Strong duality is an *if*: feasible and bounded primal $\Rightarrow$ equal optima. It does not say every LP has a dual with the same value; infeasible or unbounded problems have no value to match.
- “min of the dual equals max of the primal” is about *values*, not solutions: x^* and y^* live in different spaces ($\mathbb{R}^n$ and $\mathbb{R}^m$) and are not transforms of each other.
- Fourier–Motzkin is a proof device, not an algorithm to use by hand: the number of rows can square at every elimination.

Summary: Strong duality

- Feasible and bounded primal $\Rightarrow$ both problems have optimal solutions and $p^* = d^*$ (Theorem 4.3.1).
- Proof route: eliminate variables with nonnegative combinations (Lemma 4.3.3) $\Rightarrow$ an unsolvable system has a certificate $0 \leq \beta < 0$ (Farkas, Theorem 4.3.4) $\Rightarrow$ if no exact dual certificate existed, Farkas would produce a better plan or an improving ray.
- Simplex delivers y^* for free: the negatives of the final reduced costs of the slacks (Proposition 4.3.5).

4.4 Dualization Rules for Any Linear Program

Motivation

Definition 4.1.5 covers only max, $\leq$ rows and ≥ 0 variables. Real models have $\geq$ rows, equalities, free variables and nonpositive variables (Chapter 1). One could convert everything to canonical form first (Propositions 1.3.4 and 1.3.5) and then dualize, but that is slow and error-prone. Better: understand why each sign rule holds and write the dual directly. We first work with a maximisation primal; its dual is a minimisation. Rows become variables, columns become constraints, and the direction of each inequality and the sign of each variable determine the corresponding sign rule.

Every rule below is the answer to one question: “what sign must the multiplier have so that the weighted row still gives an upper bound?”

Rule 1: from row direction to variable sign

We want the weighted row $y_i a_i^\top x$ to be bounded above by $y_i b_i$, so that adding the rows gives an upper bound.

Primal row i	Dual variable	Reason
$a_i^\top x \leq b_i$	$y_i \geq 0$	preserves $\leq$
$a_i^\top x \geq b_i$	$y_i \leq 0$	reverses $\geq$ into $\leq$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$	an equality survives either sign

For instance, $x_1 + x_2 \geq 3$ multiplied by $y_i = -2$ gives $-2x_1 - 2x_2 \leq -6$, an inequality of the right direction.

Rule 2: from variable sign to row direction

Write $d_j = \sum_i a_{ij} y_i = (A^\top y)_j$ for the weighted coefficient of x_j . We need $c_j x_j \leq d_j x_j$ so that the objective is dominated term by term.

Primal variable j	Dual row j	Reason
$x_j \geq 0$	$d_j \geq c_j$	multiplying by a nonnegative number preserves order
$x_j \leq 0$	$d_j \leq c_j$	multiplying by a nonpositive number reverses order
$x_j \in \mathbb{R}$	$d_j = c_j$	must work for both signs

A free primal variable gives a dual *equality*.

Definition 4.4.1 (General primal–dual pair). Let the primal be $\max c^\top x$ subject to row constraints $a_i^\top x \leq b_i$ ($i \in I_\leq$), $a_i^\top x \geq b_i$ ($i \in I_\geq$), $a_i^\top x = b_i$ ($i \in I_=$), and variable signs $x_j \geq 0$ ($j \in J_\geq$), $x_j \leq 0$ ($j \in J_\leq$), $x_j \in \mathbb{R}$ ($j \in J_{\text{free}}$). Its dual is $\min b^\top y$ subject to $y_i \geq 0$ ($i \in I_\leq$), $y_i \leq 0$ ($i \in I_\geq$), $y_i \in \mathbb{R}$ ($i \in I_=$) and $(A^\top y)_j \geq c_j$ ($j \in J_\geq$), $(A^\top y)_j \leq c_j$ ($j \in J_\leq$), $(A^\top y)_j = c_j$ ($j \in J_{\text{free}}$).

Theorem 4.4.2 (Weak duality for the general pair). For every x feasible for the general primal and every y feasible for its dual, $c^\top x \leq b^\top y$. Consequently feasible x, y with equal values are both optimal.

Proof. **Step 1 (rows).** For each i , $y_i(a_i^\top x - b_i) \leq 0$: if $i \in I_\leq$, the bracket is ≤ 0 and $y_i \geq 0$; if $i \in I_\geq$, the bracket is ≥ 0 and $y_i \leq 0$; if $i \in I_=$, the bracket is 0. Summing over i : $y^\top Ax - y^\top b \leq 0$, i.e. $y^\top Ax \leq b^\top y$.

Step 2 (columns). For each j , $x_j((A^\top y)_j - c_j) \geq 0$: if $j \in J_\geq$, both factors are ≥ 0 ; if $j \in J_\leq$, both are ≤ 0 ; if j is free, the second factor is 0. Summing over j : $x^\top A^\top y - c^\top x \geq 0$, i.e. $c^\top x \leq y^\top Ax$.

Step 3. Chain: $c^\top x \leq y^\top Ax \leq b^\top y$. The optimality statement follows exactly as in Theorem 4.2.3. $\square$

Key insight

The rules are precisely the sign patterns that make each product $y_i(a_i^\top x - b_i)$ nonpositive and each product $x_j((A^\top y)_j - c_j)$ nonnegative. If you forget a rule, rederive it from that requirement.

Theorem 4.4.3 (The general dual is the canonical dual in disguise). Convert the general primal to canonical form by (i) replacing $a_i^\top x \geq b_i$ with $-a_i^\top x \leq -b_i$, (ii) replacing $a_i^\top x = b_i$ with the pair $a_i^\top x \leq b_i$, $-a_i^\top x \leq -b_i$, (iii) substituting $x_j = -x'_j$ with $x'_j \geq 0$ for $j \in J_\leq$, and (iv) substituting $x_j = u_j - w_j$ with $u_j, w_j \geq 0$ for free j . Then the canonical dual of the converted problem is equivalent to the general dual of Definition 4.4.1: feasible solutions correspond with equal objective values. In particular, strong duality (Theorem 4.3.1) holds for the general pair.

Proof. We track what each conversion does to the dual, one at a time; the conversions act on different rows and columns, so they can be composed.

(i) $A \geq \text{row}$. The converted row $-a_i^\top x \leq -b_i$ receives a canonical dual variable $y'_i \geq 0$ that multiplies $-a_i$ and contributes $-b_i y'_i$ to the dual objective. Setting $y_i = -y'_i \leq 0$, the contribution to $A^\top y$ is $y_i a_i$ and to the objective is $b_i y_i$: exactly the general dual with $y_i \leq 0$.

(ii) An equality row. The two converted rows receive $y_i^+, y_i^- \geq 0$, contributing $(y_i^+ - y_i^-)a_i$ to $A^\top y$ and $b_i(y_i^+ - y_i^-)$ to the objective. Every real y_i is of the form $y_i^+ - y_i^-$ (Proposition 1.3.5), and conversely each such difference is a real number: the general dual with $y_i \in \mathbb{R}$.

(iii) A nonpositive variable. Column j becomes $-a^{(j)}$ (where $a^{(j)}$ is column j of A) and the cost becomes $-c_j$; the canonical dual row is $-(a^{(j)})^\top y \geq -c_j$, i.e. $(A^\top y)_j \leq c_j$.

(iv) A free variable. Two columns $a^{(j)}$ and $-a^{(j)}$ with costs c_j and $-c_j$ give the two dual rows $(A^\top y)_j \geq c_j$ and $-(A^\top y)_j \geq -c_j$, together $(A^\top y)_j = c_j$.

The converted primal has the same feasible values as the general primal (Propositions 1.3.4 and 1.3.5), so it is feasible and bounded exactly when the general primal is. Theorem 4.3.1 applies to the converted pair, and the correspondence above transfers optimal solutions and values to the general pair. $\square$

Example 4.4.4 (Mixed signs). Write the dual of

$$\begin{aligned} \max \quad & 3x_1 + x_2 + 4x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq 4, \quad 2x_1 - x_2 + x_3 \geq 1, \quad x_1 + x_3 = 2, \\ & x_1 \geq 0, \quad x_2 \in \mathbb{R}, \quad x_3 \leq 0. \end{aligned}$$

Rows (Rule 1): $y_1 \geq 0, y_2 \leq 0, y_3 \in \mathbb{R}$. Objective: $\min 4y_1 + y_2 + 2y_3$. Columns (Rule 2), keeping every coefficient's sign:

$$\begin{aligned} \text{column of } x_1 = (1, 2, 1) : \quad & y_1 + 2y_2 + y_3 \geq 3 & (x_1 \geq 0), \\ \text{column of } x_2 = (1, -1, 0) : \quad & y_1 - y_2 = 1 & (x_2 \text{ free}), \\ \text{column of } x_3 = (1, 1, 1) : \quad & y_1 + y_2 + y_3 \leq 4 & (x_3 \leq 0). \end{aligned}$$

The dual of the dual

Proposition 4.4.5 (Dualizing twice returns the primal). *The dual of (D), written as a maximisation problem and dualized with the rules above, is equivalent to (P).*

Proof. (D) is $\min b^\top y$ s.t. $A^\top y \geq c, y \geq 0$. By Proposition 1.3.6 it has the same optimal solutions as $\max (-b)^\top y$ s.t. $A^\top y \geq c, y \geq 0$, and by Proposition 1.3.4 the constraints can be written $(-A^\top)y \leq -c$. This is canonical with data $(-A^\top, -c, -b)$, so its dual (Definition 4.1.5) is $\min (-c)^\top x$ s.t. $(-A^\top)^\top x \geq -b, x \geq 0$, i.e. $\min -c^\top x$ s.t. $-Ax \geq -b, x \geq 0$. Reversing the sign of the constraint and of the objective once more gives $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$, which is (P). $\square$

Interpretation & discussion

The two problems are on an equal footing: whichever one you call “primal,” the other is its dual, and the general rules are symmetric under this exchange (a $\geq$ row of a minimisation dualizes to a ≥ 0 variable, and so on). This symmetry is why the primal–dual table of Section 4.6 can be read in both directions.

Technique: The dualization checklist

1. Make sure the primal is a max (if it is a min, either dualize with the mirrored rules or negate the objective first).
2. One dual variable per row; its sign from Rule 1 ($\leq \rightarrow \geq 0, \geq \rightarrow \leq 0, = \rightarrow$ free).
3. Dual objective: $\min b^\top y$ with the primal right-hand sides.
4. One dual constraint per primal variable; read the primal column, keep every sign; its direction from Rule 2 ($\geq 0 \rightarrow \geq c_j, \leq 0 \rightarrow \leq c_j, \text{ free} \rightarrow = c_j$).
5. Always check dimensions, inequality directions and variable signs; then

verify weak duality on one feasible pair as a sanity check.

For a *minimisation* primal the rules mirror: a $\geq$ row gives $y_i \geq 0$, a $\leq$ row gives $y_i \leq 0$; a variable ≥ 0 gives a dual row $\leq c_j$, a variable ≤ 0 gives $\geq c_j$; the dual is a max. Check: applying this to (D) returns (P).

Classic problems

- Dual of $\max c^\top x$ s.t. $Ax = b, x \geq 0$? $\min b^\top y$ s.t. $A^\top y \geq c, y$ free.
- Dual of $\max c^\top x$ s.t. $Ax \leq b, x$ free? $\min b^\top y$ s.t. $A^\top y = c, y \geq 0$.
- Dual of $\min b^\top y$ s.t. $A^\top y \geq c, y \geq 0$? $\max c^\top x$ s.t. $Ax \leq b, x \geq 0$ (Proposition 4.4.5).
- A primal with 5 equality rows and 3 free variables: how many free dual variables and how many dual equalities? 5 free dual variables; 3 dual equalities.

Gotcha

- A $\geq$ row in a *maximisation* gives a *nonpositive* dual variable, not a nonnegative one. Students who “fix” the sign to ≥ 0 produce a wrong dual; verify with Theorem 4.4.2.
- Do not drop the sign of a coefficient when reading a column: in Example 4.4.4 the column of x_2 is $(1, -1, 0)$, giving $y_1 - y_2$.
- A free variable produces an *equality* in the dual, and an equality row produces a *free* variable. The two are easy to swap.
- Converting to canonical form first and then dualizing is correct (Theorem 4.4.3) but produces a dual with split variables and doubled rows; simplify it back before using it.

Summary: Dualization rules

- Rule 1 (rows $\rightarrow$ variables): $\leq \rightarrow y_i \geq 0, \geq \rightarrow y_i \leq 0, = \rightarrow y_i$ free. Rule 2 (variables $\rightarrow$ rows): $x_j \geq 0 \rightarrow (A^\top y)_j \geq c_j, x_j \leq 0 \rightarrow \leq c_j, \text{free} \rightarrow =$.
- Each rule is forced by one requirement: $y_i(a_i^\top x - b_i) \leq 0$ and $x_j((A^\top y)_j - c_j) \geq 0$, which together prove general weak duality (Theorem 4.4.2).
- The general dual equals the canonical dual of the canonical reformulation, so strong duality holds in general (Theorem 4.4.3); the dual of the dual is the primal (Proposition 4.4.5).

4.5 Complementary Slackness

Motivation

Strong duality says $c^\top x^* = b^\top y^*$ at an optimal pair. Weak duality proved this by a chain of two inequalities, $c^\top x \leq y^\top Ax \leq b^\top y$. If the ends agree, both links must be equalities. What does that force on the variables and on the slack in each constraint? The answer, complementary slackness, is both a structural insight and a practical tool: it lets you *compute* a dual solution from a primal one by

At an optimal pair, no resource is both unused and valued, and no activity is both run and overpriced. Every “and” there is a product that must vanish.

solving linear equations.

Slack measures the margin in a constraint

Definition 4.5.1 (Primal and dual slack vectors). For primal-feasible x and dual-feasible y of the canonical pair, define

$$s = b - Ax \geq 0 \quad (\text{primal slack}), \quad r = A^\top y - c \geq 0 \quad (\text{dual slack}).$$

A constraint is *tight* when its slack is 0; a strict inequality has positive slack.

In the workshop, $2x_1 + x_2 \leq 3$ leaves the margin $s_2 = 3 - (2x_1 + x_2)$, the unused capacity of resource 2; the dual constraint $4y_1 + 2y_2 + 3y_3 \geq 2$ leaves the margin $r_1 = 4y_1 + 2y_2 + 3y_3 - 2$, the excess of the weighted cost of activity 1 over its profit.

Lemma 4.5.2 (The gap is a sum of nonnegative products). For primal-feasible x and dual-feasible y ,

$$b^\top y - c^\top x = y^\top (b - Ax) + x^\top (A^\top y - c) = \sum_{i=1}^m y_i s_i + \sum_{j=1}^n x_j r_j,$$

and every term $y_i s_i$ and $x_j r_j$ is nonnegative.

Proof. Insert and remove the middle term: $b^\top y - c^\top x = (b^\top y - y^\top Ax) + (y^\top Ax - c^\top x) = y^\top (b - Ax) + (A^\top y - c)^\top x$, using $y^\top Ax = (A^\top y)^\top x$. Expanding the two inner products gives the sums. Each product is of two nonnegative numbers. $\square$

Theorem 4.5.3 (Complementary slackness). Let x be primal-feasible and y dual-feasible for the canonical pair. Then x and y are both optimal if and only if

$$\begin{aligned} y_i (b_i - (Ax)_i) &= 0 \quad (i = 1, \dots, m), \\ x_j ((A^\top y)_j - c_j) &= 0 \quad (j = 1, \dots, n), \end{aligned}$$

that is, $y_i s_i = 0$ for every i and $x_j r_j = 0$ for every j .

Proof. ($\Rightarrow$) If both are optimal, strong duality (Theorem 4.3.1; the primal is feasible and bounded since it has an optimum) gives $b^\top y - c^\top x = 0$. By Lemma 4.5.2 this is a sum of nonnegative terms equal to zero, so every term is zero: no positive term can be cancelled.

($\Leftarrow$) If every product is zero, Lemma 4.5.2 gives $b^\top y - c^\top x = 0$, i.e. equal objective values. Feasibility and Theorem 4.2.3 then prove both solutions optimal. $\square$

Key insight

A sum of nonnegative numbers is zero only if each is zero. The whole theorem is this one-line fact applied to the decomposition of the duality gap.

Interpretation & discussion

Each variable is paired with a constraint's slack on the *other* side. The conditions read, at an optimal pair:

Primal constraint i : $y_i s_i = 0$	Dual constraint j : $x_j r_j = 0$
Unused margin forces a zero dual weight: $s_i > 0 \Rightarrow y_i = 0$.	A positive activity forces a tight dual constraint: $x_j > 0 \Rightarrow r_j = 0$.
A positive dual weight forces tightness: $y_i > 0 \Rightarrow s_i = 0$.	A positive dual slack forces a zero activity: $r_j > 0 \Rightarrow x_j = 0$.

At least one factor of each product is zero; both may be zero. The economic reading: a resource with spare capacity has no marginal value ($y_i = 0$); an activity that is run must exactly pay for the resources it consumes ($r_j = 0$); an activity whose weighted cost exceeds its profit is not run. Note the direction of the implications: $s_i > 0$ forces $y_i = 0$, but $s_i = 0$ forces nothing about y_i .

Remark 4.5.4 (General pairs). The same statement and proof hold for the general pair of Definition 4.4.1, because the proof of Theorem 4.4.2 showed that each row product $y_i(a_i^\top x - b_i)$ is ≤ 0 and each column product $x_j((A^\top y)_j - c_j)$ is ≥ 0 ; the gap is $\sum_j x_j((A^\top y)_j - c_j) - \sum_i y_i(a_i^\top x - b_i)$, a sum of nonnegative terms. Equality rows and free variables contribute products that are automatically zero.

Using complementary slackness to find a certificate

Example 4.5.5 (From a primal solution to dual weights). Take the workshop point $x = (1/2, 5/4)$ and ask whether it is optimal. Read the slacks:

Resource	Used	Available	Slack
1	12	12	$s_1 = 0$
2	9/4	3	$s_2 = 3/4$
3	4	4	$s_3 = 0$

Step 1 (impose the conditions). If x is optimal and y is a matching dual optimum, Theorem 4.5.3 forces $s_2 > 0 \Rightarrow y_2 = 0$. Both coordinates of x are positive, so $x_1 r_1 = x_2 r_2 = 0$ forces $r_1 = r_2 = 0$: both dual constraints must be tight. With $y_2 = 0$ they become

$$4y_1 + 3y_3 = 2, \quad 8y_1 + 2y_3 = 3.$$

Step 2 (solve). Twice the first equation minus the second: $8y_1 + 6y_3 - 8y_1 - 2y_3 = 4 - 3$, so $4y_3 = 1$ and $y_3 = 1/4$. Substituting, $4y_1 = 2 - 3/4 = 5/4$, so $y_1 = 5/16$.

Step 3 (verify before concluding). $y = (5/16, 0, 1/4) \geq 0$ and $A^\top y = (2, 3)^\top = c$, so y is dual-feasible with $r = (0, 0)$. We also have $s = (0, 3/4, 0)$ and $y_2 = 0$, so $y_1 s_1 = y_2 s_2 = y_3 s_3 = 0$ and $x_1 r_1 = x_2 r_2 = 0$. By Theorem 4.5.3 both solutions are optimal, and $c^\top x = b^\top y = 19/4$.

Interpretation & discussion

The procedure is: read the slacks, impose the conditions, solve, verify feasibility. Step 3 is not optional. Complementary slackness was *assumed* in Step 1 (“if x is optimal”); the equations it produces always have a solution candidate, but the candidate is a certificate only if it is dual-feasible. If the computed y had a negative component or violated $A^\top y \geq c$, the correct conclusion would be “ x is not optimal,” not “the method failed.”

Example 4.5.6 (Detecting a non-optimal point). Try the same procedure on the plan $\hat{x} = (0, 3/2)$. Slacks: $s = (0, 3/2, 1)$, so $y_2 = y_3 = 0$. Only $x_2 > 0$, so only $r_2 = 0$ is forced: $8y_1 = 3$, $y_1 = 3/8$. Candidate $y = (3/8, 0, 0)$. Check dual feasibility: $4 \cdot 3/8 = 3/2 \geq 2$? No. There is no dual-feasible y complementary to $\hat{x}$, so by Theorem 4.5.3 $\hat{x}$ is not optimal, as we already knew ($9/2 < 19/4$).

Technique: Complementary slackness as an exam tool

- *Given a primal candidate x :* compute $s = b - Ax$. Every $s_i > 0$ sets $y_i = 0$; every $x_j > 0$ sets $(A^\top y)_j = c_j$. Solve the resulting linear system for the remaining y_i ; verify $y \geq 0$ and $A^\top y \geq c$. Feasible: x optimal and y its certificate. Infeasible or inconsistent: x not optimal.
- *Given a dual candidate y :* symmetric. Every $r_j > 0$ sets $x_j = 0$; every $y_i > 0$ makes row i tight.
- *Counting equations:* with k positive primal variables and ℓ positive slacks, you get k equations in $m - \ell$ unknowns. For a nondegenerate vertex, $k = m - \ell$ and the system is square.
- *Nothing is forced* by $s_i = 0$ or $x_j = 0$. Do not write $y_i > 0$ because resource i is fully used; it may still have $y_i = 0$ (degenerate case).

Classic problems

- *For max $3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$:* is $x = (2, 2)$ optimal? Both rows tight, both variables positive: $y_1 + 2y_2 = 3$, $y_1 + y_2 = 2$, so $y = (1, 1) \geq 0$. Feasible, so yes; value 10.
- *At an optimal pair, $y_3 = 2 > 0$. What do you know about resource 3?* It is fully used: $s_3 = 0$.
- *At an optimal pair, $x_2 = 0$. Is dual row 2 tight?* Not necessarily: $x_2 r_2 = 0$ holds whatever r_2 is.
- *Is x optimal if some complementary y exists with $A^\top y \geq c$ but $y_1 < 0$?* No: y is not dual-feasible, so it certifies nothing; try again or conclude non-optimality if the system forces $y_1 < 0$.

Gotcha

- Complementary slackness is a condition on a *pair*: “ x satisfies complementary slackness” means nothing by itself.
- The implications run one way: positive slack kills the dual variable; zero slack does *not* make it positive. Degenerate vertices have tight

constraints with zero multipliers.

- Skipping the verification step is the classic error. The equations assume optimality; the feasibility check tests it.

Summary: Complementary slackness

- $b^\top y - c^\top x = \sum_i y_i s_i + \sum_j x_j r_j$ with $s = b - Ax$, $r = A^\top y - c$ (Lemma 4.5.2).
- Feasible x, y are both optimal iff $y_i s_i = 0$ for all i and $x_j r_j = 0$ for all j (Theorem 4.5.3).
- Reading: spare capacity $\Rightarrow$ zero weight; positive activity $\Rightarrow$ tight dual row. Use it to compute a certificate from a candidate, then *verify feasibility*.

4.6 Three Outcomes and the Primal–Dual Table

Motivation

Not every LP has a finite optimum. This section classifies what can happen to one LP and then to a primal–dual pair, using nothing but weak and strong duality. The classification is exhaustive and each cell of the table is either proved possible by an example or proved impossible by a theorem.

Strong duality handles the “nice” case. The table tells you what the dual looks like when the primal is not nice.

Proposition 4.6.1 (Three outcomes). *For a maximisation LP, exactly one of the following occurs:*

1. Finite optimum: *a feasible solution attains a finite best objective value;*
2. Unbounded objective: *feasible solutions exist and their objective values grow without limit;*
3. Infeasible: *no point satisfies all the constraints.*

For a minimisation, “unbounded” means the objective can decrease without limit.

Proof. The feasible set is empty (case 3) or not. If not, the objective is either unbounded above on it (case 2) or bounded above. In the latter case the optimum is *attained* by Lemma 4.3.2 (via Theorem 2.5.2, after conversion to equational form), which is case 1. The three cases are mutually exclusive by definition. $\square$

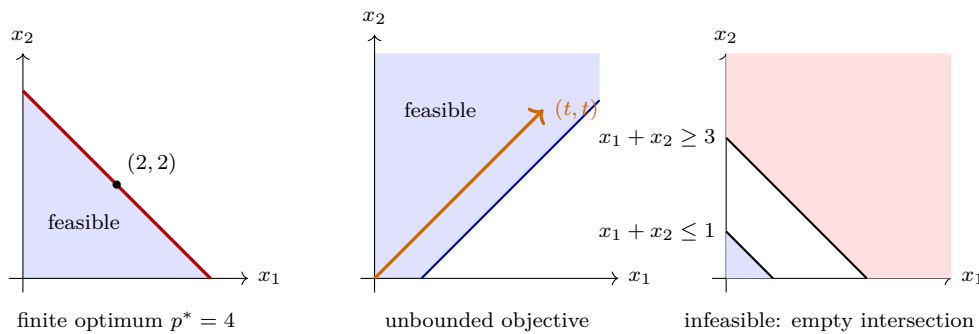
Interpretation & discussion

“Bounded, hence attained” is special to linear programming. For a general optimisation problem, sup need not be a max (max $-e^{-x}$ over $x \geq 0$ has supremum 0, never attained). For LPs, Theorem 2.5.2 rules this out: the finitely many basic feasible solutions include an optimal one.

Example 4.6.2 (Finite optimum). $\max x_1 + x_2$ s.t. $x_1 + x_2 \leq 4$, $x \geq 0$. Every feasible value is at most 4 (the constraint itself is the certificate: $y = 1$). The feasible point $(2, 2)$ attains it: $p^* = 4$. The optimum is not unique: every point of the segment from $(4, 0)$ to $(0, 4)$ has value 4.

Example 4.6.3 (Unbounded objective). $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1, x \geq 0$. For every $t \geq 0$ the point (t, t) is feasible since $t - t = 0 \leq 1$, and $x_1 + x_2 = 2t \rightarrow +\infty$. There is no finite maximum (Proposition 3.2.4 with $x^0 = 0, d = (1, 1)$). An unbounded feasible set alone does not imply an unbounded objective: the same region with objective $\max x_1 - x_2$ has optimum 1.

Example 4.6.4 (Infeasible). $\max x_1 + x_2$ s.t. $x_1 + x_2 \leq 1, x_1 + x_2 \geq 3, x \geq 0$. A feasible point would satisfy $3 \leq x_1 + x_2 \leq 1$, impossible. The feasible set is empty. (Farkas certificate: add the two rows written as $x_1 + x_2 \leq 1$ and $-x_1 - x_2 \leq -3$ with weights $(1, 1)$: $0 \leq -2$.)



What can happen to a primal–dual pair?

Theorem 4.6.5 (Primal–dual outcome table). *For the canonical pair (and, by Theorem 4.4.3, for any general pair), the possible combinations are exactly:*

Primal (maximisation)	Dual (minimisation)
finite optimum	finite optimum, same value
unbounded above	infeasible
infeasible	unbounded below
infeasible	infeasible

In particular: if one problem is unbounded, the other is infeasible; if one has a finite optimum, so does the other; and an infeasible problem's dual is either unbounded or infeasible, never finite.

Proof. There are nine combinations of the three outcomes on each side; we show which survive.

Finite primal $\Rightarrow$ finite dual, same value. Theorem 4.3.1. By symmetry (Proposition 4.4.5 and Theorem 4.4.3), finite dual $\Rightarrow$ finite primal. This settles the first row and rules out (finite, unbounded), (finite, infeasible), (unbounded, finite), (infeasible, finite).

Unbounded primal $\Rightarrow$ infeasible dual. If the dual had a feasible y , weak duality (Theorem 4.2.1) would give $c^\top x \leq b^\top y$ for every feasible x , a finite upper bound, contradicting unboundedness. Symmetrically, unbounded dual $\Rightarrow$ infeasible primal. This rules out (unbounded, unbounded) and gives rows two and three as the only options when one side is unbounded.

The remaining cells occur. (unbounded, infeasible): Example 4.6.3; its dual $\min y_1$ s.t. $y_1 \geq 1, -y_1 \geq 1, y_1 \geq 0$ is infeasible. (infeasible, unbounded): the

dual of that example, read as primal (Proposition 4.4.5). (infeasible, infeasible): Example 4.6.6 below. $\square$

Example 4.6.6 (Both problems infeasible).

$$\begin{array}{ll} \text{Primal:} & \max x_1 + x_2 \quad \text{s.t.} \quad x_1 - x_2 \leq -1, \quad -x_1 + x_2 \leq -1, \quad x \geq 0; \\ \text{Dual:} & \min -y_1 - y_2 \quad \text{s.t.} \quad y_1 - y_2 \geq 1, \quad -y_1 + y_2 \geq 1, \quad y \geq 0. \end{array}$$

Adding the primal constraints gives $0 \leq -2$; adding the dual constraints gives $0 \geq 2$. Neither side has a feasible point, and there is no finite optimum to compare.

Interpretation & discussion

Infeasibility of one problem does *not* automatically mean the other is unbounded; Example 4.6.6 shows both can fail. What *is* automatic: an unbounded problem forces its partner to be infeasible, because a single feasible point on the other side would be a finite bound. The table is the complete list of what duality allows.

A complete worked certificate

Example 4.6.7 (Prove an optimum using duality). Consider $\max 3x_1 + 2x_2$ s.t. $x_1 + x_2 \leq 4$, $2x_1 + x_2 \leq 6$, $x \geq 0$.

1. *Dual:* $\min 4y_1 + 6y_2$ s.t. $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y \geq 0$.
2. *Primal feasibility of $x = (2, 2)$:* $2 + 2 = 4 \leq 4$, $4 + 2 = 6 \leq 6$, $x \geq 0$. Value $c^\top x = 10$, so $p^* \geq 10$.
3. *A matching certificate:* both primal rows are tight and both variables positive, so complementary slackness suggests making both dual rows tight: $y_1 + 2y_2 = 3$, $y_1 + y_2 = 2$, giving $y = (1, 1)$. Check: $y \geq 0$, $1 + 2 = 3 \geq 3$, $1 + 1 = 2 \geq 2$. Value $b^\top y = 4 + 6 = 10$, so $p^* \leq 10$.
4. *Conclusion:* feasible x , feasible y , equal values, so both are optimal and $p^* = 10$ (Theorem 4.2.3, i.e. weak duality). The certificate is visible by adding the primal rows: $(x_1 + x_2) + (2x_1 + x_2) \leq 4 + 6$, i.e. $3x_1 + 2x_2 \leq 10$.

Technique: Classifying an LP and its dual quickly

- A feasible improving ray ($d \geq 0$, $Ad \leq 0$, $c^\top d > 0$ in canonical form) proves unboundedness and, at once, dual infeasibility.
- A nonnegative combination of rows reading $0 \leq$ negative proves infeasibility (Farkas); do not then assume the dual is unbounded, check it.
- One feasible point on each side proves both have finite optima (weak duality bounds both, Theorem 2.5.2 attains them).
- If you can find a feasible dual point, the primal cannot be unbounded, whatever else is true.

Classic problems

- *The primal is unbounded. Dual?* Infeasible.
- *The dual is infeasible. Primal?* Unbounded or infeasible; more information needed.
- *Both problems have feasible points. What follows?* Both have finite optima with equal values.
- $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1, x \geq 0$: write the dual and classify. Dual: $\min y_1$ s.t. $y_1 \geq 1, -y_1 \geq 1, y_1 \geq 0$: infeasible, consistent with the unbounded primal.

Gotcha

- “Unbounded feasible region” and “unbounded objective” are different statements; only the second is an outcome of the LP.
- “Infeasible $\Rightarrow$ partner unbounded” is false (Example 4.6.6). “Unbounded $\Rightarrow$ partner infeasible” is true.
- A bounded LP attains its optimum (Proposition 4.6.1); do not write “supremum” for an LP unless it is infeasible or unbounded.

Summary: Outcomes and the primal–dual table

- An LP is infeasible, unbounded, or has an attained finite optimum (Proposition 4.6.1).
- Finite $\leftrightarrow$ finite with equal values; unbounded $\Rightarrow$ partner infeasible; infeasible $\Rightarrow$ partner unbounded or infeasible (Theorem 4.6.5).
- The full verification recipe: write the dual, check primal feasibility, find a dual-feasible point with the same value (complementary slackness tells you which rows to make tight), conclude by weak duality.

Notes on the original slides

Every number in the Lecture 4 slides was recomputed: the three bounds 6, 5, $19/4$ for the workshop, the feasibility of $\hat{x} = (0, 3/2)$ and of $x^* = (1/2, 5/4)$, the dual feasibility of $(1/3, 1/3, 0)$ and $(5/16, 0, 1/4)$, the complementary-slackness derivation of y^* , the mixed-signs dual of slide 23, the dual-of-dual computation, the examples of the three outcomes, the both-infeasible pair, and the $(2, 2), (1, 1)$ certificate of slides 39–41. All are correct. Points worth a reader’s attention:

- **Strong duality (slide 18)** is stated without proof. Section 4.3 proves it from Fourier–Motzkin elimination and the Farkas lemma, following the route of Matouek and Gärtner, Chapter 6, and adds the constructive reading via the final simplex tableau (Proposition 4.3.5).
- **“Feasibility + equal values $\Rightarrow$ both optimal” (slide 17)** is a consequence of weak duality alone (Theorem 4.2.3); strong duality is only needed to know that such a pair *exists*.
- **Complementary slackness (slide 27)** is stated for the canonical pair. The “ $\Rightarrow$ ” direction uses strong duality; the “ $\Leftarrow$ ” direction uses only weak

duality. The remark after Theorem 4.5.3 extends it to general pairs.

- **Slide 32 (“read the slacks, impose, solve, verify”).** The verification step is essential: Example 4.5.6 shows the same procedure applied to a non-optimal point produces an infeasible candidate, which is how non-optimality is detected.
- **Three outcomes (slide 33).** “Exactly one occurs” silently uses that a bounded LP *attains* its optimum, which is Theorem 2.5.2; for non-linear problems this can fail.
- **Slide 37, “infeasible does not automatically mean the other is unbounded.”** Correct; the table’s fourth row is realised by slide 38, reproduced as Example 4.6.6. The missing cells (finite with non-finite, unbounded with unbounded) are impossible by Theorem 4.6.5.
- **Slide 24 (dual of the dual)** uses only the canonical rules plus $\min \leftrightarrow \max$ and sign reversal; the general dualization rules (slides 20–21) are proved in Theorems 4.4.2 and 4.4.3.

Chapter takeaways

Summary: Duality of linear programs

- **Two searches:** the primal searches for a good feasible decision; the dual searches for a good bound on what is possible. Rows become variables, columns become constraints.
- **Weak duality:** feasible decisions never beat valid bounds; a feasible pair with equal values is optimal, and that is a two-line proof.
- **Strong duality:** at a finite LP optimum the best bounds meet; simplex finds the exact certificate in its final objective row.
- **Dualization rules:** $\leq / \geq / =$ rows give $\geq 0 / \leq 0 /$ free variables; $\geq 0 / \leq 0 /$ free variables give $\geq / \leq / =$ rows. Each rule is forced by one sign requirement.
- **Complementary slackness:** spare capacity means zero weight, positive activity means tight dual row; use it to compute a certificate, then verify.
- **Outcomes:** finite pairs with finite, unbounded pairs with infeasible, and infeasible may pair with either; always check dimensions, directions and signs.

Chapter 5

The Simplex Tableau in Matrix Form and Sensitivity Analysis

5.1 A Simplex Refresher, and the Plan of the Chapter

Motivation

Chapter 3 ran the simplex method on tableaux, and Chapter 4 showed that the final objective row hides a dual solution. This chapter opens the tableau up and proves everything inside it. Part one rewrites a simplex move in matrix language: a *basic direction*, its *step length* and its *gain*, the reduced cost. Out of this come the optimality theorem, the ratio test, the basis exchange and termination, each with a complete proof. Part two uses the same matrices for *sensitivity analysis*: when a datum of an already-solved LP changes, does the old optimal basis survive, what is the new value, and how do we continue from the last tableau instead of starting over?

This chapter is self-contained. Every result it uses is stated and proved here, including those first proved in earlier chapters:

Result	First seen	Proved here as
The tableau of a basis; row operations keep solutions	Ch. 2, 3	Theorem 5.2.2
Ratio test and unboundedness	Ch. 3	Theorem 5.3.3
A pivot gives a new BFS	Ch. 3	Theorem 5.3.4
Optimality test and uniqueness	Ch. 3	Theorem 5.4.2, Corollary 5.4.3
Termination without degeneracy	Ch. 3	Theorem 5.4.5
An optimal BFS exists	Ch. 2	Theorem 5.4.8
Weak duality, matching pair	Ch. 4	Theorems 5.5.1, 5.5.2
Simplex gives the dual solution	Ch. 4	Theorem 5.5.3

Example 5.1.1 (The running example). Two divisible activities $x, y \geq 0$ earn 3 and 2 per unit and share two resources:

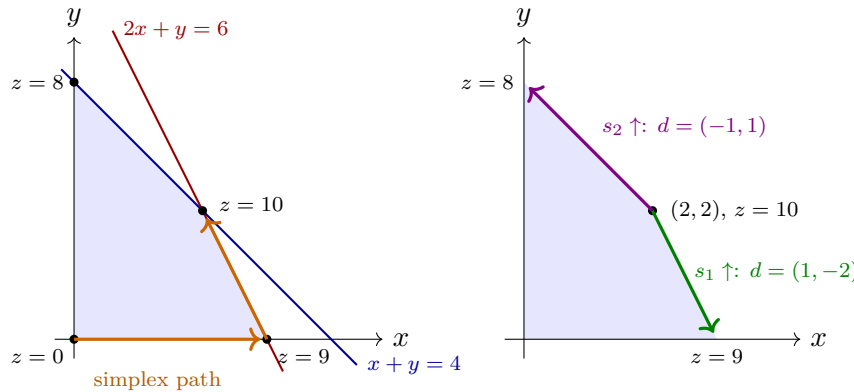
$$\max 3x + 2y \quad \text{s.t.} \quad x + y \leq 4, \quad 2x + y \leq 6, \quad x, y \geq 0.$$

With slacks $s_1, s_2 \geq 0$, simplex runs as in Chapter 3. The slides call the objective z , where Chapter 3 wrote v .

The tableau is not just bookkeeping. Every number in it is a piece of B^{-1} , and the whole chapter is about reading those numbers and proving why they mean what they mean.

Tableau 0 (basis s_1, s_2)	Tableau 1 (basis x, s_1)	Tableau 2 (basis x, y)
$s_1 = 4 - x - y$	$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2$	$x = 2 + s_1 - s_2$
$s_2 = 6 - 2x - y$	$s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2$	$y = 2 - 2s_1 + s_2$
$z = 0 + 3x + 2y$	$z = 9 + \frac{1}{2}y - \frac{3}{2}s_2$	$z = 10 - s_1 - s_2$
x in; ratios 4, 3; s_2 out	y in; ratios 6, 2; s_1 out	optimal: $z \leq 10$

The path is $(0, 0) \rightarrow (3, 0) \rightarrow (2, 2)$, with values $0 \rightarrow 9 \rightarrow 10$. In the last tableau every feasible plan has $s_1, s_2 \geq 0$, so $z = 10 - s_1 - s_2 \leq 10$, and the plan $(2, 2)$ is optimal.



The right-hand picture shows the two *basic directions* at the optimum, both downhill. Section 5.3 explains where they come from.

5.2 Bases and the Tableau of a Basis

The standing setting

Throughout, the LP is in equational form:

$$\max c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0, \quad P = \{x \in \mathbb{R}^n : Ax = b, \quad x \geq 0\},$$

where $A \in \mathbb{R}^{m \times n}$ has **full row rank** m . Redundant equations can always be deleted first (Chapter 2). For an index set S , x_S , c_S and A_S denote the entries of x and c , and the columns of A , with indices in S .

Definition 5.2.1 (Basis, BFS, nondegeneracy). Choose m column indices $B(1), \dots, B(m)$ whose columns form an invertible matrix $B = [A_{B(1)} \cdots A_{B(m)}]$ (a *basis*), and let N be the other, *nonbasic*, indices. The *basic solution* of B sets $x_i = 0$ for $i \in N$ and $x_B = B^{-1}b$. If $x_B \geq 0$ it is a *basic feasible solution* (BFS). It is *nondegenerate* if every component of x_B is strictly positive.

The slides use the same letter B for the list of indices and for the matrix; the context always tells which.

Every solution of the equations, written in the basis

The simplex tableau of Chapter 3 was built by pivots. The next theorem shows that it is really a formula in B^{-1} , and that it does not depend on the pivots used to reach it.

Once you fix the nonbasic variables, the basic ones are forced. That single fact is the whole content of a tableau.

Theorem 5.2.2 (The tableau of a basis). *Let B be a basis and define the reduced costs $r^\top := c^\top - c_B^\top B^{-1}A$. Then:*

- (a) *For every $x \in \mathbb{R}^n$: $Ax = b$ if and only if $x_B = B^{-1}b - B^{-1}A_N x_N$. In particular, x_N can be chosen freely, and x_B is then determined.*
- (b) *For every x with $Ax = b$: $c^\top x = c_B^\top B^{-1}b + \sum_{j \in N} r_j x_j$.*
- (c) *$r_j = 0$ for every basic index j .*
- (d) **Uniqueness.** *Suppose that, for some vector β , matrix Q , number z_0 and vector ρ , every solution of $Ax = b$ satisfies $x_B = \beta + Qx_N$ and $c^\top x = z_0 + \rho^\top x_N$. Then $\beta = B^{-1}b$, $Q = -B^{-1}A_N$, $z_0 = c_B^\top B^{-1}b$ and $\rho = r_N$.*

Proof. (a) Split the product by columns: $Ax = Bx_B + A_N x_N$. So $Ax = b$ is the same equation as $Bx_B = b - A_N x_N$. Because B is invertible, this holds exactly when $x_B = B^{-1}(b - A_N x_N)$.

(b) Split the objective the same way and substitute (a):

$$\begin{aligned} c^\top x &= c_B^\top x_B + c_N^\top x_N = c_B^\top B^{-1}b - c_B^\top B^{-1}A_N x_N + c_N^\top x_N \\ &= c_B^\top B^{-1}b + (c_N^\top - c_B^\top B^{-1}A_N)x_N. \end{aligned}$$

The j -th entry of the bracket is $c_j - c_B^\top B^{-1}A_j = r_j$.

(c) On the basic columns, $r_B^\top = c_B^\top - c_B^\top B^{-1}B = c_B^\top - c_B^\top = 0$.

(d) By (a), for every choice of x_N there is a solution of $Ax = b$. Take $x_N = 0$. Then (a) gives $x_B = B^{-1}b$, the hypothesis gives $x_B = \beta$, and the objective gives $c_B^\top B^{-1}b = z_0$. Next take $x_N = e_j$ (one nonbasic variable equal to 1, the others 0). Comparing the two formulas for x_B gives $\beta + Q_{\cdot j} = B^{-1}b - B^{-1}A_j$, so $Q_{\cdot j} = -B^{-1}A_j$. Comparing the two formulas for $c^\top x$ gives $\rho_j = r_j$. $\square$

Interpretation & discussion

- **Why a pivoted tableau equals this formula.** Each pivot of Chapter 3 solves one equation for one variable and substitutes the result into the other equations and the objective. Each such step is reversible (substitute back), so the system after the pivot has exactly the same solutions as before. The final tableau therefore holds for *every* solution of $Ax = b$, and by (d) its numbers must be the ones in the theorem. The tableau of a basis depends only on the basis, not on the path that reached it.
- **The identity in (b) holds for every solution of $Ax = b$,** feasible or not. This is what makes the objective row a proof of optimality, and it is used in almost every proof below.

Corollary 5.2.3 (What the tableau displays). *Write the tableau of B as $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj} x_j$ for $k = 1, \dots, m$, and $z = z_0 + \sum_{j \in N} r_j x_j$. Then:*

- (i) *the constants are $\beta = B^{-1}b = x_B$, and $z_0 = c_B^\top B^{-1}b$ is the value of the basic solution;*
- (ii) *the column of a nonbasic x_j in the basic rows is $q_{\cdot j} = -B^{-1}A_j$;*
- (iii) *the objective coefficients are the reduced costs r_j ;*
- (iv) *if the original problem has a slack s_i with column e_i and cost 0, then its tableau column is $-B^{-1}e_i$ (minus the i -th column of B^{-1}), and its*

reduced cost is $-p_i$, where $p^\top := c_B^\top B^{-1}$. When all slacks are nonbasic, the slack columns together display $-B^{-1}$.

Proof. (i)–(iii) are Theorem 5.2.2(d). For (iv), apply (ii) with $A_j = e_i$, and compute $r_{s_i} = 0 - c_B^\top B^{-1}e_i = -p_i$. $\square$

Example 5.2.4 (B^{-1} in the running example). In Tableau 2 the slack coefficients are, row by row, $(1, -1)$ for x and $(-2, 1)$ for y . Hence

$$-B^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix} \implies B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix},$$

for $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, the columns of x and y . Check: $B^{-1}b = (-4 + 6, 8 - 6) = (2, 2)$, the optimal plan. The slack reduced costs are $-1, -1$, so $p = (1, 1)$.

Gotcha

- **The sign.** Because the basic variables are isolated on the *left*, the tableau shows $-B^{-1}$, not B^{-1} , and $-p$, not p . Forgetting these minus signs is the most common error in this chapter.
- **Notation clash.** Chapter 3 called the tableau constants p_i . Here, following the slides, p is the *multiplier vector* $c_B^\top B^{-1}$, and the constants are β_k .

5.3 Moving Between Bases: Directions, Ratio Test, Exchange

Definition 5.3.1 (Feasible direction). A vector d is a *feasible direction* at $x \in P$ if $x + \theta d \in P$ for some $\theta > 0$.

One move: raise one nonbasic variable

At a BFS, pick a nonbasic $j \in N$ and let x_j rise from 0 to $\theta > 0$, keeping the other nonbasic variables at zero. So $d_j = 1$ and $d_i = 0$ for $i \in N \setminus \{j\}$. The basic variables must adjust, $x_B(\theta) = x_B + \theta d_B$, and the equations decide by how much.

You choose only one number, how much of x_j to use. The equations decide everything else.

Proposition 5.3.2 (The j th basic direction). *Staying on $Ax = b$ forces $Ad = 0$, and hence*

$$d_B = -B^{-1}A_j.$$

This $d = d^{(j)}$ is called the j th basic direction. Its basic part is exactly the column of x_j in the tableau (Corollary 5.2.3(ii)).

Proof. $A(x + \theta d) = b$ and $Ax = b$ give $\theta Ad = 0$, so $Ad = 0$ because $\theta > 0$. All nonbasic entries of d vanish except $d_j = 1$, so

$$0 = Ad = \sum_{k=1}^m A_{B(k)} d_{B(k)} + A_j = Bd_B + A_j.$$

Multiply by B^{-1} . $\square$

Theorem 5.3.3 (Moving along a basic direction: the ratio test). *Let x be a BFS with basis B , let $j \in N$ and $d = d^{(j)}$. For $\theta \geq 0$:*

(a) $A(x + \theta d) = b$;

(b) $x + \theta d \geq 0$ if and only if $\theta \leq \theta_{\max}$, where

$$\theta_{\max} = \min_{k: d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}, \quad \theta_{\max} = +\infty \text{ if } d_B \geq 0;$$

(c) $c^\top(x + \theta d) = c^\top x + \theta r_j$.

Consequently, if $r_j > 0$ and $d_B \geq 0$, the LP is **unbounded**. If $r_j > 0$ and $\theta_{\max} > 0$, moving to $x + \theta_{\max}d$ strictly increases the objective.

Proof. (a) $Ax = b$ and $Ad = 0$ (Proposition 5.3.2).

(b) Check each coordinate of $x + \theta d$.

- A nonbasic $i \neq j$: $0 + \theta \cdot 0 = 0$.
- The index j : $0 + \theta \geq 0$.
- A basic $B(k)$ with $d_{B(k)} \geq 0$: $x_{B(k)} + \theta d_{B(k)} \geq x_{B(k)} \geq 0$, so it imposes nothing.
- A basic $B(k)$ with $d_{B(k)} < 0$: $x_{B(k)} + \theta d_{B(k)} \geq 0$ if and only if $\theta \leq x_{B(k)}/(-d_{B(k)})$.

All coordinates are nonnegative exactly when θ satisfies every bound of the last kind, that is, when $\theta \leq \theta_{\max}$. If there is no such row, every $\theta \geq 0$ works.

(c) $c^\top d = c_j + c_B^\top d_B = c_j - c_B^\top B^{-1}A_j = r_j$.

Consequences. If $d_B \geq 0$, then by (a) and (b) $x + \theta d \in P$ for every $\theta \geq 0$, and by (c) its value $c^\top x + \theta r_j$ tends to $+\infty$ when $r_j > 0$. If $\theta_{\max} > 0$, the point $x + \theta_{\max}d$ is feasible and gains $\theta_{\max}r_j > 0$. $\square$

Key insight

Each basic row is a one-variable inequality $x_{B(k)} + \theta d_{B(k)} \geq 0$. The ratio test solves all of them and keeps the tightest. A zero ratio is allowed: if some $x_{B(k)} = 0$ with $d_{B(k)} < 0$, then $\theta_{\max} = 0$. This is degeneracy.

Theorem 5.3.4 (Basis exchange: the new point is a BFS). *In the setting of Theorem 5.3.3, suppose $\theta_{\max} < \infty$, and let ℓ be a row attaining the minimum, so $d_{B(\ell)} < 0$. Let B' be obtained from B by replacing the column $A_{B(\ell)}$ with A_j . Then B' is a basis, and $y := x + \theta_{\max}d$ is the BFS of B' .*

Proof. **Step 1 (B' is invertible).** Let $u := B^{-1}A_j = -d_B$. Since $B^{-1}A_{B(k)} = e_k$ for each k , the matrix $B^{-1}B'$ is the identity with its ℓ -th column replaced by u :

$$B^{-1}B' = [e_1 \cdots e_{\ell-1} \quad u \quad e_{\ell+1} \cdots e_m].$$

Row ℓ of this matrix is zero except for the entry u_ℓ on the diagonal, so expanding the determinant along row ℓ gives $\det(B^{-1}B') = u_\ell \cdot \det(I_{m-1}) = u_\ell = -d_{B(\ell)} > 0$. So $B^{-1}B'$ is invertible, and therefore so is $B' = B(B^{-1}B')$.

Step 2 (y is the basic solution of B'). The nonbasic indices of B' are $(N \setminus \{j\}) \cup \{B(\ell)\}$. For $i \in N \setminus \{j\}$, $y_i = 0$. Also $y_{B(\ell)} = x_{B(\ell)} + \theta_{\max}d_{B(\ell)} = 0$, by the choice of ℓ . So y is zero on every nonbasic index of B' , and $Ay = b$ by

Theorem 5.3.3(a). By Theorem 5.2.2(a) applied to B' , the only such vector is the basic solution of B' .

Step 3 (it is feasible). $y \geq 0$ by Theorem 5.3.3(b). $\square$

Interpretation & discussion

In tableau language this is the pivot of Chapter 3: x_j enters, $x_{B(\ell)}$ leaves, and the row of $x_{B(\ell)}$ is solved for x_j . That is possible because its coefficient $q_{\ell j} = d_{B(\ell)}^{(j)}$ is nonzero, which is exactly why B' is invertible.

Example 5.3.5 (Two basic directions at $(2, 2)$). Order the variables (x, y, s_1, s_2) , so $c = (3, 2, 0, 0)$. From Tableau 2, the basic directions are $d^{(s_1)} = (1, -2, 1, 0)$ and $d^{(s_2)} = (-1, 1, 0, 1)$.

j	$d_B^{(j)}$	$\theta_{\max}$	reaches	$r_j = c^\top d^{(j)}$	value there
s_1	$(1, -2)$	$2/2 = 1$ (y hits 0)	$(3, 0)$	$3 - 4 = -1$	$10 - 1 = 9$
s_2	$(-1, 1)$	$2/1 = 2$ (x hits 0)	$(0, 4)$	$-3 + 2 = -1$	$10 - 2 = 8$

Both moves lose 1 per unit, matching $z = 10 - s_1 - s_2$. As Theorem 5.3.4 predicts, moving along $d^{(s_1)}$ lands on the BFS of the basis $\{x, s_1\}$, which is Tableau 1, with y leaving.

5.4 Reduced Costs, Optimality and Termination

The gain along a basic direction

$$r_j = c^\top d^{(j)} = \underbrace{c_j}_{\text{direct gain}} - \underbrace{c_B^\top B^{-1} A_j}_{\text{cost of adjusting the basis}}.$$

Using one unit of x_j earns c_j directly, but forces the basic activities to shift, and that shift costs $c_B^\top B^{-1} A_j$.

Definition 5.4.1 (Reduced cost, multipliers, optimal basis). For a basis B :

- the *reduced costs* are $r^\top = c^\top - c_B^\top B^{-1} A$;
- the *multipliers* are $p^\top := c_B^\top B^{-1}$, so that $r_j = c_j - p^\top A_j$;
- B is an *optimal basis matrix* if $B^{-1}b \geq 0$ (feasibility) and $r \leq 0$ (optimality).

By Theorem 5.2.2, r_j is the coefficient of x_j in the objective row, and $r_B = 0$.

Theorem 5.4.2 (Reduced costs and optimality, maximisation). Let x be a BFS of $\max\{c^\top x : Ax = b, x \geq 0\}$ with basis matrix B .

1. If $r_j \leq 0$ for all $j \in N$, then x is optimal.
2. If x is optimal and nondegenerate, then $r_j \leq 0$ for all $j \in N$.

Proof. (1) Let u be any feasible plan. Theorem 5.2.2(b) applies to both u and x . Since $x_N = 0$:

$$c^\top u - c^\top x = \left(c_B^\top B^{-1} b + \sum_{j \in N} r_j u_j \right) - c_B^\top B^{-1} b = \sum_{j \in N} r_j u_j.$$

Each term is a nonpositive r_j times a nonnegative u_j , so the sum is ≤ 0 . Hence $c^\top u \leq c^\top x$ for every feasible u .

(2) Suppose, for contradiction, that $r_j > 0$ for some $j \in N$. Nondegeneracy means every $x_{B(k)} > 0$, so every ratio in Theorem 5.3.3(b) is positive and $\theta_{\max} > 0$. Then $x + \theta_{\max} d^{(j)}$ is feasible with value $c^\top x + \theta_{\max} r_j > c^\top x$, which contradicts the optimality of x . $\square$

Key insight

The identity $c^\top u - c^\top x = \sum_{j \in N} r_j u_j$ is the objective row of the tableau in matrix form, and it holds for every solution of $Au = b$. Part (1) only reads its signs. Part (2) is the contrapositive of “a positive reduced cost gives an improving move”, and that move needs room, which is what nondegeneracy provides.

Corollary 5.4.3 (Uniqueness). *If $r_j < 0$ for every $j \in N$, then x is the unique optimal solution.*

Proof. Let u be feasible with $c^\top u = c^\top x$. Then $\sum_{j \in N} r_j u_j = 0$, a sum of nonpositive terms, so every term is zero. Since every $r_j < 0$, this forces $u_j = 0$ for all $j \in N$. Then Theorem 5.2.2(a) gives $u_B = B^{-1}b = x_B$, so $u = x$. $\square$

Example 5.4.4 (Why part (2) needs nondegeneracy). Take $\max x_1$ subject to $x_1 + s = 0$, with $x_1, s \geq 0$. The only feasible point is $(0, 0)$, so it is optimal. The basis $\{s\}$ gives the degenerate BFS $s = 0$ and the tableau $s = 0 - x_1$, $z = 0 + x_1$, so $r_{x_1} = +1 > 0$. Yet no improvement is possible, because $\theta_{\max} = 0/1 = 0$. The basis $\{x_1\}$ describes the same point with $z = 0 - s$, which is a valid certificate. At a degenerate optimum, *some* basis certifies optimality, but not necessarily the one you are holding.

The simplex method, and why it stops

Theorem 5.4.5 (Termination without degeneracy). *Start from a feasible basis and repeat:*

- if $r_N \leq 0$, stop: the BFS is optimal;
- otherwise pick j with $r_j > 0$; if $d_B^{(j)} \geq 0$, stop: the LP is unbounded;
- otherwise exchange bases as in Theorem 5.3.4.

Each exchange changes the objective by $\theta_{\max} r_j \geq 0$. If every BFS met along the way is nondegenerate, the method stops after finitely many exchanges, with an optimal basis matrix or a proof of unboundedness.

Proof. The two stopping rules are correct by Theorem 5.4.2(1) and Theorem 5.3.3. By Theorem 5.3.3(c), an exchange changes the value by $\theta_{\max} r_j$, which is ≥ 0 . Under nondegeneracy $\theta_{\max} > 0$, so the value increases *strictly*. A basis determines its BFS, and hence its value. A strictly increasing sequence of values can never return to a basis already visited. There are at most $\binom{n}{m}$ bases, so the method stops. $\square$

Remark 5.4.6. Under degeneracy, $\theta_{\max} = 0$ is possible, the value can stall, and simplex can in principle cycle. Anti-cycling rules (for example Bland’s smallest-index rule) restore termination. Their proof is beyond this course; the theorems of this chapter never depend on them.

Optimal vertices exist

The sensitivity results about the optimal *value* need one more fact from Chapter 2: a feasible, bounded LP always has an optimal BFS.

Lemma 5.4.7 (Purification). *Suppose $c^\top x$ is bounded above on P . Then for every $x \in P$ there is a BFS x' with $c^\top x' \geq c^\top x$.*

If a feasible point is not a vertex, slide along a direction that keeps the equations and does not lose value, until a coordinate hits zero. Repeat until you reach a vertex.

Proof. By induction on the number of positive entries of x . Let $S = \{j : x_j > 0\}$.

Case 1: the columns A_S are linearly independent. Since $\text{rank } A = m$, these columns can be completed by further columns of A to m independent ones, which form a basis $B \supseteq S$. Then x is zero outside B and satisfies $Ax = b$. By Theorem 5.2.2(a), x is the basic solution of B , and it is feasible. Take $x' = x$.

Case 2: the columns A_S are dependent. Then there is $d \neq 0$ with $Ad = 0$ and $d_j = 0$ for $j \notin S$. Replacing d by $-d$ if necessary, $c^\top d \geq 0$. We may also assume that d has a negative entry:

- if $c^\top d > 0$ and $d \geq 0$, then $x + \theta d \in P$ for all $\theta \geq 0$ with value tending to $+\infty$, contradicting boundedness;
- if $c^\top d = 0$ and $d \geq 0$, replace d by $-d$.

Now let $\theta^* = \min\{x_j/(-d_j) : d_j < 0\}$. This is positive, because $d_j < 0$ only for $j \in S$, where $x_j > 0$. The point $x'' = x + \theta^* d$ satisfies:

- $Ax'' = b$;
- $x'' \geq 0$, by the same row-by-row argument as the ratio test;
- $c^\top x'' \geq c^\top x$;
- it has fewer positive entries than x , since no new positive entry appears outside S and the minimising index becomes 0.

By induction there is a BFS x' with $c^\top x' \geq c^\top x'' \geq c^\top x$. □

Theorem 5.4.8 (An optimal BFS exists). *If $P \neq \emptyset$ and $c^\top x$ is bounded above on P , the LP has an optimal solution that is a BFS. Moreover, its optimal value is the largest value $c^\top x$ among the finitely many BFSs.*

Proof. Applying Lemma 5.4.7 to any point of P shows that at least one BFS exists. There are finitely many bases, hence finitely many BFSs. Let x^* be a BFS of largest value. For any $x \in P$, the lemma gives a BFS x' with $c^\top x \leq c^\top x' \leq c^\top x^*$. So x^* is optimal. □

5.5 The Multipliers: Dual Solution and Shadow Prices

Duality, re-proved

Consider the canonical pair of Chapter 4:

$$(P) \quad \max c^\top x \quad \text{s.t.} \quad Ax \leq b, x \geq 0, \quad (D) \quad \min b^\top y \quad \text{s.t.} \quad A^\top y \geq c, y \geq 0.$$

In equational form, (P) adds slacks: $[A \ I](x, s) = b$, $(x, s) \geq 0$, with cost vector $(c, 0)$.

Theorem 5.5.1 (Weak duality). *For every feasible x of (P) and every feasible y of (D) , $c^\top x \leq b^\top y$.*

Proof. Insert the middle term $y^\top Ax$ and use one sign condition on each side:

$$c^\top x \leq (A^\top y)^\top x = y^\top Ax \leq y^\top b.$$

The first inequality holds because $A^\top y \geq c$ and $x \geq 0$. The second holds because $Ax \leq b$ and $y \geq 0$. Both use the same fact: if $u \leq v$ componentwise and $w \geq 0$, then $w^\top u \leq w^\top v$. $\square$

Theorem 5.5.2 (A matching pair is optimal). *If x is feasible for (P) , y is feasible for (D) , and $c^\top x = b^\top y$, then both are optimal.*

Proof. For every feasible x' , $c^\top x' \leq b^\top y = c^\top x$ by weak duality, so x is optimal. For every feasible y' , $b^\top y' \geq c^\top x = b^\top y$, so y is optimal. $\square$

Theorem 5.5.3 (The multipliers are an optimal dual solution). *Let B be an optimal basis matrix of the equational form of (P) , with BFS x^* . Then $p^\top = c_B^\top B^{-1}$ is optimal for (D) , and $b^\top p = c^\top x^*$.*

Proof. Step 1 (dual feasibility from $r \leq 0$). For an original column A_j , $r_j = c_j - p^\top A_j \leq 0$, so $(A^\top p)_j \geq c_j$; together, $A^\top p \geq c$. For the slack s_i , with column e_i and cost 0, $r_{s_i} = -p_i \leq 0$, so $p \geq 0$.

Step 2 (equal values). $b^\top p = c_B^\top B^{-1}b = c_B^\top x_B^* = c^\top x^*$, since $x_N^* = 0$ and slacks cost 0.

Step 3. Theorem 5.5.2. $\square$

This is Proposition 4.3.5 of Chapter 4 again, where $y^* = -r_s$. Indeed $r_s = -p$ by Corollary 5.2.3(iv). In the running example $p = (1, 1)$, with dual value $4 + 6 = 10$, equal to the primal optimum.

Corollary 5.5.4 (Complementary slackness at an optimal basis). *With B as in Theorem 5.5.3:*

- (a) *if the slack s_i is basic, then $p_i = 0$. Equivalently, $p_i > 0$ forces s_i nonbasic, so $s_i = 0$: the constraint is binding;*
- (b) *if x_j is basic, then $(A^\top p)_j = c_j$: the j -th dual constraint is tight.*

Proof. Basic reduced costs are zero (Theorem 5.2.2(c)). For s_i this reads $-p_i = 0$; for x_j it reads $c_j - (A^\top p)_j = 0$. $\square$

Interpretation & discussion

A resource with spare capacity is worth nothing at the margin. If s_i is basic, resource i is not fully used, and one more unit of it cannot help. Section 5.8 proves that p_i is exactly the value of one more unit of resource i , its *shadow price*, as long as the basis stays optimal.

Summary: Simplex in matrix form

- Every solution of $Ax = b$ satisfies $x_B = B^{-1}b - B^{-1}A_N x_N$ and $c^\top x = c_B^\top B^{-1}b + r_N^\top x_N$. The tableau is this formula, and it is unique.
- Basic direction: $d_B = -B^{-1}A_j$; step $\theta_{\max}$ by the ratio test; gain r_j per unit; the reached point is the BFS of the exchanged basis.
- $r_N \leq 0$ proves optimality, and strict signs prove uniqueness; the converse

needs nondegeneracy. Without degeneracy, simplex terminates.

- $p^\top = c_B^\top B^{-1}$ is the optimal dual solution; a basic slack has $p_i = 0$.

5.6 Sensitivity Analysis: The Framework

Motivation

A factory has its optimal plan. Then the questions arrive. Is one more hour of assembly worth buying, and at what price? A product's profit changes: is the plan still best? A new product is proposed: should we make it? Re-solving from scratch would work, but the final tableau already contains B^{-1} , p and the reduced costs. *Sensitivity analysis* reuses them to answer each question locally, around the known optimum.

We start from an optimal basis matrix B of $\max\{c^\top x : Ax = b, x \geq 0\}$, with optimum x^* ($x_B^* = B^{-1}b$, $x_N^* = 0$) and multipliers $p^\top = c_B^\top B^{-1}$.

Optimising once is rarely the end. The data move, and the question is whether yesterday's answer still holds.

Proposition 5.6.1 (Which test each change can break). (a) Changing b leaves B , B^{-1} , p and every reduced cost unchanged. Only the feasibility test $B^{-1}b \geq 0$ can fail.

(b) Changing c leaves $B^{-1}b$ unchanged, so x^* stays feasible. Only the optimality test $r \leq 0$ can fail.

(c) Adding a new variable with column a leaves B a basis, keeps x^* (extended by 0) as its BFS, and keeps the old reduced costs. It adds one new reduced cost.

In every case, B is optimal for the new data exactly when both tests hold for the new data (Theorem 5.4.2(1) and Definition 5.4.1).

Proof. (a) $r^\top = c^\top - c_B^\top B^{-1}A$ and $p^\top = c_B^\top B^{-1}$ do not involve b . (b) $B^{-1}b$ does not involve c . (c) Adding a column adds no equation, so B is still an invertible $m \times m$ matrix of columns of the new constraint matrix. Its basic solution is x^* with $u = 0$, and $r_j = c_j - p^\top A_j$ for the old columns is unchanged. $\square$

Change	What moves in the tableau	Test that can fail	If it fails
New variable (a, γ)	a new column	$r_u = \gamma - p^\top a \leq 0$	primal simplex pivot
$b \rightarrow b + \Delta b$	constants and z_0	$B^{-1}(b + \Delta b) \geq 0$	dual simplex pivot
$c \rightarrow c + \Delta c$	objective row only	$r' \leq 0$	primal simplex pivot

5.7 Case 1: A New Activity

Proposition 5.7.1 (Adding a variable). Add $u \geq 0$ with column $a \in \mathbb{R}^m$ and unit value γ . For the old basis B :

(a) $(x^*, 0)$ is still its BFS, and the new reduced cost is $r_u = \gamma - p^\top a$;

(b) if $r_u \leq 0$, then $(x^*, 0)$ is optimal for the enlarged LP;

(c) if $r_u > 0$ and x^* is nondegenerate, then $(x^*, 0)$ is not optimal, and u is

Price the new product at the shadow prices of the resources it would consume. Make it only if it earns more than that.

an improving entering variable. Its tableau column is $-B^{-1}a$.

Proof. (a) Proposition 5.6.1(c), plus $r_u = \gamma - c_B^\top B^{-1}a = \gamma - p^\top a$. (b) All reduced costs are now ≤ 0 ; apply Theorem 5.4.2(1). (c) Theorem 5.4.2(2) and Theorem 5.3.3. The column is Corollary 5.2.3(ii) with $A_j = a$. $\square$

Interpretation. $p^\top a$ is the value, at the current shadow prices, of the resources one unit of u would use. The new activity is worth considering only if it earns more than the resources it diverts, that is, $\gamma > p^\top a$.

Example 5.7.2 (A new activity using one unit of each resource). In the running example, $a = (1, 1)$ and $p = (1, 1)$, so $p^\top a = 2$. If $\gamma = 1.5$, then $r_u = -0.5$ and the old plan stays optimal. If $\gamma = 2.5$, then $r_u = +0.5$, and the basis loses its certificate.

Continue the old tableau. Append the column $-B^{-1}a = -(0, 1) = (0, -1)$ to the basic rows and $r_u = +\frac{1}{2}$ to the objective row:

$$x = 2 + s_1 - s_2 + 0u, \quad y = 2 - 2s_1 + s_2 - u, \quad z = 10 - s_1 - s_2 + \frac{1}{2}u.$$

u enters, and $y = 2 - u$ limits it to $u \leq 2$, so y leaves. Solving the y row for u and substituting:

$$x = 2 + s_1 - s_2, \quad u = 2 - 2s_1 + s_2 - y, \quad z = 11 - 2s_1 - \frac{1}{2}s_2 - \frac{1}{2}y.$$

All objective coefficients are ≤ 0 , so the new optimum is $(x, y, u) = (2, 0, 2)$ with $z = 11$. The new activity replaces y completely.

5.8 Case 2: A Resource Level Changes

Proposition 5.8.1 (Feasibility range and shadow prices). *Let $b' = b + \Delta b$, with A and c fixed.*

(a) *B is an optimal basis matrix for b' if and only if $B^{-1}(b + \Delta b) \geq 0$.*

(b) *In that case the optimum is $x'_B = B^{-1}(b + \Delta b)$, $x'_N = 0$, with value*

$$v(b + \Delta b) = c_B^\top B^{-1}(b + \Delta b) = v(b) + p^\top \Delta b.$$

So p_i is the shadow price of resource i : the change in the optimal value per extra unit of b_i , valid while B stays feasible.

Proof. (a) By Proposition 5.6.1(a), the reduced costs are unchanged and still ≤ 0 . So the only remaining condition in Definition 5.4.1 is $B^{-1}b' \geq 0$. (b) Theorem 5.4.2(1) gives optimality of the basic solution. Its value is $c_B^\top B^{-1}b' = c_B^\top B^{-1}b + c_B^\top B^{-1}\Delta b = v(b) + p^\top \Delta b$. $\square$

Proposition 5.8.2 (Ranging a right-hand side). *Let $\Delta b = \delta e_i$, and let $g = B^{-1}e_i$ be the i -th column of B^{-1} , which is minus the tableau column of s_i . Then B stays optimal exactly for $\delta_{\min} \leq \delta \leq \delta_{\max}$, where*

$$\delta_{\max} = \min_{k: g_k < 0} \frac{x_{B(k)}^*}{-g_k}, \quad \delta_{\min} = \max_{k: g_k > 0} \left(-\frac{x_{B(k)}^*}{g_k} \right),$$

with $\delta_{\max} = +\infty$ (or $\delta_{\min} = -\infty$) when no such k exists. The interval always

contains 0.

Proof. $B^{-1}(b + \delta e_i) = x_B^* + \delta g$, so by Proposition 5.8.1(a) the condition is $x_{B(k)}^* + \delta g_k \geq 0$ for every k . A row with $g_k = 0$ always holds. A row with $g_k < 0$ means $\delta \leq x_{B(k)}^*/(-g_k)$. A row with $g_k > 0$ means $\delta \geq -x_{B(k)}^*/g_k$. Intersecting all the rows gives the interval. It contains 0 because $x_B^* \geq 0$ makes every upper bound ≥ 0 and every lower bound ≤ 0 . $\square$

Example 5.8.3 (Resource 1 becomes $4 + \delta$). Here $g = B^{-1}e_1 = (-1, 2)$ and $x_B^* = (2, 2)$. The first row gives $\delta \leq 2/1 = 2$, and the second gives $\delta \geq -2/2 = -1$. So $-1 \leq \delta \leq 2$, and on that range $v(\delta) = 10 + p_1\delta = 10 + \delta$. In tableau form, only the constants and z_0 move:

$$x = (2 - \delta) + s_1 - s_2, \quad y = (2 + 2\delta) - 2s_1 + s_2, \quad z = (10 + \delta) - s_1 - s_2.$$

Beyond the range: the dual simplex step

Outside the range, some basic variable is negative, while every reduced cost is still ≤ 0 . The slides fix this with one pivot, choosing the *leaving* variable first. This is the *dual simplex* method.

The objective row is still a valid upper bound; only feasibility broke. So pivot to repair feasibility while keeping the objective row nonpositive.

Theorem 5.8.4 (The dual simplex step). Consider a tableau $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj}x_j$, $z = z_0 + \sum_{j \in N} r_jx_j$, with all $r_j \leq 0$.

- (a) **Upper bound.** Every feasible x has $c^\top x \leq z_0$. If moreover all $\beta_k \geq 0$, the basis is optimal.
- (b) **Infeasibility certificate.** If some $\beta_\ell < 0$ and $q_{\ell j} \leq 0$ for every $j \in N$, the LP is infeasible.
- (c) **The pivot.** If $\beta_\ell < 0$ and some $q_{\ell j} > 0$, let j^* minimise $(-r_j)/q_{\ell j}$ over the j with $q_{\ell j} > 0$, and pivot: x_{j^*} enters and $x_{B(\ell)}$ leaves. In the new tableau every reduced cost is still ≤ 0 , the constant of x_{j^*} is positive, and the new value z'_0 satisfies $z'_0 \leq z_0$, strictly if $r_{j^*} < 0$.

Proof. (a) The objective row holds for every solution of $Ax = b$, by Theorem 5.2.2(b). For a feasible x , each $r_jx_j \leq 0$, so $c^\top x \leq z_0$. If also $\beta \geq 0$, the basis passes both tests.

(b) Every solution of $Ax = b$ satisfies the row $x_{B(\ell)} = \beta_\ell + \sum_j q_{\ell j}x_j$. If $x \geq 0$, each term $q_{\ell j}x_j$ is ≤ 0 , so $x_{B(\ell)} \leq \beta_\ell < 0$. This contradicts $x_{B(\ell)} \geq 0$, so no feasible x exists.

(c) Write $q := q_{\ell j^*} > 0$ and $r := r_{j^*} \leq 0$. The new basis is valid, since $q \neq 0$ (Theorem 5.3.4, Step 1). Solving row ℓ for x_{j^*} gives

$$x_{j^*} = -\frac{\beta_\ell}{q} + \frac{1}{q}x_{B(\ell)} - \sum_{k \neq j^*} \frac{q_{\ell k}}{q}x_k,$$

whose constant $-\beta_\ell/q$ is positive. Substituting into the objective row:

$$z = \left(z_0 - \frac{r\beta_\ell}{q}\right) + \frac{r}{q}x_{B(\ell)} + \sum_{k \in N, k \neq j^*} \left(r_k - \frac{r q_{\ell k}}{q}\right)x_k.$$

- **New constant:** $-r\beta_\ell/q = (-r)\beta_\ell/q \leq 0$, because $-r \geq 0$, $\beta_\ell < 0$ and $q > 0$. So $z'_0 \leq z_0$, strictly if $r < 0$.

- *Coefficient of $x_{B(\ell)}$* : $r/q \leq 0$.
- *Coefficient of x_k with $q_{\ell k} > 0$* : it equals $q_{\ell k}[(-r)/q - (-r_k)/q_{\ell k}]$, which is ≤ 0 because j^* minimised the ratio.
- *Coefficient of x_k with $q_{\ell k} \leq 0$* : it equals $r_k + (-r)q_{\ell k}/q$, a sum of two terms that are each ≤ 0 . $\square$

Interpretation & discussion

The primal simplex keeps *feasibility* and climbs the value until the reduced costs are nonpositive. The dual simplex keeps *nonpositive reduced costs*, so z_0 is always an upper bound by (a), and lowers that bound until feasibility is restored. If every pivot has $r_{j^*} < 0$, the bound strictly decreases, no basis repeats, and the method stops, exactly as in Theorem 5.4.5. This is the natural tool after a change of b , because such a change never breaks the reduced costs.

Example 5.8.5 (The whole value curve). $\delta > 2$: $x = 2 - \delta < 0$. In its row, only s_1 has a positive coefficient, so s_1 enters and x leaves:

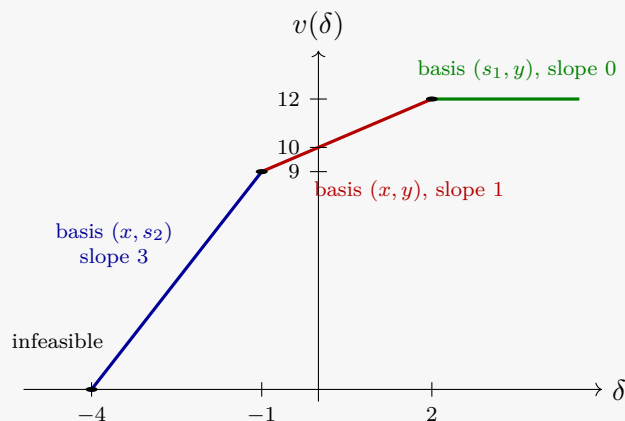
$$s_1 = (\delta - 2) + x + s_2, \quad y = 6 - 2x - s_2, \quad z = 12 - x - 2s_2.$$

This is feasible for $\delta \geq 2$, with $v = 12$. The bound fell from $10 + \delta > 12$ to 12, as part (c) predicts. Since s_1 is now basic, $p_1 = 0$ (Corollary 5.5.4): extra units of resource 1 are worthless.

$\delta < -1$: $y = 2 + 2\delta < 0$. Only s_2 has a positive coefficient in its row, so s_2 enters and y leaves:

$$s_2 = (-2 - 2\delta) + y + 2s_1, \quad x = (4 + \delta) - y - s_1, \quad z = (12 + 3\delta) - y - 3s_1.$$

This is feasible for $-4 \leq \delta \leq -1$, with $v = 3(4 + \delta)$. For $\delta < -4$, the row $x = (4 + \delta) - y - s_1$ has a negative constant and no positive coefficient, so by part (b) the LP is infeasible.



Theorem 5.8.6 (The optimal value as a function of b). Let $v(b)$ denote the optimal value of $\max\{c^\top x : Ax = b, x \geq 0\}$ when it is finite.

- (a) **Concavity.** If $v(b_1)$ and $v(b_2)$ are finite and $0 \leq \lambda \leq 1$, then the LP at $b_\lambda = \lambda b_1 + (1 - \lambda)b_2$ is feasible, and $v(b_\lambda) \geq \lambda v(b_1) + (1 - \lambda)v(b_2)$ whenever $v(b_\lambda)$ is finite.

(b) **Finitely many linear pieces.** Whenever $v(b)$ is finite,

$$v(b) = \max\{c_B^\top B^{-1}b : B \text{ a basis with } B^{-1}b \geq 0\}.$$

So at every such b , v coincides with one of the finitely many linear functions $b \mapsto p_B^\top b$, where $p_B^\top = c_B^\top B^{-1}$.

(c) **Slopes are shadow prices.** On an interval of δ where a single basis B stays optimal for $b + \delta e_i$, v is linear with slope $(p_B)_i$.

Proof. (a) By Theorem 5.4.8 there are optimal x_1, x_2 for b_1, b_2 . Let $x_\lambda = \lambda x_1 + (1 - \lambda)x_2$. It is ≥ 0 , and $Ax_\lambda = \lambda b_1 + (1 - \lambda)b_2 = b_\lambda$, so it is feasible for b_λ . Its value is $\lambda v(b_1) + (1 - \lambda)v(b_2)$, and the optimum at b_λ is at least that.

(b) Each feasible basis gives a feasible point of value $c_B^\top B^{-1}b$, so the maximum is $\leq v(b)$. By Theorem 5.4.8 some BFS is optimal, so the maximum is attained and equals $v(b)$. There are finitely many bases.

(c) Proposition 5.8.1(b) with $\Delta b = \delta e_i$. $\square$

Interpretation & discussion

A concave function made of linear pieces has *decreasing* slopes. In the example the slopes are $3 \rightarrow 1 \rightarrow 0$: each extra unit of resource 1 is worth less as it becomes more plentiful. This is the economics of diminishing returns, and here it is a theorem. A shadow price is a *local* statement, valid only on its range.

5.9 Case 3: A Profit Changes

Proposition 5.9.1 (Changing the objective). Let $c' = c + \Delta c$, with A and b fixed, and write the tableau rows as $x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj}x_j$.

- (a) x^* stays feasible, and the new reduced costs are $(r')^\top = r^\top + \Delta c^\top - \Delta c_B^\top B^{-1}A$.
- (b) **Nonbasic change.** If only c_j ($j \in N$) changes, by t , then $r'_j = r_j + t$ and every other reduced cost is unchanged. B stays optimal if and only if $t \leq -r_j$.
- (c) **Basic change.** If only $c_{B(k)}$ changes, by ε , then $z'_0 = z_0 + \varepsilon\beta_k$ and $r'_j = r_j + \varepsilon q_{kj}$ for $j \in N$. In words: add ε times the row of $x_{B(k)}$ to the objective row.

Proof. (a) Feasibility is Proposition 5.6.1(b). For the reduced costs, expand $(c + \Delta c)^\top - (c_B + \Delta c_B)^\top B^{-1}A$ and group the terms containing c into $r^\top$.

(b) $\Delta c_B = 0$, so $r' = r + \Delta c$, which changes only entry j . Then $r' \leq 0$ holds exactly when $r_j + t \leq 0$.

(c) For every solution of $Ax = b$, the new objective is $(c')^\top x = c^\top x + \varepsilon x_{B(k)}$. Substitute the objective row and the row of $x_{B(k)}$, both valid for all solutions (Theorem 5.2.2):

$$(c')^\top x = \left(z_0 + \sum_{j \in N} r_j x_j\right) + \varepsilon \left(\beta_k + \sum_{j \in N} q_{kj} x_j\right) = (z_0 + \varepsilon\beta_k) + \sum_{j \in N} (r_j + \varepsilon q_{kj}) x_j.$$

By the uniqueness in Theorem 5.2.2(d), applied to c' , these are the new constant and reduced costs. $\square$

Corollary 5.9.2 (Ranging a basic profit). *In case (c), B stays optimal exactly for $\varepsilon_{\min} \leq \varepsilon \leq \varepsilon_{\max}$, where*

$$\varepsilon_{\max} = \min_{j: q_{kj} > 0} \frac{-r_j}{q_{kj}}, \quad \varepsilon_{\min} = \max_{j: q_{kj} < 0} \frac{-r_j}{q_{kj}}.$$

Since $r_j \leq 0$, $\varepsilon_{\max} \geq 0 \geq \varepsilon_{\min}$. The plan x^ stays optimal on this interval, with value $z_0 + \varepsilon\beta_k$.*

Proof. Require $r_j + \varepsilon q_{kj} \leq 0$ for each $j \in N$. For $q_{kj} > 0$ this gives $\varepsilon \leq -r_j/q_{kj}$, and for $q_{kj} < 0$ it gives $\varepsilon \geq -r_j/q_{kj}$; the case $q_{kj} = 0$ always holds. Intersect. $\square$

Example 5.9.3 (The profit of x becomes $3+\varepsilon$). The row of x is $x = 2 + s_1 - s_2$, so $q = (1, -1)$ on (s_1, s_2) , while $r = (-1, -1)$. Then $\varepsilon_{\max} = 1/1 = 1$ and $\varepsilon_{\min} = 1/(-1) = -1$. Adding $\varepsilon \times$ the row:

$$z' = (10 + 2\varepsilon) + (\varepsilon - 1)s_1 - (1 + \varepsilon)s_2,$$

so $[-1 \leq \varepsilon \leq 1]$, with value $10 + 2\varepsilon$. This agrees with $p^\top = (3 + \varepsilon, 2)B^{-1} = (1 - \varepsilon, 1 + \varepsilon)$.

At $\varepsilon = 2$: $z' = 14 + s_1 - 3s_2$. Now s_1 enters, and $y = 2 - 2s_1$ limits it to $s_1 \leq 1$, so y leaves:

$$x = 3 - \frac{1}{2}y - \frac{1}{2}s_2, \quad s_1 = 1 - \frac{1}{2}y + \frac{1}{2}s_2, \quad z' = 15 - \frac{1}{2}y - \frac{5}{2}s_2.$$

The new optimum is $(3, 0)$, with value 15.

Interpretation & discussion

Geometry. Both x and y are basic, so Corollary 5.5.4(b) gives $c' = A^\top p'$, that is $(3 + \varepsilon, 2) = p'_1(1, 1) + p'_2(2, 1)$. This writes the objective as a combination of the normals of the two binding constraints, with weights p' . Optimality of the basis is $p' \geq 0$, so the vertex $(2, 2)$ is optimal exactly when the objective vector lies in the cone spanned by those normals. Solving gives $p' = (1 - \varepsilon, 1 + \varepsilon)$, nonnegative for $-1 \leq \varepsilon \leq 1$. At $\varepsilon = \pm 1$ the iso-profit line lies along an edge, and two vertices are optimal.

Proposition 5.9.4 (The optimal value as a function of c). *Fix A , b with $P \neq \emptyset$, and let $w(c)$ be the optimal value when finite. Then $w(c) = \max\{c^\top x : x \text{ a BFS}\}$. So w is a maximum of finitely many linear functions of c : it is **convex** and piecewise linear. On a range where one BFS x^* stays optimal, $w(c + \varepsilon e_k) = w(c) + \varepsilon x_k^*$.*

Proof. The formula is Theorem 5.4.8. A maximum of linear functions f_i is convex: $\max_i f_i(\lambda c + (1 - \lambda)c') \leq \lambda \max_i f_i(c) + (1 - \lambda) \max_i f_i(c')$, because each f_i is linear. The last claim is $(c + \varepsilon e_k)^\top x^* = c^\top x^* + \varepsilon x_k^*$. $\square$

In the example, $w = 10 + 2\varepsilon$ (slope $x^* = 2$) up to $\varepsilon = 1$, then $w = 3(3 + \varepsilon) = 9 + 3\varepsilon$ (slope $x^* = 3$, value 15 at $\varepsilon = 2$). The slopes increase, as convexity requires. The value is concave in b but convex in c .

5.10 A Complete Example: Production Planning

Example 5.10.1 (The workshop's three computers). Models A, B and C use assembly time and memory boards:

	A	B	C	Available
Assembly time	1	1	1	8
Memory boards	2	1	1	10
Profit per unit	4	3	2	

$$\begin{aligned} \max \quad & 4x_A + 3x_B + 2x_C \\ \text{s.t.} \quad & x_A + x_B + x_C + s_1 = 8, \quad 2x_A + x_B + x_C + s_2 = 10, \quad x, s \geq 0. \end{aligned}$$

Solve once. The optimal basis is (x_A, x_B) , with

$$B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}, \quad B^{-1}b = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \geq 0,$$

and $p^\top = (4, 3)B^{-1} = (2, 1)$. The reduced costs are $r_C = 2 - (2 + 1) = -1$, $r_{s_1} = -2$ and $r_{s_2} = -1$, all < 0 . So by Corollary 5.4.3, $(x_A, x_B, x_C) = (2, 6, 0)$ is the unique optimum, with profit 26. Both resources are used up ($2 + 6 = 8$, $4 + 6 = 10$), consistent with $p > 0$ (Corollary 5.5.4). The final tableau, to be reused, is

$$x_A = 2 + 0x_C + s_1 - s_2, \quad x_B = 6 - x_C - 2s_1 + s_2, \quad z = 26 - x_C - 2s_1 - s_2.$$

The shadow prices are $p = (2, 1)$: one more assembly unit is worth 2, and one more board is worth 1, locally. Each scenario below changes the *original* model independently.

Scenario	Test	Verdict	New optimum
New model D: $a_D = (1, 1)$, $\gamma_D = 3.5$	$r_D = 3.5 - (2, 1) \cdot (1, 1) = +0.5$	basis fails; x_D enters, x_B leaves ($x_D \leq 6$)	$(x_A, x_D) = (2, 6)$, $z = 29$
Assembly $8 + \delta$	$B^{-1}(8 + \delta, 10) = (2 - \delta, 6 + 2\delta) \geq 0$	basis kept for $-3 \leq \delta \leq 2$	$z = 26 + 2\delta$; at $\delta = 1$: $(1, 8, 0)$, 28
Profit of A: $4 + \varepsilon$	row of x_A added: $r = (-1, \varepsilon - 2, -1 - \varepsilon)$	basis kept for $-1 \leq \varepsilon \leq 2$	$(2, 6, 0)$, $z = 26 + 2\varepsilon$

For model D, the pivot gives $z = 29 - \frac{1}{2}x_B - \frac{3}{2}x_C - 3s_1 - \frac{1}{2}s_2$, so the certificate is immediate. All three scenarios, and the extra ones below, were checked by solving the modified LPs exactly.

5.11 Techniques, Classic Problems and Gotchas

Technique: Exam-speed sensitivity

1. From the final tableau, read B^{-1} as *minus* the slack columns, p as *minus* the slack reduced costs, and x_B as the constants.
2. **New activity:** $r_u = \gamma - p^\top a$. Its tableau column is $-B^{-1}a$.

3. **RHS change** δe_i : new constants $x_B + \delta B^{-1}e_i \geq 0$ give the range; the value is $v + p_i\delta$ inside it.
4. **Nonbasic profit change** t : keep the basis while $t \leq -r_j$.
5. **Basic profit change** ε : add $\varepsilon \times$ (its row) to the objective row; keep the basis while all coefficients are ≤ 0 .
6. **Outside a range**: an infeasible constant calls for a dual simplex pivot; a positive objective coefficient calls for an ordinary pivot.

Technique: Theory checklist: what to state and how to prove it

- **Tableau of a basis** (Thm 5.2.2): split $Ax = Bx_B + A_Nx_N$, solve for x_B , substitute into $c^\top x$. Uniqueness: plug $x_N = 0$, then $x_N = e_j$.
- **Basic direction** (Prop 5.3.2): $Ad = 0$ with $d_j = 1$ gives $Bd_B + A_j = 0$.
- **Ratio test** (Thm 5.3.3): check coordinates one by one; only rows with $d_{B(k)} < 0$ bound θ . Gain: $c^\top d = r_j$.
- **Exchange** (Thm 5.3.4): $B^{-1}B'$ is I with one column replaced by u , so $\det = u_\ell \neq 0$; the new point vanishes on the new nonbasics.
- **Optimality** (Thm 5.4.2): $c^\top u - c^\top x = \sum_N r_j u_j$. Converse: an improving step exists when $\theta_{\max} > 0$. **Uniqueness**: all terms zero with $r_j < 0$.
- **Termination** (Thm 5.4.5): strict increase, finitely many bases. **Existence** (Thm 5.4.8): purification by induction on the support.
- p is **dual-optimal** (Thm 5.5.3): $r \leq 0$ on original columns gives $A^\top p \geq c$, and on slacks gives $p \geq 0$; values match; weak duality inserts $y^\top Ax$.
- **Shadow prices and ranges** (Props 5.8.1, 5.8.2): b does not enter r ; value $c_B^\top B^{-1}(b + \Delta b)$; one inequality per row.
- **Dual simplex** (Thm 5.8.4): z_0 is an upper bound; the minimum-ratio rule keeps $r \leq 0$; a row with no positive coefficient proves infeasibility.
- **Value functions** (Thm 5.8.6, Prop 5.9.4): concave in b by mixing optimal plans; convex in c as a maximum over vertices.

Classic problems

- *Production example: boards become $10 + \delta$. Range and value?* $g = B^{-1}e_2 = (1, -1)$, so $2 + \delta \geq 0$ and $6 - \delta \geq 0$: $-2 \leq \delta \leq 6$, with $z = 26 + \delta$ (slope $p_2 = 1$). At $\delta = 6$: $(8, 0, 0)$, 32.
- *Assembly becomes 11 ($\delta = 3$, beyond the range)?* $x_A = -1 + s_1 - s_2$ is infeasible, and only s_1 has a positive coefficient, so s_1 enters and x_A leaves: $x_B = 10 - 2x_A - x_C - s_2$, $z = 30 - 2x_A - x_C - 3s_2$. The optimum is $(0, 10, 0)$ with 30. Now s_1 is basic, so $p_1 = 0$: extra assembly time is worth nothing.
- *A product E with $a = (1, 2)$ earning 3?* $r_E = 3 - (2 \cdot 1 + 1 \cdot 2) = -1$: do not make it. It breaks even at a profit of 4.
- *How profitable must model C become before it is worth making?* $r_C = -1$, so by Proposition 5.9.1(b) its profit must exceed $2 + 1 = 3$.
- *Running example: shadow price and range of resource 2?* $p_2 = 1$, and $g = B^{-1}e_2 = (1, -1)$, so $2 + \delta \geq 0$ and $2 - \delta \geq 0$: $-2 \leq \delta \leq 2$.

- *Prove: if the optimal basis has every reduced cost strictly negative, a small change of any one c_j does not change the optimal plan.* By Corollary 5.9.2 and Proposition 5.9.1(b), each range has $\varepsilon_{\max} > 0 > \varepsilon_{\min}$, because every $-r_j > 0$.

Gotcha

- The tableau shows $-B^{-1}$ in the slack columns, and $-p$ as the slack reduced costs. Two minus signs, both easy to lose.
- These are **maximisation** conventions: $r_j > 0$ means improving. Many textbooks (for example Bertsimas–Tsitsiklis) minimise, and there $\bar{c}_j < 0$ means improving. Translate before comparing.
- A shadow price holds only on its range. Beyond it, the basis changes and so does the slope.
- A change in a *basic* profit moves p , so every reduced cost must be rechecked, not just one.
- Changing b never breaks the reduced costs; changing c never breaks feasibility. Know which test each change can affect (Proposition 5.6.1).
- Part (2) of Theorem 5.4.2 fails at degenerate points (Example 5.4.4).
- The value is *concave* in b and *convex* in c ; do not mix them up.

Notes on the original slides

Every tableau, range and pivot on the Lecture 5 slides was recomputed, both by hand and by solving each modified LP exactly. All are correct. Points worth knowing:

- **Slides 10 and 20** make the same point (B^{-1} from the slack columns) twice. Corollary 5.2.3 states it once, in general, and Theorem 5.2.2 proves why it holds.
- **Slides 34–35** repair infeasibility by choosing the leaving variable first, the negative basic one. This is the *dual simplex* method, which the slides use without naming. Theorem 5.8.4 gives the entering rule and proves it.
- **Slide 15**, part 2, needs nondegeneracy, and Example 5.4.4 shows why. The slides prove it but give no counterexample.
- **Slide 24** introduces $p^\top = c_B^\top B^{-1}$ as “resource multipliers”. These are exactly the optimal dual variables of Chapter 4 (Theorem 5.5.3).
- **Slide 36**: the plot’s vertical axis marks 9 and 12. The curve also passes through (0, 10) and reaches 0 at $\delta = -4$. Its concavity is Theorem 5.8.6.

Chapter takeaways

Summary: Simplex tableau and sensitivity

- The tableau of a basis is the unique formula $x_B = B^{-1}b - B^{-1}A_Nx_N$, $z = c_B^\top B^{-1}b + r_N^\top x_N$. Everything else is read from it.
- A simplex move follows $d_B = -B^{-1}A_j$, gains r_j per unit, and stops at the ratio-test bound on a new BFS. B is optimal when $B^{-1}b \geq 0$ and $r \leq 0$; then $p^\top = c_B^\top B^{-1}$ is dual-optimal.
- New activity: $r_u = \gamma - p^\top a$. New b : keep B while $B^{-1}(b + \Delta b) \geq 0$, with value $v + p^\top \Delta b$. New c : keep B while $r' \leq 0$.
- Outside a range, pivot from the saved tableau: dual simplex for a negative constant, primal for a positive r_j . The value is concave piecewise linear in b , and convex in c .